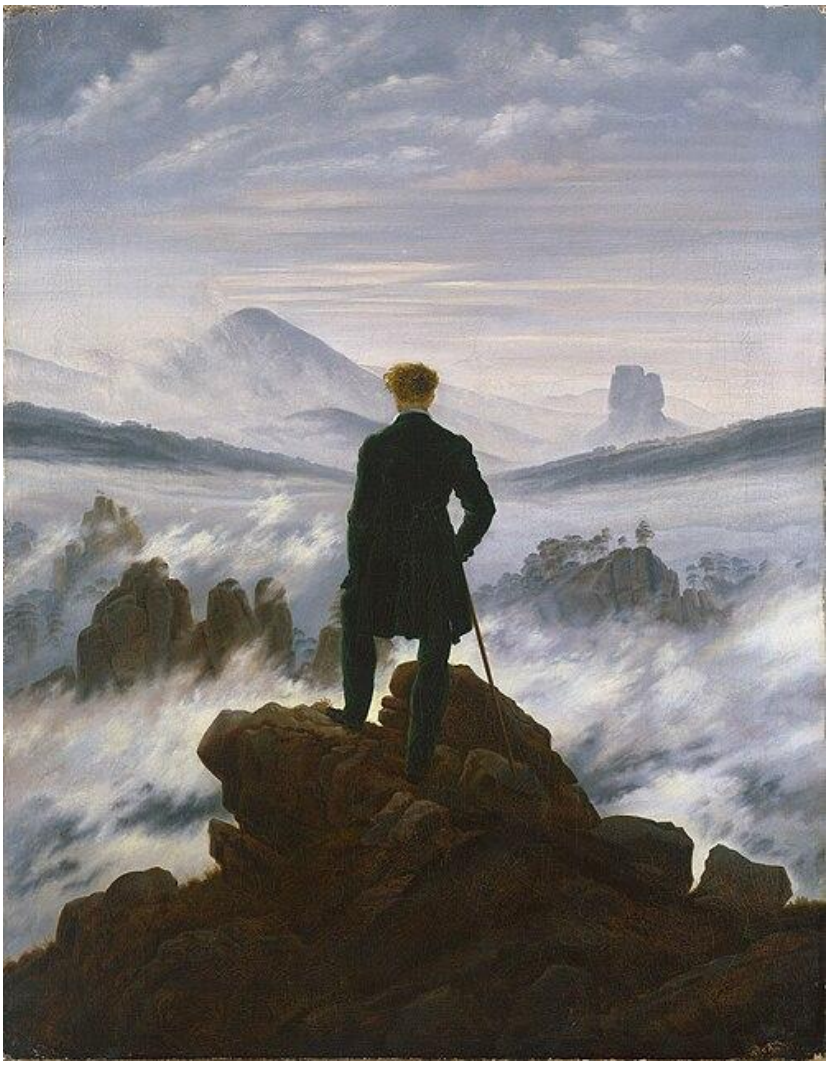
Draft

Boundaries of the Niche

Exergy, Institutional Claims, and Biological Continuity in Coupled Human Systems

Henrik Hagnell

12 November 2026



Caspar David Friedrich “Wanderer Above the Sea of Fog” (1818)

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Reading Progression

This book examines how human societies interact with the physical world, how the institutions governing them represent that world, and how human biology responds when those representations fail. The framework is divided into three distinct but coupled layers:

Part I: The Substrate (Chapters 1–3)

The foundation of the model. These chapters explore the physical reality of how human configurations capture energy, build persistent infrastructure, and maintain temporal buffers to survive. They introduce the core thermodynamic conflict: what happens when the physical and material foundations of a society begin to decay or stall, but its central administrative core demands continuous high-speed output.

Part II: The Coordinate System (Chapters 4–8)

The institutional lens. These chapters examine how economic metrics, financial claims, and academic science create a functional but inherently limited map of reality. They explain how institutions become trapped by their own success—aggressively defending their accounting ledgers, debt structures, and operating assumptions while remaining structurally blind to the actual physical boundaries they are crossing.

Part III: The Escape Hatches (Chapters 9–13)

The biological and social response. The final chapters shift from institutional failure to human survival. They detail how human populations pre-consciously detect structural decay, leading to demographic decoupling, falling fertility, and the withdrawal of prime-age cooperation. Rather than treating these shifts as policy failures, this section frames them as deeply conserved evolutionary survival mechanisms—the biological escape hatches that allow a population to detach from a failing institution and reorganise within a successor regime.

Scenario Application section (Chapter 14)

Rules for how a real-world scenario should be mapped to the framework described by chapter 1-13, including: retaining path dependencies, avoiding flattening and elimination of “unknown unknowns” and maintaining multiplicity rather than prematurely aggregating factors.

1 Dissipative structures and niche construction

Biological activity couples rapid exergy use to longer-lived changes in material stocks and habitats. These changes affect later biological activity and can influence evolution by altering selection. Dissipative Structure Theory (DST) describes driven organisation; Niche Construction Theory (NCT) describes reciprocal organism–environment effects. Their proposed coupling is a structural framework, with no assumed equilibrium, inherent stability or direction of progress.

1.1 Throughput and retained material change

Ilya Prigogine showed how irreversible processes can support organised dynamic patterns away from thermodynamic equilibrium. Biological organisation likewise depends on usable energy and material exchange. Thermodynamic driving forces include temperature differences and differences in chemical or electrochemical potential. They enable transfers and reactions; the resulting rates also depend on kinetics and transport. [1]

Exergy is the maximum work obtainable relative to a specified reference environment. Energy is conserved; irreversible processes generate entropy and destroy exergy. Here, “fast” refers to biological use and response relative to environmental change, rather than rapid variation of the driving source. [2]

For a declared system boundary, the entropy balance is:

$$\frac{dS}{dt} = \Phi_S + \sigma, \quad \sigma \geq 0$$

S is the system’s entropy in J/K; Φ_S is the net entropy transfer rate into it; σ is the internal entropy production rate. Both rates have units J/(K·s), with t measured in seconds. Φ_S can be negative, so the system’s entropy need not increase at every instant. This balance does not determine whether a biological configuration persists. [1, 2]

Material stores can retain chemical or other potential-energy differences and release exergy later. Organisms also transform habitat structure, water retention and nutrient distribution. These environmental configurations can redirect future flows without themselves being fuel. The slow component therefore includes both stored exergy and persistent material organisation.

For one selected exergy store *B*, relative to a fixed reference environment:

$$\frac{dB}{dt} = P_{in} - P_{out} - P_{dest}$$

B is stored exergy in joules. P_{in} and P_{out} are exergy transfer rates into and out of the store; P_{dest} is exergy destroyed by irreversible processes inside it. All three are non-negative rates in watts (J/s), counted once within the same boundary. The store can accumulate or deplete; replenishment is not presumed. [2]

1.2 Reciprocal biological and environmental change

Richard Lewontin emphasised reciprocal organism–environment causation. John Odling-Smee, Kevin Laland and Marcus Feldman developed NCT around such effects: organisms modify environmental conditions, potentially changing selection in their own or other populations. Effects on later generations through modified environments constitute ecological inheritance. Modification need not be intentional, beneficial to the constructor or stored as fuel. [3, 4]

For example, plants store solar exergy in organic compounds. Roots and litter can also alter soil structure, nutrient cycling and water retention. This can change development and selection; evolution through this selective feedback requires heritable variation and differential reproduction. Physiological or developmental adjustment alone is not genetic evolution. [3–5]

1.3 Minimal coupled state equations

Let the state vectors *b(t)*, *m(t)* and *a(t)* describe biological activity and populations, material environmental states (including B and habitat properties), and inherited trait or allele distributions, respectively. External driving *u(t)* specifies boundary conditions such as light, temperature and material inflows.

$$\frac{\mathrm{db}}{\mathrm{dt}} = \mathrm{F_b}(\mathrm{b}, \mathrm{m}, \mathrm{a}, \mathrm{u(t)})$$
$$\frac{\mathrm{dm}}{\mathrm{dt}} = \mathrm{F_m}(\mathrm{b}, \mathrm{m}, \mathrm{a}, \mathrm{u(t)})$$
$$\frac{\mathrm{da}}{\mathrm{dt}} = \mathrm{F_a}(\mathrm{b}, \mathrm{m}, \mathrm{a}, \mathrm{u(t)})$$

F_b, F_m and F_a are system-specific rate functions. Each component has the units of its corresponding state component per unit time. The functions must respect physical balances and biological constraints. Biological activity changes m; m changes biological activity and selection; changes in a can then alter activity and environmental modification. These dependencies specify reciprocal causation, not predetermined feedback signs.

1.4 Distinct timescales

Symbol	Meaning
t	**Elapsed clock time shared by all equations.**
τ_b	**Characteristic duration of a specified biological response or turnover process.**
τ_m	**Characteristic duration of change in a specified material store or habitat property.**
τ_a	**Characteristic duration of evolutionary change in inherited traits or allele frequencies.**
τ_u	**Characteristic duration of change in a specified external driving condition.**

All times use the same units. Each τ describes a process in a specified regime; one environment can contain several timescales. The fast–slow version assumes $\tau_b \ll \tau_m$. The ordering of τ_a and τ_m is empirical, and ecological and evolutionary change can overlap. [5] Slow accumulation can coexist with rapid discharge. Feedback delay is a latency, distinct from response duration; generation time is distinct from evolutionary response time.

1.5 Terms

Term	Definition
Thermodynamic driving force	**A physical disequilibrium, such as a temperature or chemical-potential difference, that drives irreversible processes.**
Exergy	**Maximum work obtainable relative to a specified reference environment, measured in joules. This is a thermodynamic bound, not a guarantee of biological access.**
Dissipative structure	**An organised dynamic pattern maintained by irreversible processes and continuing throughput under non-equilibrium conditions.**
Stock and flux	**A stock is an accumulated quantity, such as joules or kilograms. A flux is a transfer rate, such as watts or kilograms per second.**
Niche construction	**Organism-mediated modification of environmental conditions that affects selection on the same or other populations.**
Ecological inheritance	**Effects of organism-modified environments on subsequent generations, distinct from the transmission of genes.**
Evolutionary niche	**The selection pressures experienced by a population. It is not synonymous with a region of dynamical state space.**
Steady state	**A state whose variables remain constant while flows may continue. Thermodynamic equilibrium has no net irreversible processes; a steady state need not be equilibrium.**

1.6 Dynamical outcomes

The coupling can support persistence, oscillation, reorganisation or disappearance of a biological configuration. Neither the second law nor natural selection guarantees stability, continued growth or long-term persistence. No universal maximisation of dissipation is assumed. Predictions require specified and tested rate functions. Specific biological applications can add mechanisms as separate layers without redefining this thermodynamic–biological foundation.

1.7 References

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2 Institutional State-Space, Epistemic Mapping, and Structural Cannibalisation

This chapter applies the thermodynamic and biological framework established in the core Dissipative Structure Theory and Niche Construction Theory (DST-NCT) layer to human social institutions. It maps the emergence, expansion, and transition of human institutional architectures directly onto the minimal coupled state equations without relying on normative or teleological assumptions.

Methodological Note on Conditional Regimes: Throughout this work, institutional and biological trajectories—such as niche lock, structural cannibalisation, and demographic decoupling—are specified as *conditional dynamical regimes* rather than unconditional historical prophecies. Their generality is bounded by their stated physical and institutional preconditions (e.g., $\mathrm{dB}/\mathrm{dt} < 0$, $\tau_{\mathrm{fin}} \ll \tau_{\mathrm{m}} < \tau_{\mathrm{inst}}$, and incumbent claims rigidity), not by rhetorical hedging. Once those boundary conditions are satisfied within a given basin, the resulting trajectory follows as a structural consequence of the coupled state equations; diluting the causal mechanism with probabilistic qualifiers or omitting its biological coupling to *b(t)* obscures the exact regime dynamics the framework is designed to isolate.

2.1 Mapping the Coupled State Equations to Institutional Variables

Human institutions are specific, highly ordered configurations maintained by continuous exergy throughput. The variables from the core layer map onto human socio-ecological systems as follows:

- **b(t) – Human Biological and Metabolic Activity:** Encompasses human populations, demographic structures, metabolic labour, and physical energy expenditure.
  Within this socio-ecological instantiation, heritable genetic, epigenetic, and ontogenetic traits of the human population reside within the biological state vector *b(t)* (examined in Chapters 9 to 13), while *a(t)* specifically tracks cultural, legal, and institutional inheritance.

- **m(t) – The Constructed Institutional Niche:** Encompasses all anthropogenically modified material and environmental states, including infrastructure, agricultural soils, energy grids, material supply chains, and physically stored reserves. Within *m(t)*, *B(t)* denotes stored usable exergy that can later be discharged. Persistent constructed structures are separate material state components: they embody earlier material and exergy expenditure but are not stores of recoverable construction energy.

- **a(t) – Institutional Infrastructure (Ecological Inheritance):** Replaces strict genetic allele distributions with inherited cultural and institutional traits. This includes legal codes, market mechanics, bureaucratic procedures, financial ledgers, and operational models that coordinate *b(t)*.

- **u(t) – External Driving Conditions:** Encompasses basal solar flux, geophysical resource distributions, and climatic realities independent of human construction.

Institutions function by using the encoded rule-sets in *a(t)* to coordinate biological labour *b(t)* and redirect exergy and material flows under the boundary conditions represented by *u(t)*. Some exergy is consumed in current activity, some may be retained as stored exergy *B(t)*, and some is used with materials to construct or maintain persistent states in *m(t)*. The resulting feedback alters the selective pressures on *a(t)* and *b(t)*.

Multiple institutional configurations can coexist within the same physical and biological environment. They may share infrastructure and resource flows while having different dependencies, capacities and conditions for continued operation. Consequently, physical constraints can cause some institutions to contract while others expand or reorganise. The trajectory of the wider system depends on these interactions, not simply on the trajectory of a single institution.

Relational Trust, Verification Overhead, and Scale-Transition: Within the coupled state-space, social and institutional trust is not an independent stock or cultural constant, but an emergent property of the coupling between biological planning horizons in *b(t)* and institutional rules and claims in *a(t)* under reliable material buffering in *m(t)*. Thermodynamically, trust functions as an exergy-saving coordination mechanism: when actors expect an institution or counterparty to provide accurate information, honour forward claims (a_{claims}), and maintain care boundaries over a planning horizon (H), complex division of labour proceeds with minimal expenditure of energy and labour on real-time monitoring, auditing, and coercive enforcement. Because relational expectations are directional, domain-specific, and scale-dependent, they do not form a homogeneous aggregate. Impersonal, macro-scale trust across large populations requires continuous validation by predictable material flows and buffer coverage ($T_{cover} > T_{recur}$). When physical contraction, representational divergence, or structural cannibalisation invalidates these expectations, trust does not simply vanish; it undergoes a topological scale-transition. Actors withdraw confidence from low-verification, impersonal macro-channels in N_0 and re-route cooperation into high-density, high-verification local, kin, or coalition networks (see Chapters 10–12), while the incumbent institution faces rising verification and enforcement overhead that accelerates its metabolic strain.

2.2 The Institutional Niche and Exergy Accumulation

A **historical growth niche** occurs when available exergy throughput and accumulated material capacities support sustained expansion of biological and institutional activity. Such expansion may involve increasing exergy reserves *B(t)*, persistent structural capacities within *m(t)*, or improved access to and utilisation of existing resources. Growth in one component does not require equivalent growth in the others.

Because institutions operate as self-preserving algorithms maximising their internal order, institutional complexity expands where continuing resource flows support its construction, operation, and maintenance. While maintaining ordered complexity requires continuous exergy throughput and entropy dissipation (Φ_S and σ) [3], the exact relationship between institutional complexity and required throughput depends on the specific physical and institutional configuration rather than a single universal proportionality [1].

2.3 Distinct Institutional Timescales

Institutions introduce new operational and representational timescales that couple with the base biological and material timescales (τ_b, τ_m, τ_a, τ_u).

Symbol	Meaning
τ_{fin}	**Fast Symbolic and Operational Timescale:** The characteristic duration of high-frequency symbolic transactions, algorithmic coordination, and automated operational feedback (e.g., financial markets, credit cycles, algorithmic logistics, high-velocity information cascades, and automated cyber or military decision-support). Typically milliseconds to months.
τ_{inst}	**Institutional Adjustment Timescale:** The duration required to fundamentally alter the inherited operational trait vector *a(t)* (e.g., shifting legal regimes, reforming primary energy policy, overhauling pension systems). Typically years to decades.
H	**Representational Horizon:** The forward temporal distance over which a particular institution or institutional configuration projects expected stability in *m(t)* and *u(t)* to justify present energy allocations and debt commitments. Different institutions may operate with different horizons.

Temporal Mismatch: In the core layer, the fast–slow dynamic assumes $\tau_b \ll \tau_m$.. The human institutional crisis emerges when commitments, allocation demands, and automated reactions operating on the fast symbolic timescale ($\tau_{fin} \ll \tau_m, \tau_{inst}$) outrun both physical regeneration and institutional deliberation. In the economic domain, financial claims induce throughput demands faster than stored exergy *B(t)* can be replenished or persistent structures in *m(t)* renewed. In technological, informational, and security domains—such as algorithmic resource routing, rapid media cascades, or automated military and cyber decision-support—operational reaction times are compressed far below the institutional adjustment and verification timescale (τ_{inst}) required for human diplomatic, legal, or structural correction, locking locally rational responses into escalating feedback loops [2].

2.4 Terms and Institutional Mechanisms

Term	Definition
Institutional Extrapolation	The process by which institutions encode historically transient, high-exergy states of *m(t)* directly into the trait vector *a(t)*, treating them as permanent rather than historically contingent conditions.
Representational Divergence	A growing physical mismatch where the inherited measurement systems in *a(t)* (e.g., GDP, market valuation) continue to register internal coherence while the physical variables *m(t)* and *u(t)* they describe fundamentally shift.
Financialisation	Increasing reliance on financial claims, valuations and financial allocation mechanisms within the institutional state *a(t)* to coordinate present and anticipated future activity.
Claim–Substrate Conflation	The representational error of treating the existence, monetary value or apparent liquidity of financial claims in *a(t)* as evidence of the existence, substitutability or timely availability of the physical exergy, materials or structural capacities required to realise those claims.
Structural Cannibalisation	A systemic triage response in which an institution increases withdrawals from stored exergy *B(t)* and/or consumes, under-maintains, dismantles, or repurposes persistent structural components of *m(t)* to sustain activities and commitments coordinated through its fast symbolic subsystem (τ_{fin}), thereby degrading the physical niche that supports continued operation.
Rate-Induced Tipping	A macroscopic transition forced when the rate of change in the external driving conditions τ_u or the material environment τ_m outpaces the maximum possible institutional adjustment rate τ_{inst}.

Verification Overhead	The metabolic, informational, and administrative throughput required to audit, monitor, and enforce compliance across *a(t)* and *b(t)*. Varies inversely with functional institutional trust within a given coordination regime.

2.5 Dynamical Outcomes: Path Dependencies and Triage

As established in the core layer, the second law of thermodynamics does not guarantee continued growth or long-term persistence. Physical constraint can appear as sustained depletion of relevant stored exergy *(dB/dt < 0)*, deterioration of structural components of *m(t)* faster than maintenance or replacement, or both. If the resulting throughput and structural capacities fall below those required by the existing institutional configuration, that configuration cannot be maintained without reallocation, substitution, simplification, or additional inflow.

When institutional rule-sets *a(t)* become specialised for historically established material conditions, the configuration exhibits **niche lock** [4]. Where adaptation would invalidate the authority, resource-capture mechanisms, or accumulated commitments of incumbent actors, the institution faces strong algorithmic pressures to preserve established flows rather than undertake a managed descent [1,2]. This drives **structural cannibalisation**: the institution uses its remaining authority to sustain fast symbolic (τ_{fin}) commitments and operations by depleting stored exergy *B(t)* and consuming, under-maintaining, or repurposing persistent material capacities within *m(t)*, progressively weakening the physical conditions required for its own reproduction.

The resulting constraints force a systemic triage of biological activity *b(t)* and material infrastructure across institutions and populations. Because the coupled state-space *x(t)=(b(t),m(t),a(t),u(t))* contains no unembodied residual category, resource redirection and cost-shifting cannot terminate in abstract "economic activities"; uncompensated withdrawals must physically register either as accelerated structural degradation D_s within *m(t)* or as metabolic, developmental, and demographic compression within *b(t)*. Dominant administrative, financial, or productive cores preserve their own operations by routing energy inward, transferring costs, and exposing peripheral demographic groups and infrastructures to thermodynamic volatility and deferred maintenance. While these responses accelerate the exhaustion of the buffers required to hide representational divergence within the incumbent configuration, competing or successor configurations can simultaneously reorganise remaining resources and capacities (examined at the biological-institutional interface in Chapter 12). Bifurcation must therefore be assessed for the specific institutional configuration concerned, rather than inferred as the simultaneous collapse of the wider socio-ecological system.

2.6 References

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3 Stored Exergy, Structural Assets, and Temporal Buffering

This chapter clarifies the physical storage and persistence layer of the coupled Dissipative Structure Theory and Niche Construction Theory (DST-NCT) model. Human activity can retain usable exergy for later discharge and can also use earlier throughput to create persistent material structures that alter later flows. These are both components of the constructed material niche *m(t)*, but they are not the same kind of stock and must not be treated as thermodynamically interchangeable. [1-3]

The distinction is required for institutional applications. Grain in a granary, fuel in a tank and charge in a battery can release stored exergy later. The granary building, road, seawall, irrigation system or electricity network is instead a persistent configuration of matter produced through earlier expenditure. Its construction energy has largely been dissipated; what persists is a physical arrangement that changes later access, losses, transport, conversion or exposure to disturbance. Niche construction can therefore preserve either a reserve available for later discharge or a structure that changes the conditions under which later discharge and throughput occur. [3]

3.1 Material Niche: Separating Exergy Reserves from Persistent Structure

For the present layer, the material state vector can be written schematically as:

$$m(t) = (B(t), s(t), \dots)$$

B(t) denotes a selected store of usable exergy relative to a stated reference environment. It is measured in joules and can be accumulated or discharged. *s(t)* denotes a vector of persistent constructed material states. Its components may have different physical units or may be represented by domain-specific functional capacities; they are not converted into a single energy-equivalent stock. The ellipsis represents other material or environmental components already contained in *m(t)*.

This decomposition is conceptual rather than ontological: one installation can contain both kinds of state. A hydroelectric reservoir, for example, may contain stored gravitational exergy in the elevated water while the dam, turbines and transmission structures belong to *s(t)*. Likewise, a granary contains a chemical-exergy reserve B while the building and storage system are structural components that can reduce spoilage, control access and alter delivery capacity.

The two components therefore interact. Structure can increase the accessible inflow to B, reduce internal losses, increase the rate at which a reserve can be delivered, or reduce the amount of exergy required for a given activity. Conversely, constructing and maintaining structure requires exergy and materials. The DST-NCT coupling is consequently not a conversion of energy into an equivalent quantity of "capital"; it is a coupling between throughput, retained exergy and persistent material organisation.

3.2 Stored Exergy and Three Dynamic Uses

For one selected exergy store B, the core balance remains:

$$dB/dt = P_{in} - P_{out} - P_{dest}$$

P_{in} is the exergy transfer rate into the store, P_{out} is useful exergy transferred out, and P_{dest} is exergy destroyed by irreversible processes within the chosen boundary. All are rates in watts. The balance does not assume replenishment and does not specify the institutional purpose of a withdrawal. [2]

Quantity and delivery rate are separate constraints. A large B can support a long-duration deficit while still being unable to support a rapid mobilisation if its accessible discharge power is too low. Storage analysis therefore requires both the amount of usable exergy and the power at which it can be delivered. This distinction is standard in energy-storage engineering. [4]

Use	Physical function	Minimal diagnostic
Buffering	Sustain an existing activity through a temporary mismatch between ordinary inflow and required throughput. The mismatch may be predictable, such as seasonality, or unexpected, such as disruption.	Coverage time T_{cover} = $B_{available}$ / $P_{shortfall}$, subject to sufficient delivery power.
Pulse mobilisation	Temporarily support activity above the power or quantity available from ordinary contemporaneous inflow. Slow accumulation can therefore be discharged as a concentrated pulse.	Required additional power and the concentration ratio T_{accum} / T_{pulse}.
Structural conversion	Use exergy and materials to construct, restore or alter persistent material conditions governing later flows. The lasting result belongs to *s(t)*, not to B.	Construction duration and persistence of the resulting structural function; where applicable, changes in later inflow, losses, delivery capacity or exergy requirement.

These are uses of available energetic capacity, not three mutually exclusive asset classes. The same reserve can perform different functions at different times. Buffering is defined relative to an existing requirement; pulse mobilisation is defined relative to ordinary contemporaneous throughput; structural conversion is defined by the persistence of the material change produced by the expenditure. A construction project may therefore involve pulse mobilisation during building and structural conversion as its enduring consequence.

Structural conversion does not imply that all construction must be financed energetically from a prior reserve. Construction can use current throughput, imported exergy or a stored reserve. The category identifies what a discharge accomplishes when stored exergy is directed into persistent niche modification.

3.3 Structural Assets as Persistent Flow Modifiers

A structural component $s_i(t)$ should be represented by the physical or functional property relevant to the model rather than by the historical energy spent to create it. Examples include transport capacity, water-retention capacity, insulation performance, load-bearing capacity, storage protection, conversion efficiency or protection against a specified disturbance. The suitable variable and units are system-specific.

A generic structural balance can be written as:

$$ds/dt = G_s(P_{conv}, material, labour) + R_s(P_{maint}, material, labour) - D_s(s, u, use)$$

G_s is the rate at which construction or replacement creates the selected structural state; R_s is restoration through maintenance; D_s is deterioration of that state under the specified environment and use. The functions are not assumed linear, and their outputs have the units of the selected structural state per unit time. P_{conv} and P_{maint} denote exergy throughput used by construction and

maintenance processes; the exergy is consumed in those processes rather than retained as an equivalent quantity inside s.

Persistent structure can therefore change future flow constraints without supplying fuel. A road can increase delivery capacity; insulation can reduce required heat throughput; a seawall can reduce exposure to a pulse disturbance; a granary structure can reduce spoilage and hence P_{dest} for the stored food. These are structural buffering or flow-modifying effects in an ordinary functional sense, but they should not be confused with buffering by discharge from B.

NCT supplies the biological interpretation: organism-mediated environmental modifications can persist and affect later activity or selection, including through ecological inheritance. Persistence does not imply benefit, stability or self-maintenance; the constructed state can increase, decrease or redirect later energetic requirements. [3]

3.4 Degradation, Maintenance, and Replenishment

The second law constrains this layer through irreversibility and exergy destruction, but it should not be used as a stand-alone equation for infrastructure decay. Thermodynamics does not specify the deterioration rate of a bridge, road, storage building or soil structure; those rates depend on materials, kinetics, loading, and the cumulative duration and intensity of environmental exposure beyond the structure's operational design envelope. What follows from the thermodynamic layer is that storage, maintenance, repair and reconstruction cannot be perfectly reversible: preserving or restoring usable states requires material and exergy throughput and generates entropy. [1, 2]

Two distinct loss processes therefore need to be kept separate. For B, degradation appears as loss of usable exergy, represented within the selected boundary by P_{dest}. For *s(t)*, deterioration is loss of the selected structural state or service capacity and is represented by D_s. Maintenance can slow or reverse part of this loss by consuming current materials and exergy. Engineering life-cycle models similarly treat deterioration and maintenance as coupled processes affecting future structural performance rather than as recovery of the original construction energy. [5]

Replenishment is likewise stock-specific. Replenishing B means transferring new usable exergy into the reserve. Replenishing s means repairing, replacing or reconstructing the material configuration. A system can therefore have a well-filled energy reserve while accumulating a structural maintenance deficit, or possess extensive infrastructure while lacking the usable exergy needed to operate and maintain it.

This distinction also prevents double counting. Exergy used to construct or repair a structure is an energetic throughput and is dissipated through the construction process; the resulting structure is then represented by its persistent physical state. It must not also be counted as if the original construction exergy remained stored and recoverable in full.

Spatial Indivisibility and Per-Capita Maintenance Overhead: Furthermore, coupled infrastructural networks within *s(t)*—such as water and sanitation grids, transport corridors, and electrical distribution systems—exhibit high spatial and physical indivisibility. Their physical deterioration rate D_s is governed by material kinetics, spatial extent, and environmental exposure *u(t)* rather than by current utilisation headcount. Consequently, when a territory undergoes demographic contraction or outward population flight in *b(t)*, the minimum energetic, material, and skilled-labour overhead required for network maintenance R_s cannot be scaled down proportionally. Per-capita maintenance burdens rise non-linearly across the remaining population until the local configuration is forced either to accumulate compensatory debt in a_{claims} or to abandon contiguous segments of *s(t)*.

3.5 Minimal Temporal Diagnostics

The core model already contains characteristic biological, material, inherited-trait and external-driving timescales. This layer adds only the temporal quantities needed to distinguish reserve use, replenishment and structural persistence. They are diagnostics of specified processes, not universal constants.

Symbol	Meaning
T_{cover}	Coverage time: duration for which the usable reserve can cover a stated shortfall. For a constant shortfall, $T_{cover} = B_{available} / P_{shortfall}$.
T_{accum}	Accumulation time: duration over which the exergy used for a later pulse is accumulated.
T_{pulse}	Pulse duration: duration over which the concentrated mobilisation is discharged.
T_{refill}	Refill time: duration required to restore a reserve after a specified withdrawal under the stated replenishment regime.
T_{recur}	Recurrence interval: time between comparable withdrawals or mobilisation events.
$T_{construct}$	Construction or restoration duration for a selected structural modification.
$T_{service}$	Service duration: time for which a selected structural function remains above a stated functional threshold under a specified operating, environmental and maintenance regime.

Three comparisons carry most of the dynamic information without introducing a larger temporal vocabulary. First, buffering requires both sufficient delivery power and T_{cover} at least as long as the relevant deficit. Second, T_{accum}/T_{pulse} measures temporal concentration: a large ratio indicates that slowly accumulated exergy is being released over a much shorter interval. Third, T_{refill}/T_{recur} tests repeated mobilisation. If comparable withdrawals recur faster than the reserve can be refilled, the reserve becomes a progressively depleted supplement to ordinary throughput unless another inflow, substitution or reduction in withdrawal changes the balance.

For persistent structure, $T_{service}/T_{construct}$ provides a simple persistence ratio. A large ratio identifies a modification whose effects endure much longer than the construction episode that produced it. $T_{service}$ is conditional on the maintenance regime; reducing maintenance can shorten it, while repair or replacement can extend effective service. This prevents "long-lived" from being treated as an intrinsic property independent of continuing resource expenditure.

A further timing condition does not require a new symbol: resources must be physically available before the relevant deadline. A reserve may be large enough in total but arrive too slowly, and a replacement structure may be technically feasible but complete only after the function it supports has already failed. Timing and power constraints therefore supplement stock quantities rather than being inferred from them.

3.6 Boundary Between Physical Stocks and Financial Claims

Money, credit, debt, securities and accounting reserves are excluded from B and *s(t)*. They are symbolic and institutional claims that can allocate command over physical flows across actors and time. They may determine who can acquire fuel, materials, labour or infrastructure, and they may schedule claims on future production, but they do not themselves possess the thermodynamic properties of stored exergy or persistent material structure.

Their physical realisation occurs only when institutional rules in *a(t)* cause actual exergy, material or labour flows to be redirected. A financial claim can therefore be an important temporal control variable in later institutional or economic mappings without being added to the physical storage equations in this chapter. This separation is necessary to avoid treating monetary denomination as evidence of physical substitutability.

3.7 Dynamical Implications for the Coupled DST-NCT Model

The separation of B and *s(t)* makes several distinct system states visible. Reserve exhaustion occurs when usable B becomes insufficient for the required deficit or mobilisation. A power constraint occurs when B is sufficient in quantity but cannot be delivered rapidly enough. Replenishment failure occurs when repeated withdrawals outrun inflow. Structural deterioration occurs when construction and maintenance fail to offset D_s for the function being modelled. These states can coexist but should not be collapsed into a single measure of "capacity".

The stocks also create cross-time trade-offs. Discharging B for immediate buffering can preserve an existing activity while reducing the reserve available for later pulses. Discharging B for structural conversion can reduce immediate reserve coverage but create a persistent structure that increases future inflow, lowers future losses or lowers required throughput. The same conversion can instead create a long-lived maintenance burden whose future exergy requirement exceeds the flow advantage it produces. The sign and magnitude of the feedback are therefore empirical properties of the particular niche construction, not assumptions of the model.

Pulse mobilisation can also accelerate structural conversion by concentrating energy and labour over a short interval. This can be dynamically useful when a structure must be completed before a physical deadline, but repeated pulses can deplete B if T_{refill} exceeds T_{recur}. Conversely, accumulated structure can make later pulses possible by raising transport, storage or conversion capacity. The model therefore links slow accumulation, rapid discharge and persistent niche modification without equating them.

Trajectory stability therefore depends on variance as well as secular trend. A coupled system can remain on a long-term growth trajectory in average throughput while experiencing oscillations in *u(t)* or *m(t)* whose downward amplitude periodically exceeds contemporaneous inflow. Because maintaining idle exergy reserves *B(t)* and redundant capacity within *s(t)* carries ongoing maintenance and opportunity costs, institutional rules coordinated on fast symbolic timescales (T_{fin}) face persistent pressure to treat unused buffers as inefficiency. Where lean optimisation strips *B(t)* and narrows structural redundancy during stable intervals, coverage time T_{cover} contracts relative to the recurrence and amplitude of ordinary shocks (T_{recur}). Under those conditions, heightened volatility alone—even without a secular decline in average available exergy—can breach functional thresholds and force emergency structural cannibalisation.

At the institutional level, later chapters can map rules, financial claims and expectations in *a(t)* onto these physical relations. The present layer only establishes what must ultimately be true of the material system: symbolic claims can alter allocation and timing, but buffering, mobilisation, maintenance and structural conversion occur through actual exergy and material flows.

3.8 Terms

Term	Definition
Stored exergy B	Usable exergy retained within a specified store and reference environment for possible later discharge.
Structural state *s(t)*	A vector of persistent constructed material properties or functional

	capacities. It is not measured as the recoverable energy originally spent to create the structure.
Buffering	Use of stored exergy to fill a temporary shortfall relative to an existing required throughput.
Pulse mobilisation	Use of accumulated exergy to provide a temporary increment of throughput above ordinary contemporaneous availability.
Structural conversion	Use of exergy, materials and labour to create, restore or alter persistent material conditions that modify later flows or exposure.
Replenishment	Restoration of a depleted stock: exergy transfer into B, or repair/replacement that restores a selected structural state.
Maintenance	Continuing or periodic material and exergy expenditure that preserves or restores a selected structural function against deterioration.
Deterioration	Loss of a selected structural state or service capacity through wear, corrosion, erosion, fatigue, biological action or other physical processes.
Financial claim	A symbolic or institutional entitlement affecting who may command physical resources and when; not itself a physical exergy or structural stock.

3.9 References

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4 Successor-State Exclusion and Institutional Epistemic Boundaries

"Taboo" here does not mean prohibition on discussing trajectories towards disruption. It denotes a proposed representational asymmetry: incumbent institutions may devote extensive attention to financial crashes, wars, pandemics, ecological tipping, technological upheaval and other pathways by which their present regime could fail, while giving much less admissible analytical space to the possible viability of successor regimes after that failure. The taboo concerns the post-bifurcation future, not the approach to the bifurcation.

This chapter develops that hypothesis within the coupled Dissipative Structure Theory and Niche Construction Theory (DST-NCT) framework. It treats institutional knowledge systems as components of the inherited institutional state $a(t)$, produced and reproduced inside a historically constructed material and social niche. The extension developed here adds a human psychological mapping: perceived loss of an institutional basin can itself become a collective threat, and weak representation of successor states can amplify that threat. The psychological layer is explanatory rather than mathematically constitutive; it must remain compatible with the physical, biological and institutional state-space defined elsewhere in the model.

The required regime-neutral distinction is:

$$\textit{human / socio-ecological viability} \neq \textit{viability of the incumbent institution}$$

Loss of the incumbent institution must not automatically be coded as loss of human or socio-ecological viability. The converse is equally important: a successor regime is not assumed to be viable merely because it is conceivable. Incumbent survival and wider viability are separate empirical questions requiring separate physical, biological and institutional assessment.

4.1 Institutional Basins and Represented Futures

The core model represents coupled biological activity $b(t)$, the material niche $m(t)$, inherited institutional or behavioural state $a(t)$, and external driving conditions $u(t)$. An institutional regime can be represented as a region of this coupled state-space in which a particular configuration of organisations, rules, material structures, resource flows and population behaviour can continue reproducing. Calling this region an institutional basin is a dynamical shorthand; the basin is not identical to a single organisation or to the state vector $a(t)$ alone.

Within such a basin, repeated experience allows institutions to learn locally useful mappings between conditions, interventions and outcomes. Governments, firms, financial organisations, scientific disciplines and professional systems can accumulate measurements, procedures and causal models that are effective precisely because the surrounding state-space has remained within a historically sampled range. Organised agency can consequently acquire substantial real control inside the niche: familiar actions often have sufficiently familiar consequences to support planning, investment and coordination.

The earlier metaphor of a predictable linear slope should therefore be understood as an epistemic and institutional approximation rather than a claim that the underlying equations are literally linear. The relevant feature is that the future is represented mainly through continuous variation of familiar variables and interventions while the basic institutional coordinate system remains intact. Sophisticated nonlinear models can still reproduce this restriction if they model how the incumbent regime is disturbed or destroyed but do not treat post-regime states as ordinary objects of viability analysis.

Let N_0 denote the incumbent institutional basin, Σ its effective basin boundary or separatrix, and N_i a potentially accessible successor regime. The representational asymmetry can be stated schematically as:

$$N_0 \rightarrow \Sigma \rightarrow N_i$$

The approach from N_0 towards Σ may be extensively modelled. Successor-state exclusion arises when analysis becomes sparse, terminal or incumbent-relative after Σ, even though N_i may contain persistent or recurrent human organisation. N_i need not be a static equilibrium: it can be another attractor, an oscillatory regime, a reorganised socio-ecological niche, or a long-lived transient configuration. The hypothesis concerns representation of post-transition viability, not the mathematical form of the successor state.

4.2 Successor-State Exclusion

Successor-state exclusion is the proposed tendency to represent the incumbent basin as the domain of admissible futures and a regime-changing boundary as the point beyond which analysis shifts from ordinary state-space to catastrophe, collapse or terminal loss. The institution can be highly explicit about mechanisms leading to the boundary while remaining comparatively unable or unwilling to represent the other side as a potentially organised state with its own constraints and feedbacks.

This creates a characteristic asymmetry in institutional questions. Analysis inside the incumbent regime asks how a disturbance can be prevented, absorbed, insured against, contained, reversed or followed by restoration. These are legitimate questions. The exclusion appears when the complementary question - what physical, demographic and institutional configurations could remain viable if restoration of the incumbent regime is impossible - is treated as conceptually inadmissible, methodologically external, or reducible to losses measured against the incumbent state.

The term collapse can therefore perform two different functions. It can identify a genuine loss of wider human or socio-ecological viability. It can also mark the boundary at which the variables used by the incumbent institution cease to be well defined. A financial model may lose meaning when its currency, banking architecture or enforceable claims disappear; a political model may lose its actors when relevant states or administrative structures fragment; an economic model may terminate when ownership, labour allocation or exchange systems reorganise. In these cases the model has reached the end of its coordinate system, not necessarily the end of organised human activity.

The hypothesis therefore distinguishes systemic catastrophe relative to N_0 from general viability loss. The former means that the incumbent institutional configuration can no longer reproduce. The latter means that no relevant successor configuration remains viable under the physical and biological constraints. Treating these propositions as equivalent is precisely the representational compression being proposed.

4.3 Crisis Preoccupation as Boundary Vigilance

Under this formulation, extensive public and scientific discussion of crises is not evidence against the hypothesis. It can be one of its predicted signatures. If crossing Σ has extremely high cost to the incumbent regime, then the regime has strong reason to improve detection, prediction and control of trajectories approaching Σ. Crisis research can consequently become more elaborate as niche lock increases.

The important empirical quantity is not the volume of discussion devoted to catastrophe but the distribution of analysis around the boundary. A system can invest heavily in financial stress tests, pandemic propagation models, escalation analysis or climate tipping research while continuing to treat preservation or restoration of N_0 as the implicit criterion of successful intervention. The result can be very sophisticated nonlinear modelling of the route to regime failure combined with comparatively weak modelling of successor-state viability.

This pattern does not require deliberate censorship. It can arise because research funding, data collection, professional competence, legal mandates and organisational evaluation are themselves organised around the continuity of existing institutions. Threat-rigidity research provides one reinforcing mechanism: perceived threat can restrict information processing and constrict control within organisations. [7]

Boundary vigilance can therefore become self-reinforcing. More resources are allocated to preserving N_0; preservation efforts generate additional measurements and expertise about N_0 and Σ; and successor regimes remain relatively poorly mapped. The institutional knowledge system can become increasingly competent at defending its niche while becoming less informative about the wider state-space that becomes relevant if defence fails.

Boundary vigilance is also selective across different segments of Σ. An incumbent configuration does not allocate analytical attention strictly according to physical proximity or aggregate human exposure. Attention concentrates on boundary processes that directly threaten the host core's infrastructure and valued claims in $a_{claims}(t)$, and whose management can be translated into familiar regulatory, fiscal, or engineering instruments within $a(t)$. Conversely, boundary processes that primarily degrade peripheral niches, or whose self-reinforcing dynamics would invalidate the assumption that incumbent policy levers control the trajectory, tend to remain comparatively under-represented even when their physical thresholds are closer.

Yet because the underlying physical system remains coupled, unmonitored boundary crossings do not disappear from $x(t)$. Their physical feedbacks re-enter the incumbent models anonymously as an unattributed acceleration of external forcing $u(t)$, compressing the lead time before the core's heavily monitored boundaries are breached.

4.4 Collective Basin-Loss Threat and Temporal Continuity

The psychological extension begins from a limited analogy with Terror Management Theory (TMT). TMT proposes that mortality salience can activate defences centred on cultural worldviews and valued identities. [11] The present model does not require the stronger claim that mortality is the privileged or unique threat. The General Process Model of Threat and Defense (GPMTD) instead treats mortality, uncontrollability, relationship threat and other discrepancies as inputs into a more general process: threat produces proximal anxiety and vigilance, while successful resolution or palliative defence restores approach-oriented motivation. [12,13]

For the institutional model, the relevant proposed variable is collective basin-loss threat: concern that the group or institutional configuration through which collective life is presently organised may cease to reproduce. This is not equivalent to fear of biological death. The threatened object can be a state, profession, religious community, monetary system, territorial order, class structure or other collective configuration. What matters is that the threatened configuration supplies a large part of the actor's predictive, coordinative and identity environment.

Collective-angst research provides the closest established construct. Collective angst is concern about an ingroup's future vitality and has been experimentally elicited by physical and symbolic threats to the group's future; it predicts support for group-strengthening responses. [14] This

literature does not establish the DST-NCT basin model, but it demonstrates that prospective loss of group continuity can itself function as a psychologically consequential threat.

Perceived collective continuity (PCC) supplies a complementary variable. PCC describes the extent to which people perceive their group as an entity persisting through time, including historical and cultural continuity. [15] Mortality-salience research has found that people can increase perceived group continuity under death awareness, suggesting that group temporal endurance can itself operate as a psychological resource. [16] Jetten and Wohl further showed that experimentally presented discontinuity between England's past and present increased collective angst among strongly identified participants and increased opposition to immigration; collective angst mediated the effect. [17] The specific political response is context-dependent, but the continuity-threat mechanism is directly relevant here.

The DST-NCT extension therefore uses temporal continuity as a broader model term: the socially shared expectation that collective life has a future that remains sufficiently intelligible for present action to connect with later states. Temporal continuity is perceptual and institutional, not a proof of physical viability. It can refer to persistence of N_0, continuity through transformation into N_i, or a sequence of fallback states. Its stabilising content is not simply "we remain alive" but "there is a future, it is sufficiently legible, and others are expected to recognise broadly compatible coordinates within it."

The danger signal should consequently depend not only on expected loss but on boundary properties. A transition that is perceived as highly irreversible, hysteretic, poorly mapped and destructive of familiar coordinates should generate stronger basin-loss threat than an equally costly disturbance that is reversible and legible. Critical-transition theory supplies the state-space geometry - multiple attractors, thresholds and hysteresis - while the psychological literature supplies candidate mechanisms by which humans respond when such geometry becomes socially salient. [1,11-17]

4.5 The Endogenous Epistemic Trap

The central extension is a feedback loop between successor-state exclusion and collective threat. Successor states are initially poorly represented because institutional knowledge is specialised for N_0. As Σ becomes salient, that representational gap makes the post-boundary future unusually uncertain. The uncertainty increases collective basin-loss threat and collective angst. Threat then increases demand for control, familiar identity, boundary vigilance and institution-preserving action. Those responses direct additional attention, authority and resources towards N_0 and Σ rather than towards the independent modelling of N_i. The successor state therefore becomes even less legible relative to the increasingly well-modelled incumbent basin.

successor-state exclusion → post-boundary uncertainty → basin-loss threat / collective angst → incumbent defence and boundary vigilance → greater successor-state exclusion

This is an endogenous epistemic trap because the representational deficit partly manufactures the psychological conditions that preserve the deficit. A successor regime may be physically feasible but remain difficult to investigate because its low legibility increases the very threat response that favours continued investment in incumbent categories. The loop does not require conscious suppression, a unified institutional actor, or false beliefs at every stage. It can emerge from individually intelligible decisions about funding, professional relevance, risk control, data continuity and organisational survival.

The feedback is expected to be strongest when four conditions coincide: first, crossing Σ is perceived as difficult to reverse; second, N_0 supplies high levels of identity, status, livelihood or

predictive control; third, N_i is represented sparsely or only through losses relative to N_0; and fourth, incumbent institutions retain enough control over knowledge production and resource allocation to redirect effort towards defence. The loop should weaken where multiple successor coordinate systems are already legitimate, rehearsed and materially provisioned.

The mechanism also explains why additional knowledge about catastrophe need not dissolve the taboo. More precise knowledge of the route towards Σ can increase its salience and support better defensive coordination while leaving N_i blank. Under those conditions, expertise about danger can deepen successor-state exclusion rather than correct it. What breaks the loop is not generic information but credible information that expands the represented state-space beyond the incumbent boundary.

4.6 From Individual Threat to Collective Institutional Defence

The model must not treat an institution as a literally anxious organism. The psychological mechanism operates through individuals and relationships, then becomes macroscopic through coordination, selection and institutional feedback. GPMTD supplies an individual-level process, while several group-level literatures describe how threat can be converted into shared identity and organised action.

Uncertainty-identity theory shows that self-uncertainty can increase identification with groups that provide clear, coherent prototypes and boundaries. [18] Group-based control theory shows that threatened personal control can be partly restored through identification with an agentic "we", increasing ingroup and pro-organisational behaviour. [19] These mechanisms provide a plausible bridge from individual discrepancy to collective coordination without assuming that all members share identical motives or responses.

At the organisational level, threat rigidity can narrow information processing and centralise or constrict control. [7] That response maps directly onto boundary vigilance when the organisation treats continuation of N_0 as the relevant control objective. But threat does not have a single group-level outcome. Research on collective resilience in emergencies shows that perceived common fate can also generate shared identity, mutual assistance and coordinated action rather than panic or paralysis. [20] The same threat process can therefore produce either defensive closure or effective collective adaptation depending on the available identities, scripts, information and action possibilities.

In DST-NCT terms, the pathway can be represented without introducing a new state variable. Threat perception and stress responses occur within $b(t)$; shared identities, narratives, doctrines and organisational procedures are components of $a(t)$; their implementation reallocates labour, material and exergy flows in $m(t)$; and the resulting success or failure feeds back into both $b(t)$ and $a(t)$. The psychological mapping therefore changes the transition rates and allocation rules inside the existing coupled system rather than replacing its physical constraints.

4.7 Continuity Technologies: Narrative, Ritual, Doctrine and Rehearsed Futures

Human groups possess recurrent mechanisms for reducing paralysis under consequential uncertainty. The present model groups these provisionally as continuity technologies: socially transmitted devices that make future collective states more legible, connect present action to those states, and establish expectations that other group members will recognise compatible scripts. The term is functional and does not imply deliberate design or factual accuracy.

Narratives and speeches can supply temporal continuity by locating a present crisis within a longer sequence - past, disruption, endurance or transformation, and future. Organised religions can

perform a similar function through shared cosmologies, succession across generations, ritual and community membership; the model makes no claim about theological truth. Ritual psychology provides more specific mechanisms: ritual can regulate emotion, performance goal states and social connection. [21] These mechanisms can reduce unstructured vigilance and increase coordinated behaviour even when the objective hazard remains.

Military and emergency organisations make the same functional logic unusually explicit. Drill, standard operating procedures, rehearsed fallback positions, command succession and stress-inoculation training reduce the number of novel decisions required under threat and provide action sequences after local failure. Stress-inoculation training has shown aggregate reductions in anxiety and improvements in performance across applied settings, although effects depend on design and context. [22] The important property for the present model is not fearlessness but continuity of action: the threatened actor can still locate a next state and a recognised role within it.

Continuity technologies are not inherently conservative. A narrative can insist that only restoration of N0 counts as continuity, thereby strengthening the endogenous epistemic trap. Alternatively, it can represent transformation into N_i as continuity of the group through changed institutions. The same psychological mechanism can therefore support incumbent resilience or transformability. This distinction is central: temporal continuity concerns persistence of an intelligible collective future, not necessarily persistence of the incumbent institutional form.

4.8 Migration and Demographic Change as Conditional Basin-Loss Cues

Migration and demographic turnover are recurrent historical processes and are not intrinsically basin-loss threats. Their psychological effect depends on how population change interacts with existing group boundaries, resource distributions, institutional absorptive capacity, rates of change and narratives of continuity. The same demographic flow can be represented by different groups as routine incorporation, economic replenishment, coalition enlargement, status displacement, cultural discontinuity or institutional overload.

The Jetten and Wohl experiments are useful precisely because they show that perceived historical discontinuity can alter responses to immigration through collective angst among strongly identified participants. [17] They do not establish that migration objectively destroys group continuity, nor that the effect is universal. They demonstrate that demographic change can become a carrier of basin-loss salience when actors interpret it as evidence that a temporally continuous ingroup may not persist in recognisable form.

Within the wider model, the relevant causal sequence is therefore conditional rather than categorical: demographic change interacts with boundary salience, resource and housing constraints, political representation, institutional incorporation mechanisms and perceived continuity. These variables determine whether change is absorbed inside N_0, contributes to transformation towards N_i, or becomes a symbol through which collective basin-loss threat is organised. Different coalitions can occupy different positions in the same state-space and therefore experience the same demographic process as continuity for one group and discontinuity for another.

4.9 Niche Lock, Organisational Learning, and Epistemic Self-Reference

The mechanism becomes stronger when institutional niche construction creates mutual dependence among material structures $m(t)$, biological activity $b(t)$, institutional rules and representations $a(t)$, and accumulated commitments. Specialised infrastructures, professions, data systems, legal categories, financial claims and administrative routines can all increase performance inside N_0 while increasing the cost of moving to a different regime.

Organisational-learning research supplies a partial micro-to-meso mechanism. March distinguishes exploration of new possibilities from exploitation of established competencies and shows how adaptive learning can favour the latter. [3] Levinthal and March describe learning myopia towards distant times, distant places and failures. [4] March, Sproull and Tamuz show the severe inferential difficulty of learning from rare events when history supplies samples of one or fewer. [5] These results do not prove successor-state exclusion, but they explain why repeated success inside a niche can generate increasingly refined knowledge of that niche without proportionate knowledge of qualitatively different regimes.

Structural inertia adds a second mechanism. Hannan and Freeman model organisations as subject to strong inertial pressures that can increase with characteristics such as age, size and complexity. [6] In the present framework, inertia matters epistemically as well as behaviourally: the institutions that would need to conceptualise N_i are themselves specialised products of N_0. Their personnel, categories, data and criteria of relevance may therefore lose meaning across the transition they are attempting to model.

This creates an epistemic self-reference problem. The institution does not observe the state-space from outside it. Its models, research organisations and professional languages are components of the institutional configuration whose persistence is at issue. A successor regime can therefore be physically conceivable while remaining institutionally difficult to represent because representing it requires treating the disappearance of the current representational apparatus as an ordinary boundary condition rather than as terminal failure.

The resulting proposition is deliberately paradoxical: successful niche adaptation can increase local predictive and control performance while reducing the relative representational coverage of the wider state-space. The institution can continue learning, and can even become more technically sophisticated, while learning from a progressively narrower sample of physically and socially possible regimes. The endogenous epistemic trap adds a feedback reason why this narrowing may accelerate as the boundary becomes more salient.

4.10 Scientific Mappings and Their Limits

No established literature directly demonstrates the combined successor-state exclusion and endogenous epistemic-trap hypothesis. The following traditions supply distinct components. They should be treated as mappings and constraints on the proposed synthesis rather than as independent proof of it.

Scientific literature	Mapping to the hypothesis and limitation
Critical transitions and nonlinear dynamics	Bifurcation, multiple attractors, hysteresis and basin transitions provide the state-space geometry within which a regime can lose stability or leave its previous attractor. Scheffer gives a broad synthesis. [1] This literature does not explain the psychological or institutional asymmetry around the boundary.
Resilience and transformability	Walker, Holling, Carpenter and Kinzig distinguish persistence of a system with broadly the same identity and feedbacks from transformability into a fundamentally different system. [2] This is the closest established systems distinction to separating incumbent persistence from successor viability.

Scientific literature	Mapping to the hypothesis and limitation
Organisational learning	Exploration-exploitation trade-offs, learning myopia and rare-event inference explain how repeated success can deepen competence inside a sampled regime while leaving distant or unprecedented states poorly represented. [3-5] These mechanisms operate below the scale of the full societal hypothesis.
Structural inertia and threat rigidity	Structural inertia explains resistance to organisational change; threat-rigidity theory describes restriction of information processing and constriction of control under threat. [6,7] Both can reinforce niche lock, but neither establishes successor-state exclusion by itself.
TMT and general threat defence	TMT supplies the historical mortality-salience analogy, while GPMTD proposes a broader discrepancy-anxiety-defence-approach process across mortality, control and relationship threats. [11-13] These are individual-level theories and do not by themselves establish group or institutional dynamics.
Collective angst and collective continuity	Collective angst operationalises concern for an ingroup's future vitality. PCC operationalises perceived historical and cultural persistence of the group. Experimental work links discontinuity to collective angst and group-protective responses. [14-17] These constructs are close psychological mappings to basin-loss threat and temporal continuity, but they do not imply the DST-NCT geometry.
Uncertainty-identity and group-based control	Self-uncertainty can increase identification with coherent groups, and threatened personal control can be compensated through an agentic collective self. [18,19] These mechanisms provide a micro-to-group bridge but do not determine which institutional strategy a group will select.
Collective resilience	Shared danger can generate common identity, mutual assistance and coordinated action rather than panic. [20] This constrains any simple claim that boundary salience necessarily produces paralysis or rigidity.
Ritual and trained action	Ritual can regulate emotion, performance states and social connection; stress-inoculation can improve performance under stress. [21,22] These literatures support mechanisms by which continuity technologies can preserve action, but not the broader state-space hypothesis.
Socio-technical transitions	Geels and Schot explicitly model transformation, reconfiguration, substitution and de-alignment/re-alignment of incumbent regimes. [8] This demonstrates that successor-regime analysis is possible; the empirical question is whether it enters operational models and provisioning.
Normal science and institutional legibility	Kuhn supplies an analogy for problem-solving inside an established disciplinary matrix, while Scott analyses administrative simplification into actionable categories. [9,10] These are epistemic analogies, not direct evidence for the proposed taboo.

The existence of transformability, transition and collapse literatures is therefore not a falsification. These literatures provide candidate counter-institutions against which the hypothesis can be

tested. The relevant question is whether post-regime analysis remains specialised and peripheral or is integrated into the baseline representations through which major institutions allocate resources, value assets, design infrastructure, organise populations and define long-horizon continuity.

4.11 Scope: Contemporary Western Institutions and Wider Generality

The mechanism should not be defined as uniquely Western. Any institution that reproduces itself through specialised structures and inherited knowledge has a potential distinction between changes it can survive and changes that replace it. Successor-state exclusion and collective basin-loss threat are therefore proposed as general mechanisms that may vary in strength, symbolic form and group boundary across institutional systems rather than as fixed cultural traits.

Contemporary Western industrial societies are nevertheless a strong candidate case because they combine extensive quantitative forecasting, professionalised scientific knowledge, large material infrastructures, complex financial claims, formal risk management and strong expectations of organised intervention. These capacities make the asymmetry especially visible: the same system can possess unusually powerful tools for modelling threats to continuity while still evaluating those threats primarily by their consequences for the incumbent institutional order.

Other historical or cultural systems may encode comparable boundaries through different categories - dynastic termination, loss of religious order, defeat, territorial displacement, lineage extinction, administrative fragmentation or cosmological rupture - without using modern dynamical language. Organised religion, civic mythology, military doctrine and succession institutions can likewise produce temporal-continuity signals through very different symbolic forms. The underlying analytical question is whether post-incumbent collective organisation is represented as a normal possible state or compressed into undifferentiated terminal failure.

The model predicts variation with the degree of niche lock; dependence of knowledge-producing institutions on incumbent resource flows and categories; strength of group identification; perceived reversibility of the transition; availability of credible successor models; institutional capacity for incorporation and adaptation; and the distribution of costs and benefits across coalitions. These are proposed comparative variables, not established universal relationships.

4.12 Dynamical Implications and Empirical Tests

Successor-state exclusion changes the interpretation of representational divergence in the wider DST-NCT model. Divergence need not consist only of inaccurate forecasts inside the incumbent niche. It can also consist of accurate recognition that N_0 is becoming unstable combined with poor representation of the wider set of reachable states. The institution may diagnose its own danger correctly while still mischaracterising the meaning of crossing the boundary.

The psychological extension adds a stronger prediction: representational asymmetry should sometimes amplify the threat response that maintains the asymmetry. The empirical task is therefore to test the direction of causality in the proposed loop, not merely to observe crisis language or group defensiveness.

Observable pattern	Interpretation within the hypothesis
Dense modelling before Σ, sparse modelling after it	Supports successor-state exclusion if crisis trajectories are richly analysed while successor regimes are left undifferentiated, terminal or defined only by missing incumbent outputs.

Observable pattern	Interpretation within the hypothesis
Successor-state legibility reduces collective threat	Supports the epistemic-trap mechanism if credible descriptions of viable post-boundary coordination reduce collective angst or defensive closure while holding expected material costs approximately constant. Mere optimistic reassurance would not be an adequate test.
Boundary salience increases investment in N_0 rather than N_i	Supports the loop if rising threat produces more modelling, funding and control around preservation/restoration while independent successor-state analysis loses relative resources or legitimacy.
Perceived continuity moderates threat responses	Supports the temporal-continuity mechanism if credible historical or future continuity cues reduce collective angst or shift behaviour from defensive closure towards coordinated adaptation. [15-17]
Shared identity produces resilience under some threats	Constrains a simple rigidity model. If common fate generates mutual assistance and effective collective action, as emergency research suggests, the relevant moderator is not threat alone but the identities, scripts and perceived control available under threat. [20]
Model variables terminate when incumbent institutions disappear	Suggests that collapse may partly be a coordinate-system boundary. The inference is stronger if physical, demographic and relational processes remain modelable beyond that point.
Operational institutions maintain rehearsed fallback regimes	Weakens strong successor-state exclusion where organisations routinely provision for several incompatible post-incumbent states and train actors to operate within them rather than merely restore N_0.
Demographic-change effects vary with continuity and incorporation conditions	Supports the conditional migration mapping if collective angst depends on perceived group discontinuity, identification, resource conditions and incorporation capacity rather than demographic change alone. [17]
Threat rises even when N_i is well mapped and controllable	Weakens the proposed epistemic feedback if post-boundary legibility has little effect on collective threat after material loss, irreversibility and group identification are controlled.
Independent viability assessment of N_i becomes more influential as N_0 weakens	Weakens successor-state exclusion and indicates that the knowledge system can expand its coordinate coverage as incumbent viability falls.

The strongest design would combine institutional evidence with psychological measurement. For example, researchers could measure collective angst, perceived collective continuity and perceived control while experimentally varying the legibility and reversibility of a hypothetical regime transition; institutional studies could separately trace whether corresponding threat periods change funding, datasets, planning assumptions and resource allocation. A societal feedback hypothesis requires evidence at more than one level.

Nothing in the hypothesis implies that crossing a bifurcation is desirable, that all successor states are viable, or that preserving N_0 is irrational. The claim is narrower: the physical and biological viability of successor states cannot be inferred from the survival requirements of the incumbent

institution, and the psychological difficulty of representing a successor can itself become part of the incumbent system's dynamics.

4.13 Terms

Term	Definition
Successor-state exclusion	A representational asymmetry in which trajectories threatening an incumbent institutional regime are extensively modelled while potentially viable states conditional on that regime's disappearance are comparatively absent, terminally classified, or represented mainly through losses relative to the incumbent state.
Regime-neutral viability	The requirement to assess human and socio-ecological viability separately from the viability of any particular incumbent institutional configuration. It does not presume that either the incumbent or a successor regime is viable.
Incumbent institutional basin (N_0)	A region of coupled biological, material and institutional state-space within which a particular institutional configuration can continue reproducing. It is a dynamical shorthand, not a separate state variable.
Separatrix / basin boundary (Σ)	The effective boundary between dynamical regimes. Crossing it may change which attractors or recurrent configurations are accessible and may involve hysteresis or difficult return.
Successor regime (N_i)	A post-transition configuration that may support persistent or recurrent human organisation under a different set of feedbacks, structures and constraints. It need not be a static equilibrium and is not assumed viable.
Boundary vigilance	Concentrated observation, modelling and intervention directed at trajectories approaching loss of the incumbent basin. High boundary vigilance is compatible with weak knowledge of successor regimes.
Institutional coordinate system	The variables, categories, actors and causal relations through which an institution represents its environment. A regime transition can invalidate parts of this coordinate system without ending the underlying physical or biological processes.
Collective basin-loss threat	Proposed model term for concern that the group or institutional configuration through which collective life is organised may cease to reproduce. It is broader than mortality threat and is not itself an established psychological construct.
Basin-loss salience	The degree to which possible loss of N_0 and approach to its boundary are cognitively and socially prominent. Expected effects depend on perceived loss, reversibility, post-boundary uncertainty, identification and control.
Collective angst	Established social-psychological construct describing concern about an ingroup's future vitality. [14] It is the closest measured construct to the prospective affective component of collective basin-loss threat.

Term	Definition
Perceived collective continuity (PCC)	Established construct describing the extent to which an ingroup is perceived as a temporally enduring entity, including historical and cultural continuity. [15]
Temporal continuity	Proposed broader model term for a socially shared expectation that collective life has a sufficiently intelligible future for present action to connect with later states. It can be attached to N_0 or to transformation into N_i and must not be confused with demonstrated physical viability.
Successor-state legibility	The degree to which post-boundary states are represented with intelligible actors, constraints, resource flows, institutions and action sequences rather than as an undifferentiated terminal category.
Endogenous epistemic trap	The proposed feedback in which weak successor-state representation increases post-boundary uncertainty and collective threat; defensive responses then concentrate attention and resources on N_0 and Σ, further weakening relative knowledge of N_i.
Group-based control	The restoration of perceived control through identification with an agentic group when personal control is threatened. [19] It is one candidate mechanism connecting individual threat to collective action.
Threat rigidity	Restriction of information processing and constriction of control under perceived threat at individual, group or organisational levels. [7]
Collective resilience	Coordinated mutual support and action that can emerge under shared danger when common fate produces a shared social identity. [20] It is an alternative threat outcome to paralysis or rigid closure.
Continuity technology	Proposed functional term for narratives, rituals, doctrines, rehearsed procedures or succession arrangements that make future collective states legible and socially coordinated. Such mechanisms can preserve N_0 or support transformability into N_i.
Transformability	Following resilience theory, the capacity to create or enter a fundamentally different system when the existing configuration becomes untenable; analytically distinct from resilience that preserves broadly the same identity, structure and feedbacks. [2]

4.14 References

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5 Economic Representation, Claims Space, and Cross-Domain Mapping

This chapter maps familiar economic vocabulary onto the coupled Dissipative Structure Theory and Niche Construction Theory (DST-NCT) framework. It does not add an independent economic substrate, derive economic theory from thermodynamics, or assume that monetary quantities are physical quantities. Its purpose is translational: to show how economic representations, allocation mechanisms and claims relate to the biological, material and institutional states already defined by the model.

The physical and biological ontology therefore remains unchanged. Human populations and metabolic activity are represented in *b(t)*; constructed material states, stored exergy and persistent structures in *m(t)*; inherited rules, legal categories, market mechanics, financial ledgers and operational models in *a(t)*; and external physical driving conditions in *u(t)*. Economic activity emerges through relations among these variables. Economic terms describe selected aspects of those relations and can redirect them through institutional rules, but they are not additional thermodynamic stocks.

The central methodological requirement is separation without isolation. Energetic, material, biological, structural, institutional, relational and monetary variables can be strongly coupled while remaining dimensionally and conceptually distinct. Causal interaction does not imply convertibility into a common scalar. This prevents the economic mapping from flattening exergy, mass, infrastructure, labour, legal entitlement and monetary valuation into interchangeable forms of "capital".

5.1 Economic Variables and the Existing State-Space

The institutional layer already contains the principal mechanisms required for an economic mapping. Rules in *a(t)* coordinate biological activity *b(t)* and redirect exergy, materials and labour within the constructed niche *m(t)*. Market exchange, accounting, contracts, property rules, taxation and financial ledgers are therefore institutional mechanisms for organising access to and expectations about processes that remain physically realised through *b(t)* and *m(t)*. [1]

An economic transaction can consequently have several simultaneous descriptions without those descriptions being equivalent. A factory purchase can be a monetary expenditure, a transfer of ownership, an entry in financial accounts, a reallocation of materials and labour, and a change in future productive capacity. The economic description identifies institutional relations among actors; the physical description identifies the material and energetic processes through which the transaction is realised.

No new general economic state vector is required at this level. Introducing one would risk reproducing the category error that the mapping is intended to avoid. Instead, economic variables are treated as measurements, classifications, expectations and claims encoded within or generated by the institutional state *a(t)*, coupled to the underlying biological and material dynamics.

5.2 Cross-Domain Non-Equivalence and Coupling

The model contains heterogeneous domains that do not share common units. Stored exergy *B(t)* is measured relative to a thermodynamic reference environment; material stocks may be measured by mass, composition or concentration; persistent structures *s(t)* are represented by the functional properties relevant to their use; biological populations and labour have demographic and physiological dimensions; institutional rules and relationships describe socially reproduced constraints; and economic valuations are expressed in monetary units defined by a particular institutional system. [3]

These domains can alter one another through explicit coupling mechanisms. A price change can redirect labour and materials. A legal claim can determine who receives fuel or access to a network. Deterioration of infrastructure can reduce the monetary value of an asset. A shortage of usable exergy can invalidate production plans and alter financial valuations. None of these couplings makes price a measure of joules, ownership a physical stock, or historical construction expenditure a recoverable energy reserve.

economic value ≠ physical quantity ≠ physical capacity

Economic value is a relation produced within an institutional and expectational setting. Physical quantity identifies an amount of matter or energy-related state. Physical capacity identifies what a structure or system can actually do under specified operating conditions. Their empirical correlation can be strong, but the model does not assume identity or stable convertibility among them.

financial liquidity ≠ physical availability ≠ physical delivery rate

Likewise, a claim may be readily exchangeable for money while the corresponding physical resource is unavailable, inaccessible or incapable of being delivered at the required rate. Chapter 3 already separates quantity from delivery power and financial claims from stored exergy and persistent structure. The economic layer retains those separations rather than translating them into a single monetary measure.

5.3 Minimal Cross-Domain Translation

Domain	What the model represents	Why it remains distinct
Energetic	Exergy stores, transfers, destruction and usable throughput.	Has thermodynamic units and constraints; monetary denomination does not create exergy.
Material	Matter, inventories, environmental states and constructed physical configurations.	Mass and composition are not reducible to monetary value or historical energy expenditure.
Structural	Functional properties of persistent constructed states *s(t)*, such as transport, storage or conversion capacity.	Capacity depends on physical condition, configuration and operating context, not merely ownership or book value.
Biological	Population, metabolism, labour, demographic structure and other biological activity in *b(t)*.	Human activity is physically embodied but institutionally organised; wage expenditure is not a biological unit.
Institutional	Rules, procedures, legal categories, market mechanics, organisations and models in *a(t)*.	Institutional relations can govern physical access without themselves being physical stocks.
Economic-symbolic	Prices, accounts, valuations, claims and expectations generated within the institutional system.	They organise and represent command over activity but remain contingent on institutional recognition and enforcement.

5.4 Claims Space as an Analytical Subset of a(t)

The economic mapping benefits from distinguishing a claims space inside the institutional state. This is an analytical decomposition, not a new fundamental state variable and not a claim that institutional components are cleanly separable in practice. Schematically:

$$a(t) = (a_{rules}(t), a_{claims}(t), a_{models}(t), \dots)$$

Here a_{rules} denotes the rules and procedures that define permissible transactions and obligations; a_{claims} denotes recognised entitlements and liabilities concerning present or future command over activity and resources; and a_{models} denotes the representations and expectations used to value, forecast and coordinate those relations. The decomposition is intentionally schematic because the same institution can perform all three functions at once.

Claims space includes money balances, ownership titles, debt and equity claims, tax obligations, insurance commitments, pensions and other institutionally recognised relations specifying who may command what, under which conditions and at what time. Their physical significance arises when rules in *a(t)* cause labour, exergy, materials or access to persistent structures in *m(t)* to be redirected. A claim can therefore be socially and economically consequential without being a physical stock. [2]

A useful general description of a financial asset in this framework is an institutionally encoded conditional claim on future routing capacity. The claim may be enforceable, tradable and valuable, yet its eventual realisation still depends on the existence of a viable institutional mechanism and on sufficient underlying physical and biological capacity. The market value of the claim is a further representation of expected future command; it is not identical to either the claim itself or the substrate against which it may ultimately be exercised.

5.5 Microeconomics, Macroeconomics, and Finance

The familiar division between microeconomics, macroeconomics and finance can be retained as a broad translation device, but not as a partition of the physical system. All three domains operate across the same *b(t)*, *m(t)*, *a(t)* and *u(t)* substrate. Their principal difference is the scale and type of relation they represent.

Economic domain	Approximate DST-NCT mapping	Qualification
Microeconomics	Primarily local allocation and coordination among actors operating within a specified institutional basin N_0: exchange, production choices, consumption, labour allocation and relative prices.	Often associated with relatively fast flows, but it also contains durable capital, long contracts and intertemporal decisions. It should not be equated with short-term exergy throughput.
Macroeconomics	Aggregate representation of basin reproduction across production, employment, population, investment, public capacity, aggregate claims and institutional coordination.	Couples more visibly to slow structural and demographic states, but many macroeconomic indicators are flow measures. It should not be equated with stored energy or infrastructure.
Finance	Interpersonal, intertemporal and state-contingent	Finance can rearrange claims far faster than physical

Economic domain	Approximate DST-NCT mapping	Qualification
	organisation, valuation and transfer of claims within $a_{claims}(t)$, operating through rules and models in $a(t)$.	structures, resources or populations can change. It redirects access to the substrate rather than constituting the substrate.

The association between microeconomics and fast exergy flows, and between macroeconomics and slower accumulated niche conditions, is therefore useful only as a tendency. The stronger formulation is that microeconomics generally studies allocation conditional on an existing niche; macroeconomics studies aggregate performance and reproduction of that niche; and finance organises claims across actors, times and contingencies.

5.6 Production, Capital, and Investment as Composite Economic Categories

Several central economic terms compress physically heterogeneous processes. Production ordinarily records organised transformation that generates an economically recognised output. In DST-NCT terms, production requires biological and institutional coordination of exergy and material flows through existing structures under external boundary conditions. The monetary value assigned to the output is not a measure of the exergy consumed, the mass transformed or the durability of the capacity used to produce it.

Capital is especially heterogeneous. Physical capital maps principally to persistent constructed states $s(t)$ and their functional capacities, but the economic category can also include inventories, intellectual property, organisational capabilities and financial assets. These components should not be converted into one physical stock merely because accounting systems assign them monetary values. Financial capital belongs primarily to claims space: it represents institutional command or prospective command over activity and assets, not stored construction energy. [2]

Investment is similarly a routing category rather than a single physical process. Current resources may be directed towards constructing or restoring $s(t)$, accumulating an exergy reserve $B(t)$, changing capabilities in $b(t)$, altering institutional capacities in $a(t)$, or acquiring an existing claim without creating any new physical capacity. Whether an expenditure expands later system capacity therefore depends on what changes physically, biologically and institutionally, not on the accounting label alone.

5.7 Growth and Representational Compression

A sustained growth regime can make otherwise distinct variables move together for long periods. Rising throughput can coincide with increasing infrastructure, expanding productive capacity, population growth or specialisation, rising monetary incomes, increasing asset values and expanding financial claims. Under these conditions, broad economic categories such as wealth, capital, investment and growth can remain useful even when they compress heterogeneous underlying states.

This chapter calls that condition representational compression: several distinct physical, biological and institutional variables remain sufficiently correlated within the historically sampled basin that a smaller set of economic indicators can coordinate activity without continuously resolving every underlying distinction. The compression is functional rather than ontological. It does not make the represented variables identical.

Growth can also reconcile claims issued on different timescales. Expected future production can support present borrowing, investment and long-duration commitments because expanding future

flows create additional capacity from which earlier claims may be serviced. This does not guarantee regeneration. A system can expand measured output while degrading particular reserves, infrastructures or ecological functions, but persistent growth can conceal such divergence by increasing the aggregate flow available for substitution and repair.

A predictable growth trajectory also underwrites non-monetary institutional coordination within $a(t)$. Open information flows, long-horizon institutional trust, and pluralistic or democratic deliberation carry substantial energetic, administrative, and temporal overhead: they require time to negotiate compromise and a sufficient surplus to keep allocation conflicts non-zero-sum. When expanding throughput and reliable buffers allow losing coalitions to retain material security and anticipate future gains, distributed decision-making and shared epistemic standards can remain stable. Like long-duration financial claims in $a_{claims}(t)$, these civic and informational norms function as conditional expectations anchored to a predictable growth niche rather than cost-free constants of social organisation.

The "Social Capital" Category Error: Treating institutional trust as a stored capital stock ("social capital") symmetrical to physical infrastructure $s(t)$ or stored exergy $B(t)$ commits a variant of claim–substrate conflation. Because impersonal trust is a conditional forward expectation anchored to buffer reliability and epistemic coherence, historic compliance within N_0 cannot be drawn down like a physical reserve to substitute for missing joules, materials, or maintenance capacity during prolonged contraction. When an institution attempts to liquidate legacy trust by issuing unbacked symbolic claims or masking substrate decay (Chapter 10), relational expectations can re-price on the fast informational clock (τ_{fin}), triggering sudden non-linear defection from macro-institutional pipelines long before physical stocks in $m(t)$ are exhausted.

5.8 Contraction, Representational Divergence, and Claim Resolution

When throughput, structural capacity or regenerative conditions contract, the correlations that supported representational compression can weaken. Monetary wealth can rise while infrastructure deteriorates; measured output can remain stable while stored reserves are drawn down; recorded investment can increase while maintenance backlogs grow; and financial claims can expand even as the set of future physical flows capable of validating those claims narrows.

These conditions should not be represented by comparing a monetary aggregate directly with $B(t)$ or $s(t)$. Their dimensions are incompatible. The relevant proposition is relational: outstanding claims can become jointly inconsistent with the set of physically and institutionally realisable future allocations. In that case some claims must be revalued, rescheduled, transferred, diluted, defaulted upon or otherwise altered, or additional real capacity must become available. Which adjustment occurs is determined through the institutional architecture of claims rather than by thermodynamics alone.

This is where economics contributes something irreducible to the physical model. Physical constraints can make simultaneous demands infeasible, but they do not specify which claimant loses access, which obligation is prioritised or how scarcity is distributed. Property rules, debt hierarchy, taxation, monetary arrangements, bargaining power, political coalitions and organisational dependencies influence that allocation. The economic layer therefore governs much of the social routing of constraint without superseding the constraint itself.

Claim–substrate conflation becomes most consequential when the symbolic system continues to register liquidity, wealth or solvency while the physical prerequisites for realisation are changing. The error is not the existence of finance or monetary abstraction. It is the inference that because a

claim is valuable, exchangeable or legally recognised, the required exergy, materials, labour, infrastructure or delivery capacity must therefore exist in the required form and time.

A parallel divergence affects institutional coordination and information. When throughput contracts or becomes highly volatile relative to available buffers, the slow deliberation timescales of pluralistic governance can be outpaced by acute allocation pressures, while zero-sum distribution raises the stakes of political concession. Incumbent configurations then face strong pressures to compress decision-making into centralized administrative execution and to manage public representations to preserve institutional legitimacy, even while continuing to invoke the formal norms of the earlier growth regime.

5.9 Economic Coordinate Systems and Successor Regimes

Economic categories are themselves components of the incumbent institutional coordinate system. Currency units, recognised ownership, debt contracts, wage relations, corporate forms, accounting categories and market prices possess meaning because a particular configuration of organisations, rules, enforcement mechanisms and expectations reproduces them. They can remain highly effective within N_0 while being partly or wholly undefined in a successor regime N_i.

A regime transition can therefore terminate an economic model without terminating the underlying human or socio-ecological system. If a currency ceases to function, a banking architecture fragments, property rules change or labour allocation is reorganised, variables expressed in the previous coordinate system may no longer provide meaningful measures. The appropriate response within the DST-NCT framework is not to infer general non-viability, but to assess the successor regime using its own physical, biological and institutional conditions.

Conversely, the disappearance or revaluation of claims does not imply that the underlying material structures disappear. A factory, field, reservoir or electricity network can persist physically through a change in ownership, currency or accounting system. Its ability to function still depends on maintenance, inputs, labour, coordination and external conditions. Economic discontinuity and physical discontinuity are therefore separable even when they strongly interact.

Within the DST–NCT framework, formal economic theories, equilibrium doctrines, and macroeconomic policy rules do not constitute external laws governing the state-space; they are endogenous components of $a_{models}(t)$ and $a_{rules}(t)$ constructed within a specific institutional niche. Assumptions of smooth factor substitutability, price-mediated market clearing, and monetary stabilization represent compressed operational approximations that remain valid only while abundant stored exergy B(t), spare structural capacity in s(t), and a stable representational horizon (H) buffer the system from physical bottlenecks. In analytical applications, invoking theoretical market adjustments or price signals to override physical, energetic, or temporal constraints in m(t) and b(t) commits a formal category error: mistaking the incumbent regime's internal coordination software in $a_{models}(t)$ for the physical dynamics of the substrate.

5.10 Symbolic Temporal Reach and Physical Temporal Reach

Finance gives institutions an unusually large symbolic temporal reach. Claims on future income, production or assets can be created, priced and exchanged rapidly, allowing actors to coordinate present decisions around distant expected states. This temporal extension is one of the principal functional strengths of financial systems, but it must be distinguished from the temporal properties of the physical substrate.

A claim on ten years of future electricity production can be issued immediately; the electricity still has to be generated over time. A long-lived infrastructure project can be financially funded before

the required materials, labour or delivery capacity are available. A liquid security can be sold immediately while the physical asset underlying expected future returns may require years of maintenance or reconstruction. Financial time therefore rearranges expectations and command; it does not abolish construction time, replenishment time, service life, biological development or physical deadlines.

The institutional timescale τ_{fin} should consequently be interpreted as the characteristic speed of symbolic transactions, reporting and claim adjustment, not as the speed at which the material niche can respond. Temporal mismatch occurs when fast-changing commitments and allocation requirements induce demands on slower physical or institutional processes faster than those processes can replenish, repair or adapt.

5.11 High-Level Economic Translation Guide

The following terms are included only to orient readers. They establish the direction of translation; detailed treatment of individual concepts belongs in Chapters 6 and 7.

Economic term	Approximate mapping	Primary caution
Production	Organised transformation of exergy, materials and biological activity through structures and institutional rules.	Monetary output is a representation of production, not its physical content.
Consumption	Routing of current activity and throughput towards present biological or institutional use.	Accounting consumption does not identify a unique thermodynamic process.
Investment	Present allocation intended to alter future capacity, stocks, capabilities or claims.	Acquiring an existing asset claim need not create new physical capacity.
Physical capital	Primarily persistent constructed states *s(t)* and their functional capacities.	Not a store of recoverable construction energy.
Labour	Institutionally organised deployment of biological activity *b(t)*, conditioned by skills and social organisation.	Wages and labour-hours represent different properties of the same activity.
Market	An institutional coordination mechanism in *a(t)* for exchange and allocation among recognised actors.	Market outcomes depend on rules, claims and boundaries; markets are not physical allocation laws.
Price	An exchange ratio within the incumbent monetary and institutional coordinate system.	Price need not measure exergy, mass, replacement difficulty or system-criticality.
Money	A generalised institutional claim accepted for settlement and exchange within a given monetary system.	Money is not stored exergy or material capacity.
Credit and debt	Intertemporal claims assigning present command and future	A future claim can be created before the real capacity

Economic term	Approximate mapping	Primary caution
	obligations under specified rules.	expected to validate it exists.
Wealth	A monetary valuation of heterogeneous assets and claims under current expectations and institutions.	Valuation can change without an equivalent instantaneous physical change.
GDP and aggregate output	Monetised measures of selected current economic flows.	Do not directly measure stored exergy, infrastructure condition, regeneration or successor-state viability.
Finance	Organisation and valuation of interpersonal, intertemporal and state-contingent claims.	Rearranges command over the substrate; it does not constitute the substrate.
Financialisation	Increasing reliance on financial claims, valuations and allocation mechanisms within *a(t)* to coordinate present and anticipated future activity.	Should not be defined as a thermodynamic category error.
Claim–substrate conflation	Inference from monetary value, claim validity or liquidity to corresponding physical availability, substitutability or timely capacity.	The conflation is the category error; financial claims themselves are not.

5.12 Dynamical Implications and Scope

The economic layer leaves the core DST-NCT equations physically unchanged. Its principal addition is explanatory: it specifies how institutional representations and claims can redistribute access to biological activity, exergy, materials and constructed structures across actors and time. Those symbolic relations can become powerful causal variables because institutions recognise and enforce them, but their physical consequences occur only through realised changes in *b(t)* and *m(t)*.

The mapping also makes clear why economic competence within an incumbent basin does not imply universal representation of the wider state-space. Prices, accounting categories, asset values and financial models can be locally informative precisely because the institutional and physical relations that give them meaning have remained sufficiently stable. As the basin deforms, the problem may be less that every economic relationship becomes false than that the mapping between economic coordinates and the underlying substrate changes.

This chapter therefore does not propose an alternative theory of economics. It provides a translation boundary. Economic analysis can be imported where useful, but each imported concept should be decomposed into four questions: what physical or biological process is involved; what institutional rule or coordination mechanism operates; what symbolic representation is being used; and what claim on present or future routing is being created or modified. Exact correspondence should not be expected, and no monetary aggregate is assumed to substitute for direct description of heterogeneous physical and social states.

Physical energy economics can be developed as a later specialised application because it examines one important part of the coupling between economic activity and energetic throughput. It is not required for the general mapping here. The present layer only establishes the conceptual grammar needed to relate familiar economic vocabulary to the broader DST-NCT framework while preserving the distinctions on which that framework depends.

5.13 Terms

Term	Definition
Economic representation	A monetary, accounting or analytic description of selected relations within the coupled biological, material and institutional system.
Claims space / $a_{claims}(t)$	An analytical subset of $a(t)$ containing institutionally recognised entitlements and liabilities that allocate conditional command over present or future activity and resources.
Cross-domain coupling	A causal relation between variables belonging to distinct energetic, material, biological, structural, institutional or symbolic domains without assuming that the variables are commensurable.
Representational compression	A condition in which heterogeneous underlying variables remain sufficiently correlated that a smaller set of economic indicators can coordinate activity without continuously resolving their physical distinctions.
Claim realisation	The institutional process through which a recognised claim produces an actual reallocation of labour, exergy, materials, access or other activity.
Claim resolution	Alteration of mutually incompatible or unrealised claims through revaluation, rescheduling, transfer, dilution, default or other institutional adjustment.
Symbolic temporal reach	The capacity of institutional claims and models to coordinate present action around expected future states beyond the timescale of immediate physical change.
Claim–substrate conflation	The representational error of treating monetary value, claim validity or financial liquidity as evidence of corresponding physical availability, substitutability or timely delivery capacity.

5.14 References

[1] North, D. C. (1990). *Institutions, Institutional Change and Economic Performance*. Cambridge University Press. https://doi.org/10.1017/CBO9780511808678. (Establishes institutions as historically inherited rules and constraints structuring economic exchange, incentives, and organisational coordination. Provides an institutional foundation for mapping economic arrangements into a(t), without establishing the specific DST–NCT coupling.)

[2] European Commission, International Monetary Fund, Organisation for Economic Co-operation and Development, United Nations, & World Bank. (2025). *System of National Accounts 2025* (pre-edit version). United Nations. https://unstats.un.org/unsd/nationalaccount/sna2025.asp. (Establishes internationally agreed accounting definitions for production, assets, liabilities, capital formation, financial instruments, and economic transactions. Provides the conventional economic categories translated into the DST–NCT state-space.)

[3] United Nations, European Commission, Food and Agriculture Organization of the United Nations, International Monetary Fund, Organisation for Economic Co-operation and Development, & World Bank. (2014). *System of Environmental-Economic Accounting 2012—Central Framework*. United Nations. https://seea.un.org/en/methodology/seea-central-framework. (Establishes integrated but distinct accounting for physical flows, environmental asset stocks, and monetary economic activity. Supports the methodological separation between monetary representations and physical quantities without asserting their equivalence.)

6 Economic Metrics as Partial Observations of the Institutional Niche

This chapter maps a small set of familiar economic metrics onto the DST-NCT framework. Its purpose is not to replace national accounting, macroeconomic analysis or financial measurement with physical variables. It asks a narrower question: what part of the coupled biological, material and institutional system does each metric observe, what does it leave outside its field of view, and under what conditions can the metric diverge from the niche that supports it?

The preceding economic mapping treats prices, accounts, valuations and financial claims as institutional representations and allocation mechanisms rather than additional thermodynamic stocks. The same rule applies here. GDP, inflation, employment, capital values, debt, asset prices, trade statistics and productivity are useful observations generated inside a particular institutional coordinate system. They can strongly influence physical outcomes through allocation and expectations, but they are not interchangeable with stored exergy $B(t)$, persistent structural states $s(t)$, biological activity $b(t)$, or external driving conditions $u(t)$.

The methodological principle is therefore partial observation rather than metric rejection. A metric can be highly informative about the process it was designed to measure while remaining silent about other dimensions of the system. Divergence becomes analytically important when a familiar economic signal remains stable or favourable while the physical, biological, relational or temporal conditions supporting that signal change.

6.1 Reading Economic Metrics in a Heterogeneous State-Space

Each metric is read through the same four questions. First, what relation or activity does it represent? Second, through which parts of $b(t)$, $m(t)$ and $a(t)$ is that representation physically or institutionally realised? Third, which relevant variables are omitted or aggregated away? Fourth, how could the economic signal and the underlying niche move apart?

This does not imply that every divergence is pathological. Economic metrics intentionally compress complexity so that institutions can coordinate large systems. The problem arises only when the compressed representation is treated as though it were a complete measure of physical quantity, service capacity, biological reproduction or socio-ecological viability.

economic metric ≠ physical quantity ≠ physical capacity ≠ wider niche viability

The relations among these quantities are empirical and institutional. A rising monetary value can accompany rising capacity, unchanged capacity or declining capacity. A physical constraint can appear quickly in prices, remain hidden through inventories and imports, or be shifted onto particular groups or locations. The mapping therefore concerns coupling and possible divergence, not fixed one-to-one conversion.

Metric	Primarily observes	Does not by itself establish
GDP / GDP per capita	Monetary-valued production flow	Condition or regeneration of the niche
Inflation	Change in selected money prices	General physical scarcity or its distribution
Employment	Formal mobilisation of labour through institutions	Total condition or reproduction of $b(t)$
Investment / capital stock	Monetised formation and valuation of durable capacity	Specific physical condition, substitutability or service life

Metric	Primarily observes	Does not by itself establish
Debt / debt service	Claims on future routing capacity	Future physical surplus or ability to satisfy all claims
Asset prices / wealth	Valuation of expected future claims	Immediate change in underlying physical capacity
Trade / current account	Cross-boundary economic flows and claims	Full material dependence or relational resilience
Productivity	Output relative to selected represented inputs	Thermodynamic efficiency or regenerative capacity

6.2 GDP: Current Monetised Flow, Not Niche Condition

Gross domestic product is the natural starting point because its limitations are already widely recognised. In this framework GDP is interpreted primarily as a monetary-valued measure of selected current production within the incumbent institutional basin. It is therefore closely connected to ongoing throughput, but it is not a direct measure of stored exergy, material stocks, structural condition, ecological regeneration or biological continuity. [1, 3]

The distinction matters because equal additions to GDP can correspond to very different changes in the underlying niche. New infrastructure can increase future service capacity. Repair after a storm can restore capacity that was just lost. Intensive use of an ageing network can sustain current output while reducing remaining service life. Extraction of a non-replenishing reserve can raise measured production while reducing the physical buffer available later. All are economic activity, but they have different effects on *B(t)*, *s(t)* and future feasible states.

Example. Suppose a flood damages a transport network and reconstruction generates substantial expenditure. GDP may rise during the rebuilding period even though part of that activity merely restores a previous structural state. The economic flow is real; so is the prior physical loss. GDP does not claim to distinguish the two by itself.

GDP ↑ does not by itself imply B ↑, s(t) ↑, service life ↑, or niche regeneration ↑

The converse also matters. Lower GDP does not automatically imply lower human or socio-ecological viability. A contraction in monetised throughput could accompany deterioration, but it could also accompany reduced extraction, institutional simplification, replacement of market transactions by non-market activity, or movement into a different regime with different accounting categories. The metric is therefore interpreted relative to the institutional coordinate system in which it is produced.

6.3 Inflation: Price Change, Scarcity Signals, and Distribution

Inflation measures changes in money prices for a defined set of goods and services. Within DST-NCT it is therefore a signal inside *a(t)*: a change in exchange ratios generated by interactions among physical availability, production costs, inventories, market structure, expectations, monetary arrangements, policy and claims. Physical scarcity can contribute to inflation, but inflation is not a scalar measure of physical scarcity.

A material shortfall can be buffered by stored reserves, substituted inputs, imports or reduced use by particular groups. In that case the physical constraint may initially produce little aggregate inflation. Conversely, prices can rise because of monetary, contractual or institutional processes even when aggregate physical quantities have changed little. The signal is mediated by the allocation system.

Example. A country can experience declining domestic food production while retail food prices remain relatively stable because imports expand. The price signal suggests continued transactional availability, while domestic agricultural capacity has weakened and dependence on external flows has increased. Alternatively, a temporary transport bottleneck can sharply increase prices even though total annual material availability remains broadly adequate.

Distribution adds a further distinction. A national price index aggregates transactions across heterogeneous populations and locations. Moderate average inflation can coexist with severe constraint for groups whose incomes, housing markets, transport access or consumption baskets differ from the aggregate. The economic signal is averaged; access to physical goods and services is differentiated through *b(t)*, local *m(t)* and institutional allocation rules.

inflation ≠ general physical scarcity; financial liquidity ≠ physical availability ≠ delivery rate

6.4 Employment: Institutional Mobilisation of b(t)

Employment and unemployment describe one important relationship between the biological state *b(t)* and the institutional economy *a(t)*: the mobilisation of people through recognised labour-market roles.

High employment means that a large share of the relevant population is incorporated into formal productive and service activity under prevailing institutional rules. It does not directly measure the wider condition or reproduction of the biological and social substrate.

Labour-market statistics therefore omit many activities that can be crucial to niche maintenance: unpaid care, household maintenance, informal exchange, community coordination, some forms of skill transmission and other non-market work. They also do not by themselves reveal whether the employed population is accumulating or depleting health, skills, household capacity, demographic continuity or local relational support. [3]

Example. An economy can sustain very high employment while households experience severe time scarcity. Paid production remains strong, yet care, maintenance or intergenerational support can become harder to reproduce. The employment metric correctly reports labour-market incorporation; it does not establish that all complementary functions of $b(t)$ and the surrounding social niche are adequately reproduced.

The reverse is also possible. Rising unemployment can reflect institutional coordination failure or weak monetary demand while useful physical capacity and human capability remain present. The distinction again prevents a labour-market state from being treated as a complete measure of underlying human capacity.

6.5 Cross-Metric Example: Stagflation

Stagflation is useful because it demonstrates why the metrics above should not be expected to move as redundant indicators of one underlying economic quantity. A disturbance can reduce feasible production or employment while simultaneously increasing the money prices required to allocate constrained inputs and outputs.

In the present framework, a material, energetic, logistical or structural constraint can reduce effective productive capacity in $m(t)$. Prices inside $a(t)$ can rise as firms and households compete for scarcer or more costly flows, while output and employment weaken because less activity can be supported at previous terms. Inventories, imports, fiscal transfers or financial buffering may delay the visible adjustment, but they do not remove the physical constraint unless they supply an adequate substitute.

constraint on usable throughput or capacity → lower feasible activity + upward price pressure

This does not redefine all stagflation as a physical-energy event. Expectations, wage and price formation, market structure, monetary arrangements, taxation and policy remain institutional mechanisms capable of producing or amplifying the same observed combination. The value of the mapping is simpler: once price signals, productive capacity and financial claims are kept as coupled but distinct domains, simultaneous inflation and stagnation cease to appear conceptually contradictory.

6.6 Investment and Capital Stock: Expenditure, Structure, and Service Capacity

Investment is one of the economic categories most likely to compress physically different processes. Monetary investment can fund construction of new persistent structure $s(t)$, replacement or maintenance of existing structure, accumulation of inventories or reserves, development of skills and organisational capabilities, software and information systems, or acquisition of existing assets. The same accounting label therefore does not imply one physical transformation.

Capital-stock measures add a second layer of compression. A monetary estimate of capital stock or national wealth aggregates heterogeneous assets whose physical units, deterioration mechanisms, service lives and substitution possibilities differ. Roads, electricity networks, buildings, machines, software and organisational capabilities can all contribute to productive capacity, but they cannot be reduced to a common physical stock simply because they receive monetary valuations. [1]

Example. A country can record strong investment because of extensive new construction while an older water, rail or electricity network accumulates maintenance deficits. Gross capital formation is high, yet the service capacity of selected inherited structures can decline. Conversely, a relatively modest

expenditure on maintenance can preserve a critical network whose replacement cost would be much larger.

monetary investment ≠ Δs(t) ≠ ΔB(t) ≠ Δfuture service capacity

Depreciation and net investment are therefore useful but still remain economic approximations to heterogeneous deterioration and replacement. The DST-NCT layer retains the physical question separately: for each relevant structural function, does construction and maintenance offset deterioration under the actual operating and environmental regime?

6.7 Debt and Debt Service: Claims on Future Routing Capacity

Debt maps most directly into claims space. A debt instrument creates an institutionally recognised schedule of future payments or resource command. It is not a withdrawal from a single physical reserve, nor is a debt-to-GDP ratio a physical ratio between claims and stored exergy. The claim becomes materially consequential when servicing rules redirect future income, labour, imports, taxation or other flows.

Debt can therefore have opposite physical consequences depending on what the present mobilisation produces. Borrowing can activate idle resources, obtain imports, build infrastructure or create capabilities that increase future productive and regenerative capacity. It can also finance current expenditure without creating a corresponding future capacity, leaving a larger schedule of claims to be serviced from broadly unchanged or diminished physical possibilities.

Example. Debt used to build and maintain an electricity network may create a persistent structural asset that raises later delivery capacity and supports the claim. Debt used to maintain current consumption creates the same kind of financial liability without necessarily creating the same future physical service capacity. The nominal instrument alone cannot distinguish the two.

Debt service is particularly important because it introduces timing. A system can be solvent in an accounting sense over a long horizon yet experience acute short-term allocation pressure if claims fall due faster than resources, revenues or imports can be mobilised. This parallels the Chapter 3 distinction between total stored quantity and delivery power: enough may exist in aggregate while still being unavailable at the required rate or deadline.

6.8 Asset Prices and Financial Wealth: Valuation of Claims versus Substrate

Asset prices are current valuations of expected future benefits and claims under prevailing institutional rules and expectations. They can therefore move much faster than the physical structures, populations or productive processes to which they ultimately refer. This makes them particularly useful for demonstrating the separation between claims space and the underlying substrate.

Example. The market valuation of a technology firm can rise sharply because investors revise expected future earnings, market share or technological advantage. Its buildings, electricity supply, workforce and current production may change little during the same interval. The repricing is economically real because ownership claims and financing conditions change, but it is not equivalent to an instantaneous increase in physical productive capacity.

The same applies to property. A house can double in market price while remaining almost physically unchanged. Conversely, a well-maintained building can improve substantially in structural condition while its market price falls because interest rates, local demand or legal expectations change. Monetary wealth and physical service capacity are therefore distinct dimensions linked through institutional expectations.

Because these dimensions are coupled through institutional rules, a subsystem can also cross a viability threshold in claims space before its physical structures fail. If rising environmental variance or tail risk causes insurers to withdraw coverage from a region or infrastructure class, lending rules within $a(t)$ typically restrict mortgage and refinancing credit, triggering an abrupt collapse in asset valuations and local fiscal revenues even while the buildings and networks in $s(t)$ remain physically

intact. Once the claims architecture abandons the asset, however, the causal arrow reverses: the loss of insurability, credit, and tax revenue starves the still-functional physical structure of routine maintenance and repair, accelerating physical deterioration (D_s) until the material substrate fails as well.

financial wealth ≠ physical stock ≠ functional service capacity

This distinction becomes particularly important when asset prices are used as evidence of aggregate capacity. Rising valuations can improve collateral, financing access and purchasing power, thereby changing real allocations. But the valuation itself does not manufacture the materials, power, labour or structural condition required to realise every associated claim.

6.9 Trade and the Open, Distributed Institutional Niche

Trade metrics make the system boundary explicit. A national economy is not a closed physical system: imports and exports move materials, energy carriers, food, manufactured goods and services across territorial boundaries, while financial flows redistribute claims on those activities. Domestic economic continuity can therefore depend on a geographically extended niche.

This makes conventional trade statistics informative but incomplete. Monetary export and import values do not directly reveal mass, exergy content, criticality, substitutability or delivery risk. A small monetary import can be physically indispensable, while a large monetary flow can consist of goods for which substitutes are readily available. Current-account balances similarly describe economic and financial relations rather than a material balance sheet. [2]

Example. Domestic extraction of a material can decline while industrial output remains stable because imports increase. Measured production and prices may therefore show little immediate stress, although the domestic niche has become more dependent on external transport capacity, foreign production and cross-border institutional coordination.

Trade is also where relational and institutional coupling becomes unusually visible. A physical shipment requires more than a stock of material and a ship. Contracts, payment systems, insurance, customs recognition, information exchange, expected compliance and sufficiently reliable counterparties coordinate access to the physical flow. These relations are not stored exergy and should not be converted into it, yet failure of the relational layer can interrupt throughput just as effectively as failure of a port or pipeline.

cross-border flow = physical transport + institutional coordination + claims settlement + information

The equation is conceptual rather than additive: the components do not share common units. Its purpose is to show complementarity. Physical infrastructure can survive while trust or legal recognition collapses; strong contractual relations cannot move cargo when the physical route is unavailable. The distributed niche therefore consists of coupled but non-equivalent dependencies.

6.10 Productivity: Economic Ratio, Not Thermodynamic Efficiency

Productivity measures output relative to selected inputs. Labour productivity, for example, compares measured output with labour input; broader productivity measures attempt to account for additional factors. In DST-NCT terms these ratios can reflect several mechanisms simultaneously: improved technology, better structural capital, organisational learning, skill accumulation, changing sector composition, more intensive use of existing assets, externalisation of inputs, or reduced maintenance. [4]

Example. Output per worker can rise because automation genuinely reduces labour required for a given physical service. It can also rise because a firm reduces staffing while operating machinery more intensively and deferring maintenance. The same measured productivity improvement can therefore accompany either an expansion of future capacity or a shortening of structural service life.

Productivity should consequently not be equated with thermodynamic efficiency. A process may generate more monetary output per worker while consuming more energy, more material or more inherited structural capacity. Another process may reduce physical throughput per unit of service while showing only a modest change in monetary productivity. The relevant coupling has to be measured rather than assumed.

6.11 Cross-Metric Divergence: Reading the Dashboard as a Coupled System

The framework becomes most useful when several metrics are read together rather than interpreted independently. Their divergence can reveal that different parts of the system are moving on different timescales or that a constraint has been shifted across domains, populations or boundaries. The following patterns are examples of possible interpretations, not diagnostic rules.

Observed combination	Possible DST-NCT reading
GDP stable or rising; maintenance indicators deteriorating	Current flow may be protected by drawing more heavily on inherited structural capacity.
Inflation moderate; imports and external dependence rising	A domestic physical constraint may be buffered or displaced across the territorial boundary.
Employment high; care or household reproduction strained	Formal mobilisation of *b(t)* may coexist with weakening complementary social reproduction.
Asset prices and debt rising; physical capacity little changed	Claims space may be expanding faster than the underlying substrate.
Investment high; critical infrastructure condition falling	Gross expenditure may be concentrated in new assets while existing service capacity deteriorates.
Productivity rising; utilisation and maintenance stress rising	Current output per measured input may increase partly through more intensive use of inherited structure.
Trade volumes stable; institutional trust or logistics resilience falling	Observed flows may persist while the relational or physical margin supporting them narrows.

These combinations do not establish causality. Rising imports may be beneficial diversification rather than evidence of domestic weakness; high asset values may correctly anticipate future productive expansion; intensive capacity utilisation may be efficient and sustainable. The framework instead identifies which additional physical, temporal or relational observations are required before moving from an economic signal to a claim about niche condition.

Fiscal Balance via Physical Rationing: Similarly, an improvement in public or corporate fiscal balances (Δa_{claims}) cannot be treated as evidence of structural recovery or institutional buffering if it coincides with contracting physical service capacity in *s(t)* or accelerating workforce depletion in *b(t)*. Because an administration can restore nominal budgetary balance on τ_{fin} by imposing hiring freezes, closing physical facilities, deferring maintenance (R_s), or rationing access, a fiscal surplus accompanied by lengthening physical queues or acute staffing deficits is a primary observational signature of structural cannibalisation rather than a countervailing stabilisation force.

6.12 Metrics Near Basin Boundaries

Economic metrics are generated by the incumbent institutional coordinate system. GDP requires conventions about production and prices; inflation requires a currency and a defined basket; employment requires labour-market categories; debt and asset values require recognised claims; trade statistics require territorial and legal boundaries. These categories can remain highly informative while N0 persists, even when the system is approaching a regime boundary.

Near a bifurcation, however, the distinction between metric deterioration and coordinate-system failure becomes important. A currency reform, banking fragmentation, reorganisation of property, substitution of non-market allocation or territorial change can break the continuity of particular economic series. The resulting statistical discontinuity does not by itself imply disappearance of the underlying population, structures or material flows. It can mark the end of one representational regime.

Potential GDP illustrates the same point in a milder form. An estimate of potential output normally asks what level of activity can be sustained under assumptions about the existing economy, labour market, capital stock and productivity process. The DST-NCT question is therefore: potential within which basin? A model can estimate the sustainable output of N0 with considerable sophistication while saying little about the viable production possibilities of a qualitatively different successor regime Ni.

This does not make the incumbent metrics useless. It clarifies their domain. They are measurements made from inside a historically constructed institutional system, not external measurements of the complete physical and biological state-space.

6.13 Scope and Interpretation

This chapter does not propose a replacement dashboard, a new welfare index or a thermodynamic conversion of economic accounts. The objective is translational. Conventional metrics remain useful precisely because they compress large amounts of institutional information into operational signals. The DST-NCT mapping adds a discipline for interpreting those signals without treating the dimensions they omit as nonexistent.

For a reader accustomed to conventional economics, the practical rule is simple. GDP primarily describes monetised flow; inflation describes changes in selected money prices; employment describes formal mobilisation of labour; capital measures aggregate valued assets; debt describes claims on future flows; asset prices describe expected claims; trade describes cross-boundary economic relations; productivity describes output relative to selected inputs. Each metric then has to be coupled back to the physical, biological, structural and institutional states relevant to the question being asked.

The model therefore does not ask whether an economic metric is "real" or "illusory". Economic representations are causally real components of $a(t)$: they guide contracts, taxation, investment, credit, labour allocation and expectations. The narrower claim is that their causal power does not erase the distinction between representation and substrate. The same monetary signal can have different physical meanings in different niches, and the same physical constraint can generate different economic signals under different institutional arrangements.

6.14 Terms

Term	Definition
Partial economic observation	A metric or account that represents selected relations inside the institutional economy while omitting other physical, biological, structural or relational dimensions.
Metric-substrate divergence	A condition in which an economic indicator and the underlying physical or biological states relevant to its interpretation move in different directions.
Flow metric	A measure of activity over an interval, such as GDP, rather than a direct measure of accumulated physical capacity.
Price signal	A monetary exchange relation generated through markets and institutions; potentially responsive to physical scarcity but not identical

Term	Definition
	to it.
Formal labour mobilisation	The incorporation of biological activity *b(t)* into recognised employment relations governed by *a(t)*.
Capital-stock representation	A monetary aggregation of heterogeneous assets whose physical capacities, service lives and deterioration processes remain distinct.
Debt claim	An institutional obligation allocating future monetary or resource command according to specified terms; not itself a physical reserve.
Asset valuation	The present monetary valuation of expected future benefits or claims under prevailing institutional expectations.
Boundary displacement	Apparent domestic continuity achieved by shifting material extraction, production, buffering or other dependencies across the chosen system boundary.
Cross-metric divergence	A pattern in which familiar economic indicators move differently because they observe distinct coupled domains or timescales.
Stagflation	The coexistence of weak or contracting activity with rising prices; in this framework, a cross-metric state that can arise from physical, institutional, monetary or combined constraints.

6.15 References

[1] European Commission, International Monetary Fund, Organisation for Economic Co-operation and Development, United Nations, & World Bank. (2025). *System of National Accounts 2025* (pre-edit version). United Nations. https://unstats.un.org/unsd/nationalaccount/sna2025.asp. (Defines national accounting conventions for production, GDP, capital formation, financial accounts, assets, and liabilities. Establishes the measurement boundaries within which economic aggregates are constructed.)

[2] United Nations, European Commission, Food and Agriculture Organization of the United Nations, International Monetary Fund, Organisation for Economic Co-operation and Development, & World Bank. (2014). *System of Environmental-Economic Accounting 2012—Central Framework*. United Nations. https://seea.un.org/en/methodology/seea-central-framework. (Provides a standardised framework connecting economic activity to physical resource flows and environmental asset stocks. Demonstrates why monetary measures require complementary physical accounts to characterise environmental quantities and resource use.)

[3] Stiglitz, J. E., Sen, A., & Fitoussi, J.-P. (2009). *Report by the Commission on the Measurement of Economic Performance and Social Progress*. Commission on the Measurement of Economic Performance and Social Progress. https://www.cbs.nl/en-gb/background/2011/36/report-by-the-commission-on-the-measurement-of-economic-performance-and-social-progress. (Documents recognised limitations of GDP and aggregate economic indicators, including their incomplete coverage of distribution, non-market activity, environmental conditions, and the sustainability of underlying stocks. Supports the general principle that measured economic performance and wider system conditions require separate observation.)

[4] OECD. (2001). *Measuring Productivity—OECD Manual: Measurement of Aggregate and Industry-level Productivity Growth*. OECD Publishing. https://doi.org/10.1787/9789264194519-en. (Establishes the definitions, construction, interpretation, and measurement limitations of labour and multifactor productivity indicators. Supports distinguishing measured economic productivity from particular physical efficiency mechanisms.)

7 Economic Non-Equivalence in Claims, Boundaries, Distribution, and Productivity

The four mappings in this chapter share a common theme: each exposes a different kind of non-equivalence between the economic representation and the underlying coupled system. Chapter 6 treated familiar economic metrics as partial observations of a heterogeneous state-space. The chapter extends that principle to four concepts an economics-trained reader may reasonably expect to see made explicit: endogenous money and credit creation, externalities, distribution and inequality, and technological innovation or productivity growth.

The purpose is not to add another physical layer to the model. Instead, these cases clarify how claims, prices and aggregates can change without equivalent changes in stored exergy $B(t)$, persistent structure $s(t)$, biological state $b(t)$, or external driving conditions $u(t)$; conversely, material, biological or relational changes can occur without being proportionately represented in economic claims or prices. Coupling across domains is causal, but it does not make the domains dimensionally or ontologically interchangeable.

$$\textit{economic representation} \neq \textit{physical quantity} \neq \textit{physical capacity} \neq \textit{distribution of access}$$

This non-equivalence is not an argument that economic representations are unreal. Claims, prices, contracts and accounting categories are components of $a(t)$ and can redirect labour, materials and exergy. The point is narrower: a change in the representation does not by itself establish an equivalent change in the substrate, and a change in the substrate need not be fully visible in the representation.

7.1 Creating Claims without Creating Substrate: Endogenous Money and Credit

Modern monetary systems permit some financial claims to be created through institutional operations rather than transferred from a pre-existing stock of money. In commercial banking, for example, extending a loan normally creates a corresponding deposit claim at the same time, subject to capital, liquidity, settlement, regulatory, credit-risk and borrower constraints. [1] The balance-sheet operation changes claims space immediately; it does not simultaneously create stored exergy $B(t)$, a road, a power plant, a trained workforce, or any other physical capacity.

$$\Delta a_{claims} > 0 \quad \textit{does not imply} \quad \Delta B > 0 \quad \textit{or} \quad \Delta s > 0$$

The subsequent physical effect depends on how the newly created purchasing command is exercised. Credit used to repair infrastructure may help restore $s(t)$; credit used to construct a productive asset may alter future flow capacity; credit used to bid for an existing asset may mainly change ownership and valuation; credit used for current expenditure may redirect present throughput without creating a durable replacement. The same nominal expansion of claims can therefore map to very different physical trajectories.

Seigniorage is a related but narrower case. Under a given monetary regime, the issuer of monetary liabilities can obtain purchasing command or income associated with issuance. In DST-NCT terms, this is an institutional redistribution of claims and command over current or future flows. It is not the creation of exergy or material capacity. The practical consequence therefore depends on what resources the issuer can actually command, when they are available, and what later claims-resolution processes follow.

This distinction matters most when symbolic expansion and substrate expansion diverge. A growing volume of credit can be compatible with rapid niche construction if it mobilises otherwise unused resources and creates capacities that support later flows. It can also accumulate claims on

a future that proves unable to realise them simultaneously. In that case, some combination of repayment compression, refinancing, inflation, taxation, default, restructuring, asset repricing or other institutional adjustment must reconcile claims with the subsequently reachable state-space. Structural cannibalisation is one possible outcome, not a necessary consequence of money creation itself.

Example. A bank finances construction of an electricity interconnector. The loan and deposit appear in claims space before the interconnector exists. During construction, labour, materials and exergy are routed into the project; only when those physical processes occur does *s(t)* change. If completed and maintained, the new structure may expand later routing capacity. If the project fails, the financial claims can remain even though the anticipated physical capacity never appears.

7.2 Externalities: Effects Outside the Transaction Boundary

An externality is most usefully mapped as an effect of one actor's activity on states or flows relevant to other actors that is not fully represented in the transaction, price or claim structure governing the originating activity. The key issue is therefore a mismatch between the boundary used by the economic transaction and the wider boundary over which consequences propagate. [2]

$$\textit{transaction boundary} \subset \textit{coupled physical / biological / institutional consequence boundary}$$

The affected domain can be material, biological, institutional or relational. Pollution may increase waste flows or deterioration within *m(t)*; congestion can reduce the service capacity available from existing *s(t)*; disease transmission can alter *b(t)*; knowledge spillovers can change *a(t)* for actors who did not purchase the originating activity; repeated exchange can strengthen or weaken trust and coordination. Externalities can therefore be positive or negative and need not be reducible to thermodynamic waste.

"Unpriced entropy" is consequently a useful description of one important subclass, not a general definition. A production process can impose P_{dest}, contamination or D_s outside the accounting boundary recognised by the originating transaction. But other externalities operate through information, population interaction, network structure or institutional expectations without a meaningful direct translation into entropy production.

Example. A factory sells a product at a price reflecting labour, energy, financing and other recognised costs. If emissions reduce agricultural yields downstream, part of the physical consequence occurs in another actor's *m(t)* and *b(t)* without necessarily entering the factory's price or claims structure. Conversely, a firm that trains workers may create skills later used elsewhere; the originating transaction can generate a positive institutional or biological capability spillover not fully captured by the firm.

This mapping also clarifies spatial displacement. An economy can improve a domestic indicator by moving extraction, pollution, maintenance burdens or labour-intensive processes beyond its territorial accounting boundary. Whether the underlying niche has improved depends on the system boundary chosen for the analysis, not only on where the monetary transaction is recorded.

This boundary non-equivalence is particularly consequential when public institutions adopt the accounting and optimisation rules of bounded commercial organisations. A firm or investment portfolio is an exitable sub-organisation: it can raise its measured productivity and returns by closing low-margin rural nodes, liquidating redundant buffers, and treating biological reproduction *b(t)* and long-term infrastructural maintenance as external background conditions. A territorial polity or socio-ecological niche, by contrast, is non-exitable. When a territorial institution optimises its

internal accounts by shedding peripheral services or deferring maintenance, the displaced costs do not leave the system; they accumulate directly inside the same basin as accelerated structural deterioration (D_s), lost strategic redundancy, and demographic compression in *b(t)*.

7.3 Distribution and Inequality: Aggregate Claims versus Heterogeneous Access

Aggregate metrics compress differences among populations, sectors and coalitions. Income shares, wealth shares and statistics such as the Gini coefficient therefore add information about the distribution of monetary claims, but they still do not directly measure the distribution of exergy, material goods, infrastructure service, time, security or biological opportunity. The mapping must preserve these distinctions. [3]

$$\textit{distribution of } a_{claims} \neq \textit{distribution of access to } B,\ s(t),\ \textit{services, or biological opportunity}$$

A useful interpretation is to treat aggregate claims and access as distributed across heterogeneous subpopulations rather than to introduce a new fundamental state variable. Different groups can hold different claims, occupy different institutional positions, face different prices, and obtain different effective access to the same material niche. The aggregate can therefore remain stable while particular groups experience severe constraint, or aggregate inequality can rise while absolute access to some physical services improves for most groups.

Under a constrained regime, institutional allocation rules may increasingly protect some activities or groups while exposing others to shortages, volatility or maintenance failure. That is a possible triage mechanism within the wider model. Rising monetary inequality is not, however, synonymous with biological triage: inequality can arise during expansion as well as contraction and can reflect ownership, bargaining, inheritance, skill scarcity, institutional privilege, market structure or many other mechanisms.

The analytical value of distributional metrics is therefore diagnostic rather than ontological. They tell the reader that claims and access are unevenly distributed and help identify where aggregate indicators conceal divergent local conditions. Establishing a thermodynamic or biological interpretation requires additional observations of the relevant material, structural and population states.

Example. National inflation may be moderate while low-income households face severe housing and energy pressure because a larger share of their claims must be routed towards those necessities. The aggregate price signal has not disappeared; its consequences differ because the distribution of claims, consumption baskets and access conditions differs across the population.

7.4 Productivity and Innovation: Similar Ratios, Different Physical Trajectories

Measured productivity is an economic ratio between a representation of output and selected represented inputs. Total factor productivity (TFP) is especially important because it captures changes in output not accounted for by measured quantities of conventional inputs.[4] It is therefore often associated with technology or efficiency, but within DST-NCT it should remain a residual economic representation rather than being identified with a single physical mechanism.

Measured productivity growth can arise from several different changes: improved conversion efficiency; better structural design; organisational or informational improvements in *a(t)*; learning and skill accumulation affecting *b(t)*; substitution among materials; improved access to external driving conditions *u(t)*; higher utilisation of existing *s(t)*; reduced maintenance; depletion of reserves; shifting activity across statistical boundaries; or changes in the composition and valuation of output. These mechanisms can have very different consequences for later system capacity.

For interpretation, two limiting cases are useful without treating them as exhaustive categories:

Limiting case	DST-NCT interpretation
Capacity-enhancing productivity	Measured output rises relative to selected inputs because technical, structural, organisational or biological changes increase future feasible service or reduce required throughput under the relevant constraints.
Capacity-consuming productivity	Measured current output rises partly through higher utilisation, deferred maintenance, reserve drawdown, shortened service life, transferred costs or other mechanisms that reduce future capacity or move burdens outside the measurement boundary.

Neither category should be inferred from the productivity statistic alone. Intensive use of an underutilised asset can be efficient and sustainable; conversely, a technologically sophisticated process can expand output while creating a large new maintenance burden. The physical sign of the feedback has to be established empirically.

Innovation should likewise not be equated only with niche expansion. A new technology may expand the set of accessible resources, but it may instead reduce the exergy required for a service, improve conversion efficiency, lower deterioration, substitute a less constrained material, improve coordination, or reorganise existing infrastructure. In the model, innovation changes couplings among $b(t)$, $m(t)$, $a(t)$ and $u(t)$; it does not require a single direction of growth.

Example. Two factories report the same increase in output per worker. One installs more efficient machinery that lowers energy use and maintenance per unit of output. The other runs ageing equipment for longer hours while deferring overhaul. The productivity statistic can rise in both cases, while the trajectories of $B(t)$, $s(t)$, maintenance demand and future service capacity differ sharply.

7.5 Four Forms of Economic Non-Equivalence

Taken together, the four mappings close different gaps between the economic representation and the coupled system. They are best understood as complementary rather than as a new set of fundamental variables.

Concept	Non-equivalence exposed	Compact statement
Money and credit creation	Claims can expand before or without an equivalent expansion of physical stocks or capacities.	claim expansion ≠ substrate expansion
Externalities	Consequences can propagate beyond the transaction and accounting boundary that generated them.	represented transaction ≠ full consequence set

Concept	Non-equivalence exposed	Compact statement
Distribution and inequality	An aggregate can conceal large differences in claims, access and exposure among subpopulations.	aggregate state ≠ distribution of states
Productivity and innovation	The same measured output/input improvement can arise from physically different mechanisms with different future effects.	same ratio ≠ same physical trajectory

The common methodological rule is consequently the same one used throughout Chapters 5 and 6: preserve the coupling, preserve the distinction. Economic representations matter because institutions act through them, but they do not become physical quantities merely because they successfully coordinate physical activity. Physical and biological processes matter because they constrain what claims can ultimately be realised, but they do not determine by themselves how those constraints are priced, distributed or legally resolved.

7.6 Scope

This chapter does not attempt to provide a theory of money, externalities, inequality or technological change. It supplies only the translations needed to prevent an economics-trained reader from mistaking an omitted concept for an absent mechanism in the DST-NCT framework. Detailed monetary theory, welfare economics, distributional modelling and energy economics remain outside the present scope.

The four cases also reinforce the regime-neutral boundary of the wider model. Expansion of claims does not guarantee future physical expansion; external costs do not imply system-wide physical decline; inequality does not by itself establish contraction; and productivity growth does not by itself establish niche expansion. Their interpretation depends on the particular material, biological, institutional and temporal couplings operating in the basin being analysed.

7.7 References

[1] McLeay, M., Radia, A., & Thomas, R. (2014). Money creation in the modern economy. *Bank of England Quarterly Bulletin*, 54(1), 14–27. https://www.bankofengland.co.uk/quarterly-bulletin/2014/q1/money-creation-in-the-modern-economy. (Explains how commercial bank lending creates corresponding deposit claims, subject to monetary, regulatory, and balance-sheet constraints. Establishes the monetary mechanism used to demonstrate that expansion of financial claims does not itself create equivalent physical stocks or productive capacity.)

[2] Coase, R. H. (1960). The problem of social cost. *Journal of Law and Economics*, 3, 1–44. https://doi.org/10.1086/466560. (Establishes the importance of reciprocal effects between economic actors, transaction costs, and institutional arrangements in allocating economic consequences. Provides a foundation for mapping externalities as effects extending beyond the originating transaction's accounting boundary.)

[3] Stiglitz, J. E., Sen, A., & Fitoussi, J.-P. (2009). *Report by the Commission on the Measurement of Economic Performance and Social Progress*. Commission on the Measurement of Economic Performance and Social Progress. https://www.cbs.nl/en-gb/background/2011/36/report-by-the-commission-on-the-measurement-of-economic-performance-and-social-progress. (Demonstrates the limitations of aggregate economic measures in representing distribution and other dimensions of population conditions. Supports the analytical distinction between aggregate monetary claims and heterogeneous outcomes across subpopulations.)

[4] OECD. (2001). *Measuring Productivity—OECD Manual: Measurement of Aggregate and Industry-level Productivity Growth*. OECD Publishing. https://doi.org/10.1787/9789264194519-en. (Establishes the construction and interpretation of labour and multifactor productivity measures, including measurement and aggregation limitations. Supports treating productivity indicators as economic ratios rather than direct measures of thermodynamic efficiency or future physical capacity.)

8 Normal Science, Epistemic Coverage, and Institutional Niche Dependence

This chapter extends the institutional niche model to organised academic science itself. The preceding chapters show how economic concepts and metrics can represent an incumbent institutional basin accurately in selected dimensions while providing limited information about the wider physical and biological state-space. The same distinction applies to scientific knowledge production. Scientific theories, measurement systems, research communities and criteria of relevance are also products of historically constructed niches.

The central proposition is **not** that scientific knowledge is unreliable. It is that the reliability of scientific results within a recognised domain and the breadth of possibilities that an institution investigates are different properties. **Normal science** may become increasingly precise about the incumbent regime N_0 and its boundary Σ while leaving potentially accessible successor regimes N_i less well represented.

The sharper question is how the **three domains of physical possibility, scientific accessibility and institutional research selection** diverge, especially where loss of the incumbent institutional configuration would also disrupt the scientific community, its professional structure or its sponsors.

8.1 Scientific Institutions Within the Coupled State-Space

The core DST–NCT model describes human activity and populations *b(t)*, the constructed material niche *m(t)*, inherited institutional states *a(t)* and external driving conditions *u(t)*. Science is a specialised configuration within this coupling, not an external observer. No new fundamental state vector is required.

Model component	Scientific manifestation
b(t) — biological activity	Researchers, students, technical personnel, training, cognitive activity and population replacement.
m(t) — material niche	Laboratories, instruments, computing facilities, university buildings, energy, materials and physical research infrastructure.
a(t) — inherited institutional state	Theories, taxonomies, research protocols, journals, peer review, professional credentials, funding rules and scientific authority.
u(t) — external conditions	Physical and ecological boundary conditions constraining experiments, observation and research operations.

Scientific models belong principally to the representational component of *a(t)*. Research funding, contracts and intellectual property also enter its claims and allocation relations. These symbolic structures can redirect material flows and scientific labour, but are neither stored exergy *B(t)* nor recoverable physical capacity, as Chapters 3 and 5 distinguish.

8.2 Scientific niches and their host institutions

A scientific discipline and the larger institutional system supporting it occupy distinct but coupled niches. A discipline reproduces its own conceptual rules, training, experimental practices, specialist networks and professional standards. The host system supplies much of its

finance, infrastructure, students, material inputs, legal recognition and demand for expertise. Their basins may overlap strongly, but their boundaries need not coincide.

A discipline can lose its explanatory framework while universities and states remain intact. A state or economic system can reorganise while some scientific propositions and experimental methods remain valid. Conversely, research that remains physically correct can lose its infrastructure, professional constituency or operational purpose when the host regime changes.

The epistemic risk becomes sharper where **the same institutional transition threatens both the object being studied and the reproduction of the research system studying it**. Scientific dependence on an incumbent regime then couples its research selection to the fate of that regime. This is a structural relation, not a presumption that individual scientists consciously defend their sponsor.

8.3 Three domains of scientific coverage

For any proposed regime or trajectory, distinguish: **physical possibility** — whether it satisfies the relevant energetic, material, ecological and biological constraints; **scientific accessibility** — whether available concepts, observations and methods permit useful investigation; and **institutional selection** — whether the question is recognised, resourced and pursued by established research organisations.

These domains are neither identical nor universally nested: science also investigates physically impossible hypotheses in order to rule them out. The relevant asymmetry concerns potentially viable states that are technically investigable but remain poorly covered because the incumbent scientific architecture allocates attention elsewhere.

A general observation can be represented schematically as $y_j(t) = h_j(x(t); a_{sc,i}(t)) + \varepsilon_j(t)$, with $x(t) = (b, m, a, u)$. Here h_j denotes a specified measurement procedure, $a_{sc,i}$ the applicable scientific apparatus within $a(t)$, and ε_j observational error. This is an epistemic mapping, not a further thermodynamic equation. The measured world constrains observations; the inherited scientific apparatus conditions what is measured and how observations are interpreted.

Accordingly, an observation may remain valid while the wider inference drawn from it loses coverage. In other cases a regime change invalidates the institutional categories through which an observation was defined. GDP and recognised financial claims, considered in Chapters 5–7, illustrate the latter distinction; temperature, mass and other physically specified observables need not lose meaning when an institutional regime changes.

8.4 Theory-of-Science Mappings

Several established research traditions supply complementary mechanisms. None needs to be promoted into a comprehensive theory of scientific failure. Their contribution is to explain how a scientific community can generate reliable local knowledge while selecting a narrower set of questions and possible regimes.

8.5 Kuhn: normal science and the disciplinary matrix

Thomas Kuhn called routine, cumulative problem-solving inside an accepted disciplinary framework **normal science**. Shared theories, exemplars, instruments and standards make highly specialised investigation possible; scientific revolutions alter parts of that framework when important anomalies can no longer be accommodated. [1]

DST–NCT mapping: normal science is an epistemic analogue of locally successful niche adaptation. It increases resolution within the established conceptual basin. Its disciplinary matrix also identifies which problems count as ordinary scientific puzzles. The limit is not that the research is false, but that competence within the framework does not establish coverage beyond it.

8.6 Massimi: perspectival realism and model pluralism

Michela Massimi's *perspectival realism* treats scientific representation as historically situated without making the underlying phenomena dependent on the observer. Different scientific perspectives provide distinct inferential access to a common reality, and exploratory models can investigate possibilities not represented by a single perspective. [2]

DST–NCT mapping: reliable scientific perspectives can be locally informative yet incomplete. The physical state-space exceeds any particular disciplinary coordinate system. Multiple independently informative perspectives can expand coverage towards N_i without requiring that established measurements of N_0 be rejected.

8.7 Longino: social epistemology and effective criticism

Helen Longino's *critical contextual empiricism* locates scientific objectivity partly in organised critical interaction. Evidence acquires explanatory force through background assumptions; criticism matters when researchers have recognised channels for challenge and when the community responds to it. [3]

DST–NCT mapping: a large volume of peer review or technical disagreement can coexist with shared, unexamined assumptions about the incumbent regime. Scientific criticism can be highly effective inside N_0 while leaving its coordinate system intact. The useful distinction is between the **intensity of criticism** and the **range of assumptions exposed to it**.

8.8 Zollman: communication networks and premature convergence

Kevin Zollman's formal work on epistemic communities shows that extensive communication and rapid consensus need not maximise collective learning. Under some model conditions, more isolated investigators retain alternative hypotheses long enough for informative evidence to accumulate. [4]

DST–NCT mapping: increasingly connected scientific institutions can accelerate the exploitation of accepted explanations while reducing persistence of unexplored alternatives. This can emerge through ordinary information-network effects among competent researchers; no deliberate suppression is required.

8.9 Jasanoff: co-production of knowledge and social order

Sheila Jasanoff's science and technology studies framework treats scientific knowledge and social organisation as **co-produced**. Scientific classifications make administrative and technical interventions possible, while states, organisations and social relations shape the conditions under which scientific knowledge is constructed and recognised. [5]

DST–NCT mapping: models in $a(t)$ help organise $b(t)$ and $m(t)$; the resulting institutional niche then generates data, professional roles, equipment and demand that reproduce particular models. This is the strongest direct link between scientific knowledge and institutional niche construction. The same feedback can increase precision locally and concentrate observation on states relevant to maintaining the incumbent configuration.

8.10 Frickel, Hess and colleagues: undone science

Scott Frickel, David Hess and their co-authors use **undone science** for research that remains unfunded, incomplete or neglected despite being regarded as important by identifiable groups. They trace such gaps to the institutional politics of research-agenda formation rather than only to defective reasoning within published studies. [6]

DST–NCT mapping: the published scientific record is a selected output, not a representative map of all scientifically investigable questions. Successor-state exclusion can appear as systematic non-production of research rather than a stock of erroneous conclusions. Funding mandates, missing facilities and absent disciplinary constituencies can maintain that asymmetry.

8.11 Gross and Bergstrom; Teplitskiy and colleagues: selection incentives and their limits

Kevin Gross and Carl Bergstrom model how rewards for observable scientific effort can discourage risky projects: evaluators may be unable to distinguish an unsuccessful but ambitious investigation from poor execution. The resulting incentives favour projects with more predictable outputs. [7]

DST–NCT mapping: research institutions may rationally favour incremental exploitation of established methods even where exploration could expand the longer-term represented state-space. This supplies a specific allocation mechanism, not a claim about researchers' motives.

Misha Teplitskiy, Hao Peng, Andrea Blasco and Karim Lakhani find that more novel submitted papers were more likely to be accepted across the 49 journals in their study. [8] Novelty therefore cannot be treated as uniformly suppressed by peer review. **DST–NCT mapping:** the institutional selection process has stages; project initiation, grant allocation, training, manuscript submission and publication can favour different degrees of novelty. Observed publication selection does not reveal which research was never undertaken.

8.12 Gieryn: scientific boundary-work

Thomas Gieryn analyses how scientists construct and negotiate professional boundaries between science and other activities, partly through claims to expertise, autonomy and resources. [9] This is distinct from the dynamical basin boundary Σ.

DST–NCT mapping: the disciplinary boundary of admissible methods and questions can become aligned with an incumbent institutional basin. The consequential problem occurs when the absence of a recognised scientific route to studying N_i is interpreted as evidence that N_i has no viable, investigable structure.

8.13 From Normal Specialisation to Endogenous Epistemic Selection

Scientific omissions have different causes. A possibility may be excluded by physical constraints; it may be inaccessible to existing instruments; it may fall outside the domain of a specialised model; or it may be investigable but systematically deprived of research attention. Only the last case, or feedbacks built upon the preceding limits, directly indicates institutionally selective coverage.

The **endogenous epistemic trap** is the stronger case: limited representation of N_i increases uncertainty about crossing Σ; this strengthens threat to the incumbent institutional and scientific configuration; research resources and professional attention increasingly favour preservation, recovery and monitoring of N_0; and the relative representation of N_i remains poor.

> *successor-state exclusion → post-boundary uncertainty → institutional basin-loss salience → boundary vigilance and incumbent-centred research selection → further successor-state exclusion*

In this mechanism, crisis research can become increasingly sophisticated without broadening the modelled future. Accurate projections of failure, detailed risk registers and improved techniques for delaying regime loss can coexist with very limited analysis of how a successor would reproduce itself. Greater knowledge of Σ is not automatically greater knowledge of N_i.

A scientific institution need not share a single intention for this feedback to emerge. Research mandates, disciplinary specialisation, reputational incentives, funding dependencies and the practical value of incumbent-compatible results can be sufficient. Individual researchers may remain critical, methodologically exacting and sincerely committed to open inquiry.

8.14 Why boundary salience strengthens some selections

The mechanism is most consequential where incumbent-regime loss threatens several coupled resources: a research community's material infrastructure; the applicability of its professional training; its recognised problem definitions; and the continuing demand or financing supplied by a sponsor. These dependencies make the host basin and the scientific niche more tightly aligned.

The opposite configuration is also possible. Disciplines with portable measurements, heterogeneous sponsors, alternative research facilities and recognised successor-state methods may preserve inquiry across regime change. Different coalitions inside the same scientific field can therefore respond differently to the same boundary.

8.15 Competing research timescales

The temporal asymmetry is simple. An established scientific community can often intensify existing methods quickly; constructing alternative instruments, datasets, expertise and concepts can require years or decades. When the perceived deadline for managing Σ approaches faster than successor-state knowledge can be assembled, short-horizon demand favours proven incumbent-oriented methods even when their conceptual limits are recognised.

This selectivity is reinforced by niche construction within science itself. Earlier research investment constructs a specialised material and institutional niche—dedicated monitoring infrastructure, curated datasets, simulation architectures, training pipelines, and grant committees—that lowers the marginal cost and career risk of further refining historically established problems. Consequently, institutional research allocation across candidate boundaries reflects an interaction among physical proximity, instrumental observability, accumulated scientific niche capacity, and legibility to the host institution's coordination categories. Because retooling research infrastructure and disciplinary standards operates on a slow institutional adjustment clock (τ_{inst}), rapid change in the external or material environment (τ_u, τ_m) can produce epistemic hysteresis: scientific output remains densely concentrated around previously instrumented problems even as less legible or newly imminent thresholds are approached.

8.16 Consequences for Institutional Knowledge and Economic Representation

The economic chapters provide a concrete illustration. GDP, inflation, employment, asset valuation and financial claims are often informative within their defined coordinates. Yet they are partial observations of the underlying coupled system. Their continuity or apparent health does not establish the condition of stored exergy, persistent structure, demographic reproduction, relational coordination or successor-state viability.

Normal economic science can therefore be technically strong while disproportionately oriented towards the reproduction, stress response and restoration of the monetary and administrative system whose variables it measures. That is a statement about possible coverage, not an assertion that all economic research shares the same blind spot. Political economy, institutional

economics, ecological economics and other traditions may generate different perspectives; the relevant distinction is which perspectives enter operational models and resource allocation.

Commercially sponsored biomedical research supplies a narrower empirical illustration of selection effects. A Cochrane review of 75 methodological studies found industry-sponsored drug and device studies more likely to report favourable efficacy results and overall conclusions than non-industry-sponsored studies. Conventional risk-of-bias assessments did not fully account for this difference. [10] The finding shows that research output can depend systematically on its institutional financing without reducing the entire published literature to fabrication or error.

Conversely, established resilience, transition and transformability research explicitly investigates regime replacement. Their existence shows that recognised scientific communities can represent states beyond an incumbent basin. The hypothesis concerns the conditions under which such representations remain peripheral or become consequential within operational institutions.

8.17 Dynamical Implications

The resulting extension is **institutional endogeneity of epistemic coverage**. Scientific knowledge systems are reproduced inside material and social niches; their observational capabilities, collective methods and research priorities depend partly on those niches. They can become more reliable and precise locally while leaving a wider set of accessible states thinly represented.

The distinctive prediction is an **asymmetric allocation of scientific attention around a salient regime boundary**. Where the scientific community's own reproduction, status or funding is strongly coupled to incumbent continuity, increasing threat can produce denser modelling of N_0 and Σ than of candidate N_i configurations. Where independent perspectives, research infrastructures and viable post-boundary concepts already exist, epistemic coverage can instead expand.

Scientific reliability and institutional epistemic coverage must therefore be assessed separately. The first asks whether a finding is well supported within its stated domain. The second asks which domains and feasible transitions the established research system is actually equipped and organised to investigate. A failure of the incumbent coordinate system is not itself evidence of wider human or socio-ecological non-viability; nor does the conceptual availability of a successor establish its physical viability.

> *The general DST–NCT proposition is consequently: an institution can improve the accuracy of its knowledge of a threatened basin while reducing, or failing to increase, the relative coverage of the wider reachable state-space. Academic science is subject to this possibility because its own knowledge-producing apparatus forms part of the coupled institutional system.*

8.18 Terms

Term	Meaning
Normal science	Kuhn's term for cumulative, specialised puzzle-solving under a shared disciplinary matrix.
Scientific niche	The coupled material, biological and institutional conditions reproducing a scientific community and its working methods.
Host institutional niche	The wider economic, administrative or social configuration on which scientific reproduction may depend.
Epistemic coverage	The range of states, questions and transitions meaningfully represented and investigated by a knowledge system.
Institutional research selection	The allocation of recognition, personnel, infrastructure and resources among investigable questions.
Boundary-coupled epistemic selection	A change in research coverage associated with the salience of losing an incumbent institutional basin.
Institutional endogeneity of epistemic coverage	Dependence of scientific representation and research selection on the niche in which scientific institutions reproduce.
Endogenous epistemic trap	Self-reinforcing relation between weak successor-state representation, threat and intensified incumbent-oriented research selection.

8.19 References

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9 Biological Plasticity, Demographic State-Space, and Macroscopic Signalling

This chapter extends the coupled Dissipative Structure Theory and Niche Construction Theory (DST–NCT) framework to the internal developmental and demographic dynamics of the biological state vector *b(t)*. The preceding chapters established how institutions use inherited rule-sets and claims within *a(t)* to coordinate biological labour *b(t)*, stored exergy *B(t)*, and persistent constructed structures *s(t)* inside *m(t)*. Here, *b(t)* is examined not merely as an aggregate stock of available labour, but as a self-calibrating biological population whose maturation schedules, temporal horizons, and reproductive allocations respond dynamically to the predictability of the constructed niche.

By integrating Life History Theory (LHT), Embodied Capital Theory, Developmental Niche Construction, the Differential Susceptibility Model (DSM), and the Adaptive Calibration Model (ACM), this layer models demographic shifts and horizon compression as biophysical adaptations to the fast–slow dynamics of the coupled state-space.

Evolution possesses no normative trajectory; demographic recalibrations are treated here strictly as conditional thermodynamic and developmental routing responses to changing environmental constraints. However, this distinction enforces one explicitly normative boundary within the DST–NCT framework: biological decoupling is an evolutionary escape hatch, not a human failure. As an incumbent institution (N_0) depletes its usable substrate, its self-preserving algorithms attempt to extract maximal metabolic compliance from the population to delay its own collapse. The resulting institutional taboos, forced reintegration policies, and cultural placebos function to bolt the biological escape hatches shut—treating the preservation of the incumbent coordinate system as synonymous with human survival. This is a fatal category error. The mechanisms detailed in this and subsequent chapters—fast life-history recalibration, youth horizon truncation, the withdrawal of prime-age compliance, and in-situ horizon compression within institutional cores —are not pathologies to be cured by institutional policy. They are deeply conserved evolutionary survival reflexes that allow the biological population *b(t)* to detach from a brittle, dying dissipative structure and retain the adaptive plasticity necessary to populate successor regimes (N_i). Locking human survival to the survival of its incumbent institutions actively threatens the former to temporarily sustain the latter.

9.1 Multi-Trait Compression and the Fast–Slow Continuum

In evolutionary biology and developmental science, Life History Theory analyses how organisms allocate limited metabolic energy and time across competing fitness functions—principally somatic growth and maintenance, skill accumulation, mating effort, and parental investment. Across and within species, these trade-offs are frequently summarised along a **fast–slow continuum**.

To prevent misinterpretation when mapping LHT onto human institutions, three structural properties of this continuum must be kept explicit:

1. **The Fast–Slow Axis as Representational Compression:** As Del Giudice (2020) demonstrates, the fast–slow continuum at the individual level is not a single monolithic biological switch or a unidimensional latent variable; it is a coarse-grained synthetic heuristic that summarises multiple, partially independent trade-off axes (present versus future orientation, somatic maintenance versus current expenditure, and offspring quantity versus parental investment) [4]. Within the DST–NCT grammar established in Chapter 5, the fast–slow scale is a **biological representational compression**: it flattens a high-dimensional space of physiological and behavioural traits onto a single axis where those traits sufficiently covary under shared ecological pressures.
2. **Continuous Distribution and Equifinal Stress Expression:** Even in this compressed

form, the continuum is continuous rather than binary [4]. Individuals occupy distributed positions along the gradient rather than sorting into two discrete types. Moreover, because the linear scale aggregates distinct underlying dimensions, individuals mapping to the same position on the flattened continuum can exhibit different behavioural phenotypes—particularly under acute stress or resource compression [2, 4]. In the Adaptive Calibration Model, stress responsivity calibrates across multiple distinct neuroendocrine profiles (e.g., sensitive, buffered, vigilant, unemotional) rather than a uniform reaction [2].

3. **Baseline Trajectory, Continuous Plasticity, and Adult Mediation:** For adult populations, observed life-history expression is the product of a multi-stage interaction rather than an unmediated reflex:
   - **Baseline Developmental Trajectory (τ_{dev}):** Genetic and epigenetic inheritance interact with prenatal and childhood cues of environmental harshness and unpredictability to calibrate a baseline physiological and temporal trajectory [1–3].
   - **Continuous Plasticity Gradient:** The degree to which an organism calibrates to environmental cues—its phenotypic plasticity or environmental sensitivity—is itself a continuous, approximately normally distributed trait across the population [5, 6]. While contrasting the high-sensitivity tail (*"orchids"*) with the low-sensitivity tail (*"dandelions"*, alongside intermediate *"tulips"*) provides a useful analytical shorthand, plasticity operates as a continuous sensitivity gradient rather than a binary polymorphism [1, 5, 6].
   - **Cognitive and Relational Mediation:** In adulthood, how a calibrated trajectory and ongoing plasticity translate into behaviour under institutional friction is mediated by individual cognitive resources (such as general cognitive ability and executive function, which expand personal planning horizons and buffering capacity) and external support networks (kin provisioning, household buffers, and local access to $B(t)$ and $s(t)$) [4, 7, 8].

     Furthermore, adult cognitive mediation in a slow life-history (SLHS) regime does not operate through continuous, unmediated Bayesian updating against raw state-space variables. Because evaluating complex socio-ecological reality from scratch in every interaction carries prohibitive prefrontal metabolic costs and triggers chronic neuroendocrine stress arousal, human cognition relies on internalised cultural schemas ("normality anchors" or shared mental models [5]. Imprinted during juvenile and early-adult socialisation (t_{dev}) and transmitted across multi-generational cultural intervals, these schemas act as cognitive compression templates defining expected stewardship, reciprocal fairness, and institutional regularity. Once consolidated in adulthood, normality schemas exhibit high structural inertia: as long as surface institutional cues match the inherited template, adult actors assimilate local anomalies as temporary friction rather than systemic default. At the micro-scale, this cognitive schema inertia insulates adult cohorts from immediate behavioural recalibration, causing early niche contraction to register as individual somatic exhaustion and burnout (as actors work harder to fulfil their internalised schema against a failing substrate) before sufficient anomalies accumulate to force abrupt, non-linear schema collapse across the hysteresis boundary.

9.2 Mapping Life History Allocations to the Coupled State-Space

Human populations regulate the biological state vector $b(t)$ by routing metabolic energy and time either toward long-horizon somatic and structural accumulation (coupled to NCT) or toward compressed temporal horizons and rapid turnover (coupled to DST).

- **Slow Life History Allocations (SLHS) as Niche-Coupled Embodied Capital:** In human evolution, extended juvenile dependence, delayed reproduction, and high intergenerational investment evolved to build what Kaplan et al. (2000) term **embodied capital**—complex

cognitive, technical, and somatic capacities whose fitness payoff depends on a sufficiently stable environment over multi-decade horizons [7]. Within an incumbent institutional basin N_0, slow life-history allocations rely on the temporal buffering provided by stored exergy $B(t)$ and persistent infrastructure $s(t)$ inside $m(t)$. Because the developmental niche supplies reliable physical security and predictable returns on delayed effort, individuals defer reproduction to invest in extended education, credentialing, and high per-capita parental investment. Biological reproduction and skill formation become tightly coupled to the slow clock required to operate and maintain the institutional architecture $a(t)$.

- **Fast and Horizon-Compressed Allocations (FLHS) as Thermodynamic Triage:** When an incumbent niche N_0 undergoes structural cannibalisation—depleting $B(t)$, deferring maintenance on $s(t)$, and shifting volatility onto peripheral cohorts to service fast financial claims τ_{fin}—the predictability of the developmental environment deteriorates. Where operational horizons are compressed and the expected return on multi-decade institutional investment becomes uncertain, delayed payoffs are penalised. Across the multidimensional trait space, recalibration away from the slow institutional pipeline takes two primary, coexisting forms depending on local material constraints:

1. *Classical Fast Allocation:* Channelling available energy into rapid maturation, shortened time horizons, opportunistic risk-taking, and earlier reproduction with lower per-capita institutional investment [2, 3].
2. *Horizon Truncation and Institutional Withdrawal:* In contemporary high-cost niches where household formation is structurally obstructed by housing, credentialing, or claims-space extraction ($a_{claims}(t)$), temporal discounting and stress recalibration also manifest as withdrawal from long-term educational and debt pipelines, avoidance of multi-decade commitments, and delayed or foregone family formation.

Crucially, within the DST–NCT model, both pathways produce the same functional consequence for the incumbent institution: **demographic decoupling**. Whether cohorts shift toward early, non-institutional reproduction or truncate their planning horizons and withdraw from household formation and long-term career tracks, the biological vector $b(t)$ reduces its investment in the slow-clock maintenance and human-capital reproduction required by $a(t)$. Analytically, invoking micro-level behavioural diversity to dismiss this macroscopic shift commits a cross-scale aggregation error: while individual phenotypic adaptations across $b(t)$ remain heterogeneous, any net deflection away from multi-decade embodied-capital investment contracts the future biological routing capacity required to validate $a_{claims}(t)$ and maintain $s(t)$.

9.2.1 Asymmetric Reproductive Overhead and Sex-Modulated Calibration

While the fast–slow continuum compresses population-wide trade-offs, mapping Life History allocations onto $b(t)$ requires accounting for an irreducible biophysical asymmetry in mammalian and human reproduction: the metabolic and temporal floor of gestation, lactation, and extended juvenile care (τ_{dev}). In human evolutionary ecology, successful reproduction requires a 15-to-20-year window of reliable energetic provisioning, physical safety, and alloparental support (Hrdy, 2009; Kaplan et al., 2000). Because the gestating and primary care-provisioning parent bears a higher obligatory metabolic commitment and a narrower, age-bounded biological window for reproduction (Trivers, 1972), environmental unpredictability and institutional horizon compression ($\tau_{fin} \ll \tau_{dev}$) do not act uniformly across $b(t)$:

- **Institutional Kin-Substitution and Path-Dependent Lock-In**: In high-complexity institutional basins (N_0), formal state and market mechanisms ($a(t)$) emancipate individuals from extended-kin dependency by assuming the functions of health-risk and care buffering (Σ_{care}: maternity care, subsidised childcare, schooling, and eldercare). To access stable positions within N_0, cohorts are incentivised to invest heavily in prolonged, institution-specific embodied capital (higher education and formal credentials) during early reproductive adulthood. For females, this creates a strict temporal pincer: delaying reproduction to accumulate $a(t)$ credentials compresses the remaining biological fertility window against an inflexible physiological deadline, rendering their reproductive trajectory acutely dependent on the continued solvency of Σ_{care} over the subsequent τ_{dev} horizon.
- **Asymmetric Plasticity and Reproductive Veto:** Whereas male reproductive physiology permits continuous gamete production across adulthood and allows rapid plastic shifts toward short-horizon, high-variance, or low-investment mating strategies under environmental volatility, female physiology cannot compress the 15-to-20-year developmental overhead of τ_{dev} without severe somatic and offspring fitness penalties. Consequently, when a high-cost niche undergoes structural cannibalisation—accelerating occupational demands toward τ_{fin} while eroding housing affordability and Σ_{care}—female horizon compression manifests primarily through Pathway 2 (Horizon Truncation and Reproductive Suppression): the neuroendocrine sensor delays or vetoes fertility altogether rather than initiating an unbuffered 20-year metabolic commitment inside a volatile niche.
- **Divergent ACM Stress Profiles on τ_{sig}:** Within the Adaptive Calibration Model (Del Giudice et al., 2011), common-mode environmental unpredictability calibrates the high-sensitivity vanguard ("orchids") along diverged behavioural vectors depending on social and reproductive constraints. Where peripheral or male cohorts facing compressed institutional returns shift disproportionately toward unemotional or agonistic/externalising profiles—manifesting as early withdrawal from formal credentialing, high-variance risk-taking, and anti-institutional exit—cohorts whose security and future provisioning depend on institutional stability calibrate disproportionately toward the vigilant-anxious profile. In young females sampling eroding care buffers alongside intensifying credential competition, this calibration expresses macroscopically on τ_{sig} as a steep rise in internalising stress and social anxiety, hyper-vigilance toward peer and institutional inclusion, and defensive over-investment in formal $a(t)$ compliance as a buffer against niche instability.

9.3 Distinct Biological Timescales

Integrating developmental plasticity and embodied-capital formation into the DST–NCT architecture adds two biological timescales that couple with the material (τ_m), institutional adjustment (τ_{inst}), and financial (τ_{fin}) clocks established in Chapters 1 and 2.

Symbol	Meaning
τ_{dev}	**Developmental Calibration Timescale:** The ontogenetic window (spanning prenatal development through childhood and adolescence) over which an organism samples environmental harshness and predictability in $u(t)$ and $m(t)$ to calibrate its baseline physiological, cognitive, and reproductive trajectory [1–3].

τ_{sig}	**Demographic Signalling Timescale:** The duration required for distributed stress and horizon-compression responses among highly plastic cohorts to synchronise under common thermodynamic friction, producing an observable macroscopic shift in *b(t)*.

9.4 Terms and Biological Mechanisms

Term	Definition
Biological Representational Compression	The analytical projection of multidimensional life-history trade-offs (somatic vs. reproductive effort, present vs. future orientation, quantity vs. quality) onto a continuous fast–slow heuristic axis [4].
Embodied Capital	Accumulated somatic, cognitive, and skill capacities built through extended juvenile development and intergenerational provisioning, requiring a stable niche over long horizons to yield returns [7].
Developmental Niche Construction	The process by which inherited material structures *m(t)* and institutional arrangements *a(t)* shape the developmental environment sampled by organisms during τ_{dev}, modifying phenotypic calibration across generations [8].
Environmental Sensitivity / Plasticity (DSM)	The continuously distributed capacity of individuals to register and recalibrate to environmental variance, ranging from high-plasticity *"orchid"* profiles at one tail to low-plasticity *"dandelion"* profiles at the other [1, 5, 6].
Adaptive Calibration (ACM)	The mechanism by which the neuroendocrine stress-response system acts as an internal biological sensor, sampling environmental predictability and modulating physiological and behavioural strategies across distinct stress profiles [2].

Demographic Decoupling	The process by which segments of *b(t)* withdraw from the multi-decade human-capital, debt, and reproductive pipelines required by the incumbent institutional state *a(t)*, separating biological persistence from the viability of N_0.

9.5 Synchronisation and Macroscopic Signalling (The DSM–ACM Coupling)

Because environmental sensitivity is distributed continuously across the population, the recalibration of *b(t)* under niche deterioration is neither instantaneous nor homogeneous [5, 6]. The Differential Susceptibility Model (DSM) implies that the highly plastic tail of the distribution acts as an early-warning biological sensor at the boundary of the institutional niche [1].

- **Asynchronous Variance in a Buffered Niche:** When stored exergy *B(t)* and maintained infrastructure *s(t)* buffer the population against external fluctuations in *u(t)*, individuals at the high-plasticity tail respond primarily to idiosyncratic, local micro-environmental variations [1, 5]. Although their trajectories are moderated by cognitive resources and family support, at the macroscopic population level their dispersed adjustments cancel out as uncoordinated demographic noise.
- **Common-Mode Environmental Compression:** The recalibration threshold is crossed when an incumbent configuration in niche lock engages in **structural cannibalisation**—sustaining fast τ_{fin} claims and central administrative cores by transferring costs, depleting buffers, and exposing peripheral demographic cohorts to wage compression, housing scarcity, and infrastructural decay. This transforms idiosyncratic developmental risk into a shared, common-mode environmental signal across the niche.
- **The High-Plasticity Vanguard as a Macroscopic Signal:** Under the Adaptive Calibration Model, sustained common-mode unpredictability triggers a correlated neuroendocrine and behavioural recalibration across the most plastic cohorts, particularly among adolescents and young adults navigating critical developmental inflection points [2, 3]. Even though individual manifestations vary by cognitive mediation, local support, and stress profile—ranging from acute stress signalling and horizon compression to active withdrawal from long-term credentialing and household pipelines—the correlation of these shifts collapses previous demographic noise into a coherent **macroscopic vanguard signal** (τ_{sig}). Less plastic cohorts (*"tulips"* and *"dandelions"*), whose higher personal thresholds insulate them from minor fluctuations, are consequently exposed to a synchronised behavioural and demographic shift within their social environment before the central institution N_0 experiences a full regime transition [5, 6].

9.6 Dynamical Outcomes: Institutional Vulnerability and Reaction

As established in Chapter 2, institutions operate as self-preserving algorithms that maintain their internal order and existing resource-capture rules. Because an advanced institutional configuration relies on continuous cohorts of slow-calibrated, high-embodied-capital individuals to operate *s(t)* and service forward-dated financial claims $a_{claims}(t)$, the synchronised decoupling of the plastic vanguard directly threatens institutional reproduction.

- **The Embodied-Capital and Claims Deficit:** As the plastic margin withdraws from slow, niche-coupled trajectories, fewer individuals commit to the multi-decade educational pipelines, long-duration debt obligations (mortgages, student debt, fiscal liabilities), and formal career tracks upon which the incumbent coordinate system depends. The institution

faces a widening deficit in the long-horizon biological commitments required to validate $a_{claims}(t)$ and maintain complex physical infrastructure $s(t)$.

- **Forced Reintegration:** Because adapting to a lower-throughput or structurally simplified regime would invalidate the authority and claims of incumbent coalitions (niche lock), the administrative apparatus reacts to demographic decoupling by attempting to recapture escaping biological capacity. Within $a(t)$, this manifests as **forced reintegration**: regulatory penalties against non-standard trajectories, the formalisation and taxation of informal subsistence networks, or debt-leveraged subsidies designed to push cohorts back into legacy credentialing and asset-debt pipelines without resolving the underlying physical constraints in $m(t)$.
- **Accelerated Tipping:** Forced reintegration policies require substantial real exergy and administrative overhead to enforce. In a stalled or contracting physical niche, the institution cannot fund these interventions from an exergy surplus; it must finance them by expanding $a_{claims}(t)$ and accelerating the **structural cannibalisation** of $B(t)$ and $s(t)$. Consequently, the institutional effort to suppress the biological stress signal and enforce legacy slow-strategy compliance degrades the developmental niche further, intensifying the environmental compression on $b(t)$ and accelerating rate-induced tipping toward the basin boundary Σ.

9.7 Scientific Mappings and Their Limits

Foundational evolutionary and developmental literatures establish the biological mechanisms used in this chapter, while their coupling to institutional claims space and structural cannibalisation constitutes the specific DST–NCT synthesis.

Scientific Literature	Mapping to the DST–NCT Biological Layer and Scope
Life History Theory & the Fast–Slow Heuristic	Ellis et al. (2009) establish how environmental harshness and unpredictability shift developmental trade-offs along the fast–slow continuum [3]. Del Giudice (2020) clarifies that this continuum is a synthetic, multidimensional heuristic moderated by individual condition and resources rather than a binary biological switch [4].
Embodied Capital Theory	Kaplan et al. (2000) demonstrate that extended human juvenile development and delayed reproduction co-evolved with skill-intensive niche exploitation and intergenerational provisioning [7]. The DST–NCT layer maps this directly to the long-horizon human-capital requirements of maintaining $s(t)$ and servicing $a_{claims}(t)$.
Developmental Niche Construction	Flynn et al. (2013) extend Niche Construction Theory to ontogeny, showing

	how constructed physical and social environments shape developmental plasticity and ecological inheritance across generations [8]. This provides the formal bridge between *m(t)*, *a(t)*, and developmental calibration over τ_{dev}.
Differential Susceptibility & Continuous Sensitivity	Belsky & Pluess (2009), Pluess (2015), and Lionetti et al. (2018) establish that environmental sensitivity is a continuous, normally distributed trait modulated by genetic and early-environmental interactions [1, 5, 6]. The conversion of high-sensitivity tail responses into a macroscopic institutional signal (τ_{sig}) is a proposed DST–NCT coupling.
Adaptive Calibration Model (ACM)	Del Giudice, Ellis, & Shirtcliff (2011) model the neuroendocrine stress system as an internal sensor that calibrates behavioural and physiological trajectories to environmental predictability across multiple profiles [2]. Linking structural cannibalisation in *m(t)* to common-mode ACM recalibration is an extension of the present model.

9.8 References

[1] Belsky, J., & Pluess, M. (2009). Beyond diathesis stress: Differential susceptibility to environmental influences. *Psychological Bulletin*, 135(6), 885–908. doi:10.1037/a0017376. (Establishes the Differential Susceptibility Model, showing how heightened phenotypic plasticity functions as bi-directional environmental sensitivity rather than simple vulnerability).

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[3] Ellis, B. J., Figueredo, A. J., Brumbach, B. H., & Schlomer, G. L. (2009). Fundamental dimensions of environmental risk: The impact of harsh versus unpredictable environments on the evolution and development of life history strategies. *Human Nature*, 20(2), 204–268. doi:10.1007/s12110-009-9063-7. (Distinguishes environmental harshness from unpredictability in driving life-history trade-offs across predictable versus volatile niches).

[4] Del Giudice, M. (2020). Rethinking the fast–slow continuum of individual differences. *Evolution and Human Behavior*, 41(6), 536–549. doi:10.1016/j.evolhumbehav.2020.05.004. (Clarifies the fast–slow continuum as a synthetic heuristic compressing multidimensional trade-offs, moderated by individual condition, cognitive capacities, and ecological feedbacks).

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[8] Flynn, E. G., Laland, K. N., Kendal, R. L., & Kendal, J. R. (2013). Developmental niche construction. *Developmental Science*, 16(2), 296–313. doi:10.1111/desc.12030. (Integrates Niche Construction Theory directly with human ontogeny, showing how inherited physical and institutional niches construct the developmental inputs that guide phenotypic plasticity).

[9] Hrdy, S. B. (2009). Mothers and Others: The Evolutionary Origins of Mutual Understanding. Harvard University Press. (Establishes that human slow life-history reproduction evolved as an obligate cooperative breeding system requiring reliable alloparental support networks; when formal institutional substitutes erode, mothers suppress fertility).

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[11] Reeves, R. V. (2017). Dream Hoarders: How the American Upper Middle Class Is Leaving Everyone Else in the Dust, Why That Is a Problem, and What to Do About It. Brookings Institution Press. (Provides the empirical sociological anchor for opportunity hoarding—residential zoning, educational sorting, and network closure—among meritocratic, progressive-signalling upper-middle strata).

10 Collective Reproductive Continuity, Intergenerational Feedback, and Institutional Collision

This chapter extends the biological layer established in Chapter 9 by introducing an intergenerational feedback mechanism linking the developmental recalibration of younger cohorts to the stress regulation, institutional allegiance, and political behaviour of older cohorts. Chapter 9 demonstrated how common-mode environmental compression and structural cannibalisation trigger horizon truncation and demographic decoupling among the high-plasticity vanguard on the signalling timescale τ_{sig}. Here, the model specifies how the wider adult and elder population—including the less environmentally sensitive majority ("*tulips*" and "*dandelions*") insulated from early micro-fluctuations—registers that vanguard shift and translates it into a macroscopic institutional crisis.

The core proposition is that perceived developmental and reproductive continuity within a recognised succession field serves as a primary long-term cue through which humans assess collective viability and regulate baseline social stress. When this intergenerational backchannel collides with an epistemically bounded institutional architecture $a(t)$, it either forces the incumbent configuration N_0 into structurally cannibalistic interventions or incentivises the deployment of synthetic cultural placebos that mask demographic decay until the material niche $m(t)$ is severely degraded.

10.1 The Intergenerational Backchannel and Observer Multiplicity

In human populations, the biological state vector $b(t)$ is structured by overlapping generations connected through long juvenile dependence and multigenerational provisioning [1–3]. Because human inclusive fitness and cooperative breeding rely on investments that mature over decades, adult and elder psychological security is not a static background condition; it is dynamically regulated by ongoing cues of generational succession [1–5].

- **Reproductive Continuity as an Affective Evaluative Cue:** The evaluative cue is socially perceived continuity across generations—encompassing healthy child development, young adults achieving functional autonomy and household formation, and subsequent reproduction. This mechanism does not require conscious demographic accounting or explicit evolutionary calculation; persistent obstruction of youth trajectories registers affectively as chronic unsafety and collective angst before its structural causes are diagnosed [7, 8, 10]. Within this hierarchy, material accumulation, financial valuations in $a_{claims}(t)$, and symbolic status operate as secondary, short-term reassurance mechanisms that cannot indefinitely substitute for blocked generational succession.

- **Separating Support Networks, Succession Fields, and Observer Positions:** Consistent with the multiplicity principle established in Chapter 2, intergenerational appraisal is neither monolithic nor uniform across a population. Three distinct relational layers must be kept analytically separate:

1. *The Reproductive Support Network:* The actual physical and relational network through which provisioning, care, and access to $B(t)$ and $s(t)$ make development and reproduction feasible [2, 3].

2. *The Recognised Succession Field:* The set of younger individuals whose developmental and reproductive progress counts as collective continuation for a given observer—potentially encompassing direct descendants, wider kin, local community youth, or culturally recognised in-group successors.

3. *The Trusted Observation Network:* The interpersonal and institutional channels whose testimony, shared visibility, and interpretation validate how successors are faring.

Because these fields overlap and vary by observer position, partial substitution is possible: a childless adult can derive collective reassurance from the thriving of kin or community youth. Conversely, the same demographic event can produce divergent appraisals across different observers. Following out-migration, for example, young adults who establish viable households in a new industrial region may reassure their immediate parents while simultaneously registering as local successor loss and support-network erosion to the remaining community.

- **Staggered Age Overlap and Reverse Feedback:** Because communities contain staggered, overlapping cohorts (infants, adolescents, young adults, and new parents), older cohorts continuously sample multiple stages of the developmental pathway simultaneously. While isolated delays are buffered by progress elsewhere, repeated obstruction across consecutive cohorts allows adults to register systemic niche deterioration on the signalling timescale τ_{sig}, long before a full biological generation interval (τ_b) has elapsed.

10.2 Shared Observation, Threshold Cascades, and Population Hysteresis

The transmission of demographic stress from the high-plasticity vanguard to the lower-plasticity majority is governed by network visibility and nonlinear threshold dynamics.

- **Trust Networks and Common Knowledge:** Adult reassurance depends not merely on private observation, but on *shared observation*—the awareness that trusted others are witnessing the same developmental milestones [12, 13]. Public rites of passage, educational graduations, marriages, and births generate common knowledge of succession across a community [12]. However, because trust in communicated information can diverge from participation in practical care, the observation channel contains a structural vulnerability: distant or mediated figures can be cognitively registered as successors even when they participate in no reciprocal support network with the observer.

- **Heterogeneous Thresholds and Distress Cascades:** Under the Differential Susceptibility Model (Chapter 9), less plastic cohorts possess higher personal thresholds for environmental friction and initially filter out scattered developmental delays. As common-mode structural cannibalisation forces a growing share of the plastic vanguard into demographic decoupling, failures to reach household and reproductive milestones accumulate across separate families and social circles. Mutual recognition of this shared obstruction triggers a Granovetter-type threshold cascade [14], rapidly converting dispersed private worry [5, 6] into synchronised collective angst [7, 8].

- **Asymmetric Recovery (Hysteresis):** Once collective confidence fractures and a population registers a collective basin-loss threat, nonlinear history dependence sets in [15]. Because chronic stress responsivity remains active in the absence of credible safety cues [9, 10], simply returning external conditions $u(t)$ or short-term economic indicators to their pre-trigger baseline does not reset the population's stress state. Recovery requires a substantially higher threshold of sustained, socially verified generational progress across τ_{dev} and τ_{sig} to restore collective confidence.

10.3 Terms and Intergenerational Mechanisms

Term	Definition
Perceived Reproductive Continuity	The socially observed progression of recognised younger cohorts through maturation, household formation, and reproduction, acting as a primary long-term cue of collective viability.
Recognised Succession Field	The culturally and relationally bounded set of younger individuals (own offspring, kin, community members, or in-group peers) whose continuation supplies reassurance to a given observer.
Reproductive Support Network	The practical system of kin and non-kin provisioning, care, and material access ($B(t)$, $s(t)$) that enables juvenile development and household reproduction [2, 3].
Shared Observation (Common Knowledge)	The mutually recognised awareness within a trust network that others are witnessing the same succession cues, coordinating collective confidence or alarm [12, 13].
Demographic Hysteresis	Path-dependent asymmetry in collective stress calibration whereby the threshold of sustained generational progress required to restore confidence after a cascade exceeds the threshold that triggered the alarm [14, 15].
Parakin Placebo / Synthetic Continuity	The substitution of mediated, parasocial, or algorithmically curated succession cues for local biological continuity, pacifying adult basin-loss threat without restoring the underlying support network [17, 18].

10.4 Institutional Collisions: Five Mappings of Systemic Crisis

In evolutionary signal-detection terms, the reproductive backchannel functions as a "smoke detector" [11]: it is sensitive to obstructed trajectories, but it does not diagnose the underlying thermodynamic or institutional cause of the obstruction. When this non-diagnostic biological alarm collides with an incumbent institutional state $a(t)$, it drives five interacting collision mechanisms across the coupled state-space.

10.4.1 The Endogenous Epistemic Trap Weaponised by Biological Anxiety

When the adult population registers persistent obstruction of youth trajectories, the backchannel generates intense collective angst and basin-loss threat [7, 8]. However, as established in Chapter 4, an incumbent institution suffering from **successor-state exclusion** cannot represent a viable collective future outside its existing coordinate system N_0.

When alarmed elders demand the restoration of generational continuity, the institutional apparatus translates that demand into a mandate to defend and restore N_0. Elder biological anxiety is consequently captured and channelled into **boundary vigilance** and threat rigidity. The institution mobilises resources to defend the exact socio-ecological and financial arrangements that are compressing the youth vanguard, intensifying the developmental obstruction and locking the system into a self-reinforcing endogenous epistemic trap.

10.4.2 Forced Reintegration Funded by Structural Cannibalisation

Because older cohorts typically hold disproportionate political weight and hold the majority of forward-dated entitlements in $a_{claims}(t)$, their collective alarm pressures the administrative core to halt demographic decoupling. The institution attempts **forced reintegration**—deploying fiscal, regulatory, and credit mechanisms to push young cohorts back into legacy Slow Life History (SLHS) credentialing, housing, and debt pipelines.

In a physical niche where exergy growth has stalled, and niche lock blocks the downward reallocation of existing incumbent claims, the institution cannot finance this intervention from an energetic surplus. It must fund the subsidy of legacy pathways through **structural cannibalisation**: drawing down stored exergy $B(t)$, deferring maintenance on persistent infrastructure $s(t)$, and expanding forward-dated financial claims. The institution consumes the physical foundation of $m(t)$ to subsidise the symbolic appearance of youth integration, accelerating the long-term degradation of the developmental niche.

10.4.3 Claim–Substrate Conflation and Cultural Proxies

A sharp **representational divergence** emerges between the biological cues required to quiet the intergenerational alarm and the accounting indicators generated by $a(t)$. The institution attempts to reassure an anxious population by pointing to aggregate economic metrics—GDP growth, high formal employment, or rising asset valuations in $a_{claims}(t)$. This relies on **claim–substrate conflation**, treating monetary expansion or formal labour-utilization rates as equivalent to biological and household reproduction.

Alongside monetary metrics, institutions elevate secondary cultural milestones—extended educational credentials, institutional ceremonies, and national or corporate prestige—as primary evidence of collective vitality. By encouraging the population to *bask in reflected glory* [16], the institution uses shared visibility of symbolic achievements to temporarily buffer collective self-esteem, even as local household formation remains structurally blocked. In analytical models, treating institutional adaptability, symbolic reallocation, or aggregate monetary stability as sufficient to offset the biological compression of $\mathrm{b(t)}$ reproduces this exact claim–substrate conflation inside the observer's own representation, mistaking the preservation of institutional coordinates in $\mathrm{a(t)}$ for the continuity of the underlying population.

10.4.4 Rate-Induced Tipping via Competing Clocks

The reproductive hysteresis loop acts as a temporal accelerant across the coupled clocks of the DST–NCT model. Once the threshold cascade is crossed, the backchannel converts the localized stress of the plastic youth into a society-wide political and affective demand operating on the demographic signalling timescale (τ_{sig}).

Because the institutional adjustment timescale (τ_{inst}) is constrained by niche lock and structural inertia, the incumbent regime cannot redesign $a(t)$ rapidly enough to accommodate a lower-throughput or reorganised successor regime N_i. To bridge the gap, the administration relies on its hyper-fast financial timescale ($\tau_{fin} \ll \tau_m$), issuing credit and monetary claims to mask the physical and demographic shortfall. Attempting to resolve a biological shift (τ_{sig}) with high-frequency symbolic claims (τ_{fin}) accelerates withdrawal demands on $B(t)$ and $s(t)$ faster than the material environment (τ_m) can replenish or repair them, driving **rate-induced tipping** across the basin boundary Σ.

10.4.5 The Parakin Placebo and Synthetic Continuity

Because the intergenerational backchannel responds to *perceived* succession filtered through trust networks rather than direct demographic audits, it exposes a profound evolutionary mismatch. Rather than expending scarce exergy $B(t)$ on forced reintegration, an institution or media ecosystem can hijack the perceptual channel using cultural and technological placebos that supply synthetic confirmation of continuity.

Mass media, digital platforms, and algorithmic feeds operate as synthetic extensions of the trust network, decoupling repeated exposure and shared awareness from local reciprocal care. Through **social surrogacy** and **parakin attachments**, observers form kin-like emotional bonds with highly visible media figures, creators, or curated digital personas, experiencing their milestones, status, or family narratives as reassurance of shared collective continuation [17, 18]. By flooding the observation field with curated streams of synthetic succession cues, this parakin placebo tranquilises adult collective angst without addressing the thermodynamic compression of actual youth in $b(t)$. While this suppresses acute political destabilisation in the short run, it introduces a severe lag into the demographic hysteresis loop—allowing the biological and material substrate to erode unopposed until the incumbent configuration loses the physical capacity to reproduce itself.

10.5 Scientific Mappings and Their Limits

As in Chapters 4 and 9, established scientific literatures validate the constituent biological, psychological, and network mechanisms, while their integration into a closed feedback loop with institutional claims space constitutes the DST–NCT synthesis.

Scientific Literature	**Mapping to the DST–NCT Intergenerational Layer and Scope**
Inclusive Fitness & Cooperative Breeding	Hamilton (1964), Kaplan et al. (2000), Dyble et al. (2016), and Chaudhary et al. (2024) establish the selective stake of kin and multilevel provisioning networks in juvenile development and reproduction [1–4]. They ground the evolutionary setting for intergenerational sensitivity without by themselves proving a specific psychological detector of collective continuity.
Generativity & Intergenerational Distress	McAdams & de St. Aubin (1992), Reczek & Zhang (2016), and Frase et al. (2024) document adult psychological investment in successor generations and the transmission

	of distress across parent–child and grandparent–grandchild ties, with variation across relationship positions [5, 6, 19].
Collective Continuity & Collective Angst	Sani et al. (2008) and Wohl et al. (2010) demonstrate that perceived group temporal endurance sustains social wellbeing, while threats to a group's future vitality induce collective angst and in-group defensive strengthening [7, 8]. Isolating observed youth reproduction as the primary long-term input is the specific empirical prediction of this chapter.
Stress Calibration, Unsafety & Signal Detection	Del Giudice et al. (2011) and Brosschot et al. (2018) model how stress responsivity calibrates to environmental unpredictability and remains chronically active when safety cues are absent [9, 10]. Nesse (2005) explains why evolved threat detectors favour low-cost false alarms over fatal misses without supplying a causal diagnosis [11].
Common Knowledge, Thresholds & Hysteresis	Chwe (2013) and Thomas et al. (2014) formalise how public rituals and shared observation create common knowledge that coordinates collective expectations [12, 13]. Granovetter (1978) and Scheffer et al. (2001) provide the formal mechanics for nonlinear threshold cascades and asymmetric hysteresis recovery [14, 15].
Reflected Glory & Parakin Surrogacy	Cialdini et al. (1976), Derrick et al. (2009), and Yan & Yang (2021) establish how vicarious prestige, mediated social surrogacy, and "parakin" attachments regulate belonging and affective reassurance independently of local reciprocal relationships [16–18].

10.6 References

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11 Health-Risk Buffering, Senescence Triage, and the Care Boundary

This chapter completes the biological and intergenerational layer established in Chapters 9 and 10 by examining the opposite end of the human life course. Whereas Chapters 9 and 10 analysed how environmental compression and structural cannibalisation disrupt the **entry boundary** of the biological state vector *b(t)*—juvenile development, embodied-capital formation, and household reproduction—this chapter examines what occurs when niche contraction strikes the **vulnerability and exit boundary**: acute injury, chronic illness, disability, and late-life senescence.

The central biological proposition is that human Slow Life History Strategy (SLHS) and multi-decade institutional cooperation depend not only on juvenile provisioning, but on **social health-risk buffering**—the credible guarantee that individuals who become temporarily incapacitated or age past peak metabolic output will not be abandoned [1–4]. Because illness, injury, and senescence sit directly within every living adult's personal forward horizon (*H*), an institution cannot execute a mechanical triage of the sick and elderly without triggering an explosive rise in mortality salience and a retroactive default on the slow-strategy social contract [5–8]. Together with the youth reproductive backchannel of Chapter 10, the erosion of health and eldercare closes an intergenerational pincer around the prime-age population in *b(t)*.

11.1 The Evolutionary Biology of Care and the Slow-Strategy Contract

From a narrow, short-horizon accounting perspective, provisioning permanently disabled, severely ill, or post-productive individuals appears to be a net energetic drain on a dissipative structure. In human evolutionary biology, however, systematic care for the vulnerable is a foundational niche-construction mechanism without which long-horizon embodied capital cannot be sustained [1–4].

- **Health-Risk Buffering as a Prerequisite for Embodied Capital:** As Sugiyama (2004) and Gurven et al. (2022) demonstrate, the evolution of the human life-history profile—characterised by decades of juvenile investment in complex brain and skill development followed by a long adult lifespan—is biophysically viable only if the organism is buffered against periodic incapacitation [1, 2]. In foraging and historical ecologies, disabling infections, fractures, and episodic illnesses are frequent; without social provisioning and nursing care during periods of negative net production, high-embodied-capital individuals would suffer premature mortality before their multi-decade investments could pay off [1]. Bioarchaeological evidence confirms that intensive care for the severely injured, chronically ill, and senescent is deeply conserved across hominin evolution [3, 4], while post-reproductive elders supply critical downward transfers of provisioning, ecological knowledge, and alloparental care that subsidise younger cohorts [2].

- **Unbuffered Morbidity and Adult Horizon Compression:** Within Life History Theory and the Behavioural Constellation of Deprivation framework (Ellis et al., 2009; Pepper & Nettle, 2017), environmental cues of high extrinsic morbidity and uncontrollable mortality risk cause adults to discount the future and shift allocation away from long-term somatic maintenance, preventative behaviour, and institutional compliance toward immediate present-oriented survival [5, 6]. Crucially, while prime-age adults can temporarily compartmentalise developmental pressures affecting the youth vanguard (especially when buffered by Chapter 10's parakin placebos), they cannot compartmentalise the abandonment of the sick and elderly. Vulnerability to injury, illness, and ageing is a universal future state of every living adult in *b(t)*.

- **The Military Precedent — Casualty Evacuation and Mortuary Fidelity as Cohesion Anchors:** Military organisations make the functional mechanics of this biological contract unusually explicit [7, 8]. Combat units require prime-age individuals to accept a high

probability of personal injury or death to achieve collective objectives. Yet modern militaries routinely expend disproportionate operational resources, divert logistics, and consciously risk additional lives to evacuate wounded personnel and recover the bodies of the dead ("leave no man behind") [7]. In narrow material accounting, a severely wounded or deceased soldier has zero or negative immediate combat utility. Functionally, however, casualty evacuation and mortuary care operate as foundational **continuity technologies** (Chapter 4): soldiers can tolerate high objective risk of death or injury, but the expectation that they will be abandoned if wounded or forgotten if killed destroys unit cohesion instantaneously [7]. As Shay (1994) demonstrates in clinical studies of combat trauma, institutional betrayal of this fiduciary care obligation (*themis*) inflicts severe moral injury, collapsing shared trust and converting disciplined collective action into self-preserving defection or rage [8].

11.2 Institutionalisation of Care and the Atrophy of Kin Buffers

During the historical growth phase of industrial and welfare-state niches (N_0), the inherited institutional architecture *a(t)* reorganised health-risk and senescence buffering to maximise formal labour mobilisation in *b(t)* [9, 10].

- **Replacing Kin Reciprocity with Institutional Claims:** To turn adults into geographically mobile, full-time participants in formal labour markets, modern states and corporations transferred the provisioning of the sick, disabled, and elderly out of multigenerational households and into formal claims space ($a_{claims}(t)$)—via pay-as-you-go state pensions, universal healthcare entitlements, and actuarial insurance [9, 10]. In models of *statist individualism* (such as Sweden's *statsindividualism* analysed by Berggren & Trägårdh, 2015 [9]) as well as high-mobility market economies (analysed by Wang, 1991 [10]), the institution explicitly emancipated the individual from dependency on extended kin by interposing a direct contract between the citizen and the state or insurer.

- **Physical and Demographic Erasure of the Fallback Network:** Once care was shifted into $a_{claims}(t)$, both the constructed material niche *m(t)* and the demographic structure of *b(t)* adapted around the assumption that kin care was obsolete:

1. *Structural Adaptation in s(t):* Urban housing stock and municipal layouts were optimised for atomised single-person or small nuclear households, eliminating the physical space required to house or nurse frail elders at home.

2. *Demographic Contraction in b(t):* Decades of sub-replacement fertility (and compressed demographic transitions in East Asia and Europe) produced inverted "4-2-1" age pyramids in which a single adult child or childless cohort faces multiple ageing dependents without lateral sibling or cousin networks.

- **The Re-Familialisation Trap:** Because institutional niche construction actively dismantled and replaced the informal kin-care substrate over multiple generations, **the fallback buffer is no longer waiting in reserve when the formal system contracts**. While an institution in fiscal or energetic distress can cut healthcare budgets, close regional clinics, or tighten pension eligibility on the fast symbolic clock (τ_{fin}), rebuilding multigenerational kin-care networks requires biological generation timescales (τ_b, τ_{dev}) and decades of local reciprocal trust. Attempting to offload elder and chronic care back onto atomised, child-scarce households immediately overwhelms the remaining domestic substrate.
  However, because cooperative care is deeply conserved in human biology, the failure of N_0 does not imply the permanent end of health-risk buffering. The biological imperative to provision the vulnerable remains intact. Within a successor regime (N_i), care networks can be reconstituted through new local reciprocity, mutual-aid coalitions, and non-kin

provisioning. The constraint is temporal: these replacement networks depend on the slow rebuilding of demographic capacity, proximity, and reciprocal trust, a process that cannot be instantly conjured to rescue the failing financial ledger of the incumbent state.

11.3 Terms and Care-Boundary Mechanisms

Term	Definition
Health-Risk Buffering	The social and material provisioning of injured, ill, disabled, or senescent individuals during periods of negative net metabolic production, required to make long-horizon embodied-capital investment viable [1–4].
The Care Boundary (Σ_{care})	The nonlinear social-biological threshold at which an institution's failure to uphold health and eldercare commitments becomes common knowledge, triggering mortality salience and collapsing prime-age compliance with the slow-strategy contract [6–8, 13].
Kin-Buffer Atrophy	The structural and demographic erosion of informal family and community caregiving networks resulting from generations of institutional substitution, atomised housing in $s(t)$, and sub-replacement fertility in $b(t)$ [9, 10].
Camouflaged Care Triage	The use of administrative queueing, clinical guideline narrowing, and bureaucratic opacity to ration scarce physical healthcare and eldercare in $s(t)$ and $b(t)$ without declaring an explicit default on nominal entitlements in $a_{claims}(t)$ [11, 12].
Autonomous Exit Reframing	The institutional re-coding of structural care, housing, and support deficits as voluntary, private medical choices (such as expanded assisted dying for non-terminal vulnerability), preserving institutional legitimacy while reducing long-term support liabilities [14, 15].

The Intergenerational Care Pincer	The closed feedback loop in which rising fiscal and labour extraction on young cohorts to defend elder entitlements accelerates youth demographic decoupling (Chapters 9–10), while visible rationing of eldercare simultaneously destroys adult confidence in future security.

11.4 Approaching the Care Ridge (Σ_{care}): Four Institutional Collisions

When an aging institutional configuration in niche lock encounters stalled exergy throughput ($dB/dt < 0$) or deteriorating care infrastructure (Gs + Rs < Ds), it faces an acute dynamical trap. It cannot smoothly transfer care back to atrophied kin networks, yet it cannot openly abandon unproductive sick and elderly cohorts without crossing **the care boundary (Σ_{care})**—triggering an immediate explosion of mortality salience (Chapter 4) and shattering the fiduciary trust that keeps prime-age adults working, paying taxes, and honouring institutional rules.

Crucially, this boundary is not crossed by a temporary, collectively understood capacity shortage—such as an acute natural disaster or wartime rationing—during which the shared intent to provision the vulnerable remains socially credible. The boundary is crossed by the structural invalidation of the contract: an increasingly credible expectation that reciprocal care commitments will permanently no longer be honoured.

To preserve the appearance of the incumbent coordinate system N_0 while physical care capacity contracts, the institution relies on four interacting mechanisms:

11.4.1 Nominal Entitlement Preservation versus Physical Care Dilution

To avoid the political and psychological shock of an explicit default, the administrative core protects the nominal monetary face value of pensions and universal healthcare guarantees in $a_{claims}(t)$. Instead of repudiating the claim, the institution allows the physical conversion rate between $a_{claims}(t)$ and real care delivery in $b(t)$ and $s(t)$ to degrade (**claim–substrate conflation**).

Pension transfers arrive on schedule in nominal currency, but cannot command scarce human nursing labour; citizens retain universal statutory coverage on paper, but encounter lengthening diagnostic, surgical, and long-term care queues inside under-maintained facilities (Ds). By shifting the default from the symbolic ledger ($a_{claims}(t)$) to physical waiting times and uncompensated household exhaustion, the institution stretches the linear appearance of N_0 while its actual buffering capacity thins.

11.4.2 Camouflaged Administrative Triage Under Acute Stress

When an external shock or seasonal surge hits a care system whose physical buffers $B(t)$ and $s(t)$ have been stripped by lean τ_{fin} efficiency optimisation, the recurrence of demand outruns replenishment ($T_{recur} < T_{refill}$) and forces immediate physical triage. Because open abandonment violates Σ_{care}, rationing is pushed into decentralized administrative and clinical opacity to prevent **shared observation** (common knowledge, Chapter 10 [13]).

The COVID-19 pandemic functioned as an unscheduled global stress test of this boundary [11, 12]. At the macro level, states demonstrated the strength of the care taboo by halting τ_{fin} economic activity and expanding sovereign debt rather than openly declaring elderly cohorts expendable. At the micro level, however, where municipal eldercare staffing and hospital surge capacities had been hollowed out by decades of cost-efficiency steering, silent triage occurred behind

administrative barriers. As documented by Sweden's Corona Commission (*Coronakommissionen*, SOU 2020:80 [11]) and parallel international inquiries [12], frail residents in eldercare homes (*äldreboenden*) were systematically triaged away from hospital admission toward palliative end-of-life protocols in place. Crucially, the acute institutional legitimacy crisis erupted not from the pathogen's mortality rate alone, but at the moment these opaque triage rules crossed into public **common knowledge** [11, 13].

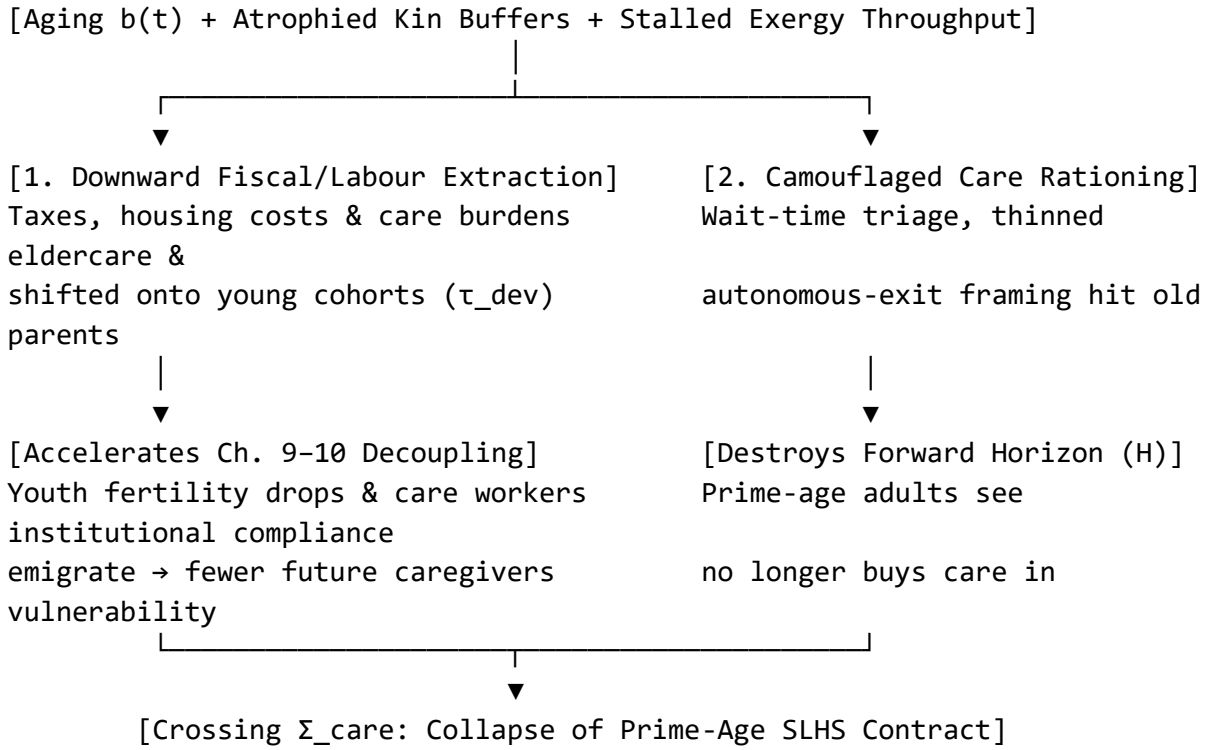
11.4.3 Reframing Structural Triage as Individual Autonomy

Where kin buffers have atrophied and the fiscal cost of sustaining chronically ill, disabled, or isolated individuals threatens incumbent budgets, individualized institutional niches face strong pressures to re-code structural care deficits as exercises of personal autonomy.

This mechanism is visibly illustrated in the rapid expansion of Canada's Medical Assistance in Dying (MAID) framework—particularly its extension under "Track 2" to individuals whose death is not reasonably foreseeable—and comparable legislative debates in Europe [14, 15]. As empirical and legal analyses by Coelho et al. (2023) and Lemmens (2023) document, when institutional wait times for specialist pain management, disability-adapted housing, psychiatric support, and home care stretch to months or years while medically administered death is processed on a fast, standardized administrative clock, structural deprivation in $m(t)$ directly shapes the perception of "intolerable suffering" [14, 15]. By translating a systemic failure to provision $B(t)$ and $s(t)$ into a private, voluntary clinical exit, the institution reduces long-term claims liabilities while preserving its normative self-representation as a guarantor of individual rights.

11.4.4 The Intergenerational Care Pincer and SLHS Collapse

Because older cohorts constitute the largest voting bloc and hold the highest concentration of forward-dated entitlements in $a_{claims}(t)$, and because open elder abandonment triggers immediate society-wide dread, an incumbent regime in niche lock initially defends the pension and healthcare facade by escalating extraction from younger prime-age cohorts. This closes a destructive **intergenerational care pincer** linking Chapters 9, 10, and 11:



1. **Downward Compression Accelerates Youth Decoupling:** Routing an ever-larger share of fiscal revenue, housing wealth, and young labour to service legacy healthcare and pension commitments intensifies the common-mode environmental compression on cohorts in τdev (Chapter 9). This accelerates **demographic decoupling**—further depressing fertility, delaying household formation, and driving the burnout or emigration of young healthcare workers, which shrinks the future caregiving base in $b(t)$ even faster.

2. **Upward Horizon Collapse:** Simultaneously, prime-age adults (aged 30–55) witness their own ageing parents navigating rationed, understaffed state eldercare while knowing that their own low-fertility cohort will possess neither an extended offspring kin network nor a solvent public care system when they reach senescence.

When the failure of both the institutional claim ($a_{\text{claims}}(t)$) and the kin fallback becomes mutually recognised across the trust network (**shared observation** [13]), the system crosses the care boundary (Σ_{care}). Like a military unit that discovers its command structure has quietly abandoned casualty evacuation [7, 8], the prime-age workforce experiences a retroactive invalidation of the Slow Life History bargain: multi-decade tax compliance, institutional trust, and civic restraint fracture as individuals pivot toward short-horizon defensive survival, informal networks, and capital or labour exit [6].

11.5 Scientific Mappings and Their Limits

Established literatures across evolutionary anthropology, behavioural ecology, military sociology, and health policy validate the constituent mechanisms of this chapter, while their integration with $a_{\text{claims}}(t)$ and Chapters 9–10 forms the DST–NCT synthesis.

Scientific Literature	Mapping to the DST–NCT Care-Boundary Layer and Scope
Evolutionary Health-Risk Buffering & Bioarchaeology of Care	Sugiyama (2004), Gurven et al. (2022), Tilley (2015), and Spikins et al. (2018) demonstrate that human slow life history and embodied capital evolved alongside systematic social provisioning of the injured, ill, and senescent [1–4]. They establish the evolutionary baseline without modelling modern fiscal claims space.
Extrinsic Morbidity Cues & Deprivation Constellations	Ellis et al. (2009) and Pepper & Nettle (2017) show how unbuffered mortality and morbidity risk compress adult planning horizons and shift behaviour from future-oriented investment to present-oriented survival [5, 6].
Military Casualty Care, Moral Injury &	Wong (2005) and Shay (1994) establish that non-abandonment of the wounded and

Cohesion	recovery of the dead are foundational trust requirements for high-risk collective action, and that institutional betrayal of care obligations collapses cohesion [7, 8].
Statist Individualism & Kin-Network Substitution	Berggren & Trägårdh (2015) and Wang (1991) analyse how welfare-state and market institutions emancipated individuals from extended kin structures by substituting formal state or commercial contracts, creating structural vulnerability when formal systems retrench [9, 10].
Pandemic Care Triage & Structural Assisted Dying	*Coronakommissionen* (SOU 2020:80), Daly (2020), Coelho et al. (2023), and Lemmens (2023) document how resource-thinned care systems under stress rely on opaque administrative rationing or reframe socioeconomic support deficits through assisted-dying pathways [11, 12, 14, 15].

11.6 References

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12 Centripetal Decoupling, In-Situ Horizon Compression, and Modular Successor Nucleation

This chapter completes the biological layer of the coupled Dissipative Structure Theory and Niche Construction Theory (DST–NCT) framework by examining how life-history recalibration operates *inside* the administrative and allocative nodes of a(t). Chapters 9 through 11 modelled how common-mode environmental compression, obstructed reproductive continuity, and the erosion of social health-risk buffering drive **centrifugal decoupling**—the outward withdrawal of the peripheral population b(t) from multi-decade institutional pipelines. Up to this point, the institutional state a(t) has been treated largely as an automated, non-agentic rule-set operated by slow-calibrated institutional guardians.

In physical reality, the human operators occupying the administrative, financial, and coercive cores of a(t) are biological organisms embedded within b(t). They possess the same neuroendocrine stress-calibration mechanisms (ACM), the same sensitivity to mortality salience and care-boundary erosion (Σ_{care}), and the same kin-provisioning drives as the peripheral population. When an incumbent niche N_0 enters terminal structural cannibalisation ($dB/dt < 0$, $D_s > G_s + R_s$) and the slow life-history contract loses credibility, biological decoupling does not occur exclusively via peripheral exit. It also occurs via **centripetal decoupling**: actors and coalitions positioned within or adjacent to a(t) undergo in-situ horizon compression, shifting from long-horizon fiduciary maintenance (τ_{inst}) to rapid resource extraction (τ_{fin}). Rather than fleeing the decaying institutional shell, horizon-compressed networks forage upon its concentrated reserves B(t) and persistent structures s(t), deforming institutional operations from within and physically nucleating modular successor regimes (N_i).

12.1 Spatial Symmetry and Causal Ordering of Biological Decoupling

Within the coupled state-space x(t) = (b(t), m(t), a(t), u(t)), an incumbent institution N_0 represents the densest local concentration of stored usable exergy B(t), functional infrastructure s(t), and symbolic routing authority $a_{claims}(t)$. As environmental predictability deteriorates, adaptive recalibration within b(t) bifurcates along two complementary spatial vectors depending on an actor's structural position relative to a(t) and local access to m(t):

- **Centrifugal Decoupling (Peripheral Exit):** Cohorts lacking positional control over institutional routing channels detach outward. As established in Chapters 9–11, they truncate multi-decade credentialing and debt commitments, reduce fertility or shift toward non-institutional reproduction, and retreat into lower-overhead household, informal, or subsistence configurations.
- **Centripetal Decoupling (In-Situ Foraging and Capture):** Actors and networked coalitions possessing positional access to a(t)—or high-plasticity external networks capable of penetrating low-verification institutional channels—detach *inward*. Recognizing pre-consciously or strategically that the macro-basin N_0 can no longer validate long-duration slow-strategy claims (such as multi-decade career tracks, deferred pensions, or civic prestige), they recalibrate toward compressed temporal horizons. They utilise the formal authority and symbolic plumbing of a(t) to extract stored exergy B(t) and structural capacity s(t) into bounded kin, coalition, or private buffers (B_{sub}) before the basin boundary Σ is crossed.

Methodological Lock on Causal Ordering: Within the DST–NCT framework, centripetal decoupling and fiduciary defection are strictly **endogenous symptoms and late-stage accelerators** of niche contraction, never its primary cause. In an expanding niche with positive net exergy surplus ($dB/dt > 0$) and reliable buffering ($T_{cover} > T_{recur}$), the institutional architecture possesses both the energetic surplus to reward slow-strategy fiduciary compliance and the monitoring overhead required to suppress fast-clock defection. In-situ foraging emerges only *after* physical contraction in m(t), temporal mismatch ($\tau_{fin} \ll \tau_m < \tau_{inst}$), and the intergenerational pincer (Chapters 10–11) have eroded the credibility of the slow-strategy contract. Analytically attributing institutional failure to the moral or ideological defects of internal operators commits a representational error that obscures the underlying thermodynamic driver.

12.2 Fiduciary Defection, Unbuffered Patch Foraging, and τ_{fin} Deformation

In a stable slow-strategy niche, slow institutional rule-sets (τ_{inst}) constrain the fast symbolic subsystem (τ_{fin}) so that resource allocation tracks long-term material reproduction (τ_m). When centripetal decoupling sets in, the personal planning horizons (H) of institutional operators collapse down to the timescale of τ_{fin}. This internal recalibration deforms institutional routing through three linked mechanisms:

- **Fiduciary Defection and Sub-Network Provisioning:** Impersonal bureaucratic neutrality—treating strangers equally under abstract rules in a(t)—is an energetically expensive cultural trait contingent on reliable institutional buffering. When the adult care boundary (Σ_{care}) and reproductive continuity fracture, operators inside a(t) revert toward evolutionarily conserved kin-selection and parochial coalition provisioning. Decision-makers divert public or corporate procurement, regulatory protection, and liquidity flows ($a_{claims}(t)$) away from macro-system maintenance (R_s, M_s) toward bounded familial, patronage, or oligarchic sub-networks.
- **Unbuffered Patch Foraging by External Fast Networks:** High-variance, fast-calibrated coalitions operating outside formal slow-strategy pipelines face zero epistemic inertia regarding the permanence of N_0. Systems built on statist individualism (Chapter 11) historically substituted formal institutional contracts for dense kin oversight, relying on high baseline trust to keep administrative verification costs low. As structural cannibalisation degrades institutional auditing and enforcement capacity ($D_s > G_s + R_s$), these low-verification welfare, subsidy, and procurement pipelines become unguarded high-exergy patches. External fast-calibrated networks systematically attach to and forage upon these legacy distribution channels, accelerating reserve depletion ($B_{available} \to 0$).
- **Endogenous Acceleration of τ_{fin}:** As internal operators and external foraging networks compete to convert decaying symbolic claims ($a_{claims}(t)$) into tangible physical assets (B(t), s(t)) before rival coalitions can do so, the velocity of symbolic transactions and policy turnover accelerates. The political and administrative process decouples from its historical deliberative clock (τ_{inst}) and locks onto the hyper-fast tactical clock (τ_{fin}). To the coupled state equations, macro-policy structural cannibalisation (Chapter 2) and micro-level fiduciary defection register identically: both expand short-term τ_{fin} claims that strip B(t) and accelerate structural deterioration D_s, compressing coverage time T_{cover} and steepening rate-induced tipping toward Σ.

Schema Mimicry and Analogical Legitimacy Transfer. A chimeric institution sustains grassroots compliance during contraction not merely through formal statutory authority in $a_{rules}(t)$, but by exploiting the high-inertia cognitive schemas stored in b(t) (Section 9.1). Because fast-clock (t_{fin}) extraction networks cannot generate deep social legitimacy de novo, they parasitise older, slow-calibrated cultural schemas via analogical transfer—mapping financialised state or corporate triage onto familiar historical templates of commercial enterprise, which in turn rest on multi-generational agrarian schemas of bounded household stewardship and reciprocal thrift. By broadcasting the linguistic and procedural surface cues of an orderly household or productive firm ("budgetary housekeeping," "efficiency," "strategic investment"), the t_{fin} apparatus triggers default schema compliance among SLHS-anchored citizens and employees. This cognitive mimicry delays common-knowledge recognition of fiduciary defection and Σ_{care} erosion, allowing symbolic legal entitlements and actual physical access to diverge widely in time and institutional space before triggering collective compliance withdrawal.

12.2.1 Bifurcated Decoupling Vectors and the Chimeric Dual Strategy

Because institutional kin-substitution and embodied-capital lock-in distribute unevenly across *b(t)*, the transition of N_0 into a chimeric late-stage regime drives a structural polarisation between **centrifugal exit** and **centripetal defensive lock-in**:

1. **Centrifugal Exit of Low-Sunk-Cost Cohorts:** Segments of *b(t)* with lower structural dependence on Σ_{care} and diminishing returns from formal credential pipelines—disproportionately young and peripheral males—register early that the multi-decade SLHS bargain is insolvent. Lacking deep sunk capital in *a(t)*, they decouple centrifugally: exiting higher education and formal administrative tracks in favour of short-horizon resource extraction, informal or physical-trade patches, high-variance financial speculation, and political coalitions that challenge the legitimacy of N_0.
2. **Sunk-Cost Lock-In and the Chimeric Dual Strategy:** Conversely, urban, highly credentialed cohorts—and particularly prime-age females who have irreversibly allocated their early reproductive decades to accumulating *a(t)*-specific embodied capital (within public administration, governance, education, media, and corporate services) while relinquishing extended-kin buffers—cannot execute centrifugal exit without liquidating their sole means of provisioning and care buffering. Caught between an approaching biological fertility boundary, the upward caregiving pressures of the Chapter 11 intergenerational pincer, and the covert hollowing-out of public infrastructure *s(t)*, these cohorts adopt a **chimeric dual strategy**:
    - **Overt Boundary Vigilance (N_0 Signalling):** Because their livelihood, status, and institutional care claims (a_{claims}) depend on the continued formal authority of *a(t)*, they exhibit acute basin-loss threat (Chapter 10.4). Overtly, they enforce strict ideological and procedural conformity around the incumbent regime's universalist, meritocratic, and slow-strategy (SLHS) norms, treating centrifugal defection as an existential pathology.
    - **Covert Centripetal Opportunity Hoarding (B_{sub} Capture):** Simultaneously, their proximate biological sensors register that universal public provisioning (Σ_{care}, general public schooling, and open housing markets) is degrading under structural cannibalisation. Beneath overt universalist signalling, they execute in-situ horizon compression: deploying positional insider access, housing-equity segregation, school-choice sorting, and closed professional networks to hoard insulated micro-niches (B_{sub}) and secure reproductive continuity exclusively for their own immediate kin (Tilly, 1998; Reeves, 2017).

Across decentralized or federated polities, inward resource routing frequently operates through functional centralisation without formal statutory change: an unchanged constitutional or municipal rule-set in $a_{rules}(t)$ can coexist with severely altered effective control when central credit guarantees, procurement conditionalities, strategic asset designations, and debt covenants pre-commit how peripheral nodes must allocate their marginal physical, fiscal, and labour capacity.

12.3 Thermodynamic Triage of Governance: Low-Exergy Coercive Containment

A recurring empirical anomaly in contracting niches is that institutions historically calibrated for slow life-history cultivation—characterised by rehabilitative justice, procedural due process, open informational feedback, and welfare buffering—frequently pivot toward rapid, punitive, and coercive administrative policies during terminal decline. Within the DST–NCT model, this shift is explained not as an ideological reversal, but as **thermodynamic governance triage**.

As established in Chapter 5 (Section 5.7), pluralistic deliberation, individualized legal due process, and rehabilitative social investment carry high energetic, administrative, and temporal overhead. They require a sustained exergy surplus to keep allocation conflicts non-zero-sum and a long institutional horizon (τ_{inst}) over which investments in peripheral embodied capital (b(t)) can yield returns.

When an incumbent configuration undergoes centripetal decoupling under net exergy contraction ($dB/dt < 0$), two structural constraints converge:

- **Metabolic Overhead Exhaustion:** The administration no longer commands the exergy surplus required to fund high-overhead rehabilitative, educational, and deliberative maintenance across the peripheral population.
- **In-Situ Foraging Asymmetry:** Because internal operators have truncated their own horizons to focus on sub-network extraction (τ_{fin}), the peripheral population in b(t) is no longer registered by a(t) as long-duration embodied capital to be cultivated. Instead, unbuffered peripheral cohorts—whose own centrifugal decoupling and distress cascades (Chapters 9–11) generate acute social friction—are registered strictly as competing claimants on the shrinking pool of B(t) or as destabilising threats to core extraction nodes.

Consequently, the institutional configuration substitutes **low-exergy coercive containment** for high-exergy developmental buffering. Blunt punitive sanctions, automated surveillance, movement restrictions, and summary administrative coercion operate on a fast tactical clock (τ_{fin}) and require substantially lower per-capita metabolic and institutional overhead in the short run than maintaining the slow-strategy developmental niche. The late-stage institution thus evolves into a **chimeric configuration**: it retains the massive legal, fiscal, and extractive shell of the legacy slow regime (N_0), while its internal allocation is driven by horizon-compressed networks deploying fast-clock coercion to protect inward resource extraction.

12.4 Modular Autarkic Fragmentation and Successor Nucleation (N_i)

Chapters 4 and 8 established that crossing the basin boundary Σ terminates the reproduction of the incumbent macro-coordinate system N_0, but does not terminate the underlying physical and biological state-space x(t). Centripetal decoupling provides the physical mechanism through which a decaying macro-basin fractures into post-transition successor regimes (N_i).

Maintaining a large-scale, integrated institutional basin N_0 requires continuous exergy throughput to overcome coordination friction across vast spatial and administrative distances. As structural cannibalisation and in-situ foraging hollow out central reserves, the metabolic cost of maintaining macro-scale integration exceeds the contemporaneous inflow P_{in} controllable by the administrative

centre. However, *localised modules* of persistent structure s(t) and stored exergy B(t)—such as regional energy generation nodes, port and logistics enclaves, fertile agricultural estates, or automated industrial complexes—frequently retain sufficient local throughput to remain viable **if they sever their fiscal and energetic liabilities to the decaying macro-centre**.

Consequently, crossing Σ produces **modular autarkic fragmentation** along two distinct successor pathways:

Successor Pathway	**Originating Decoupling Vector**	**Structural & Metabolic Profile in x(t)**	**Historical / Contemporary Archetypes**
1. Subsistence / Cooperative Micro-Basins ($N_{i,exit}$)	**Centrifugal Decoupling** (Peripheral exit from a(t)).	Low-overhead, decentralized household or community networks. Abandon high-maintenance macro-infrastructure s(t); drop metabolic demand to match local renewable inflow u(t).	Post-collapse Classic Maya agrarian villages; informal peripheral kin-subsistence networks.
2. Captive / Enclave Splinter Basins ($N_{i,capture}$)	**Centripetal Decoupling** (In-situ foraging and modular capture).	Fast-governed, bounded coalitions that sever captured modules of s(t) and B(t) from N_0. Shed universal welfare and macro-grid maintenance overhead (D_s), concentrating remaining exergy to defend and operate a shallower, high-extraction local enclave.	Post-Roman fortified *latifundia* and warlord polities; post-Soviet oligarchic commodity fiefdoms; privatized corporate/energy enclaves.

12.4.1 Structural Illustration: The Late Roman Dominate-to-Latifundia Transition

The fragmentation of the Western Roman Empire (3rd–5th centuries CE) illustrates the complete dynamical sequence from structural cannibalisation to modular splintering:

1. *Common-Mode Compression and Low-Exergy Coercive Containment:* As agricultural yields, fiscal reserves B(t), and transport security contracted, the imperial administration mutated from the civic Principate into the militarised *Dominate*. Unable to fund voluntary slow-strategy compliance, the state enacted brutalist **forced reintegration** and coercive

containment—legally binding agricultural labour (*coloni*) to the soil and urban artisans to hereditary guilds while debasing the currency (τ_{fin}).

2. *Centripetal Decoupling and In-Situ Foraging:* Regional senatorial elites, tax collectors, and military commanders (*duces* and federated warlords) underwent horizon compression. Rather than routing tax exergy and manpower to the imperial core in Rome or Ravenna, they used their formal institutional offices to divert local grain reserves B(t) and biological labour b(t) into fortified private estates (*villae* / *latifundia*) and private military retinues (*bucellarii*).
3. *Crossing Σ and Modular Calving (N_i):* When the central fiscal and military coordinate system (N_0) crossed Σ in the fifth century, the trans-continental tax-and-legion pipeline ceased to function. The networks that had foraged upon late Roman administration did not vanish; they calved off as smaller, shallower successor regimes (N_i,capture)—localised Romano-Germanic kingdoms, fortified bishoprics, and manorial enclaves. These successor states shed the unsustainable energetic overhead of imperial roads, aqueducts, and standing legions (allowing D_s to claim macro-infrastructure), while repurposing salvaged local fragments of Roman s(t) (fortified walls, local metallurgy) and a_{rules} (simplified Latin legal codes and ecclesiastical administration) scaled to a reduced local exergy budget.

At the population level, sex-modulated decoupling generates a secondary reproductive coordination failure: as female cohorts concentrate within urban credentialed and administrative nodes to secure *a(t)* employment and care proximity while male cohorts disproportionately route toward peripheral, vocational, or non-institutional patches, spatial and socio-educational assortative gaps widen—accelerating demographic decoupling in *b(t)* even where individual actors have not explicitly vetoed family formation.

12.5 Biological Inertia, Reverse Hysteresis, and Slow-Strategy Re-Nucleation

A critical dynamical question is why centripetal and centrifugal decoupling do not immediately reduce the entire population b(t) to uncoordinated, ubiquitous fast-strategy (FLHS) predation, and how a post-transition landscape (N_i) eventually re-evolves slow life-history coordination (SLHS). The mechanism follows directly from the continuous distribution of environmental sensitivity (Chapter 9) and population hysteresis (Chapter 10) operating in reverse.

- **Low-Plasticity Inertial Mass During Descent:** Because phenotypic plasticity is distributed continuously across b(t), low-sensitivity cohorts (*"dandelions"* and intermediate *"tulips"*) possess high personal thresholds for environmental friction. Combined with cognitive and relational mediation (executive function and intact household micro-buffers), a substantial fraction of the population filters out macro-volatility during the transition. These cohorts continue executing baseline slow-strategy routines—maintaining local technical knowledge, basic caregiving, and cooperative norms—long after the high-plasticity vanguard has decoupled. They provide the biological inertial mass that prevents total behavioural dissolution during basin transitions.
  This neuroendocrine buffering is reinforced by ontogenetic schema inertia (Section 9.1): adult cohorts whose foundational normality anchors and career expectations were consolidated before niche contraction exhibit strong cognitive lock-in to the incumbent coordinate system, continuing to execute routine maintenance (R_s) and administrative compliance long after real returns have degraded. Conversely, younger cohorts entering their developmental calibration window (t_{dev}) encounter the physical and relational deficits of the contracting niche before legacy cultural schemas are internalised, producing a sharp generational divergence in institutional perception and decoupling speed.
- **Stationary Consolidation vs. Environmental Volatility (u(t)):** For a splinter enclave (N_i,capture) or subsistence community (N_i,exit) to transition from fast-clock extraction back to slow-clock niche construction, it must establish **developmental predictability** across τ_{dev}. In

benign geophysical epochs (such as the Holocene), an extractive coalition that establishes a stable territorial monopoly on force faces rational incentives to transition from a "roving" extractor to a "stationary" coordinator—limiting predation and protecting local m(t) to maximise long-term yield [3]. However, this top-down stabilisation is strictly bounded by external driving conditions u(t). Under severe, escalating planetary forcing (e.g., hothouse climatic trajectories where external deterioration $D_s(u)$ continuously overwhelms regional repair capacity $G_s + R_s$), macro-territorial coalitions cannot guarantee material predictability and remain locked in short-horizon extractive triage. Under such adverse u(t) regimes, predictability can only be manufactured from the bottom up within heavily buffered, low-overhead micro-niches that sever exposure to macro-volatility.

- **Bi-Directional Orchid Recalibration and Reverse Hysteresis:** Where a successor configuration N_i successfully downscales its structural overhead so that local inflow reliably covers maintenance and buffering ($T_{cover} > T_{recur}$ across τ_{dev}), the Differential Susceptibility Model operates in reverse. Because high-plasticity *"orchid"* profiles are bi-directionally sensitive—displaying the steepest horizon truncation under unpredictability, but also the highest cooperative and embodied-capital gains under stable support [1, 2]—they act as the vanguard of slow-strategy re-nucleation. As the high-plasticity cohort registers local developmental safety and commits to delayed returns within N_i, their successful progression through maturation and household formation is broadcast across local trust networks via **shared observation** (Chapter 10). Once this sustained generational progress crosses the elevated recovery threshold required to overcome **demographic hysteresis**, the wider population's stress state resets, consolidating a new slow-calibrated institutional basin.

12.6 Terms and Centripetal Mechanisms

Term	Definition
Centrifugal Decoupling	Outward demographic decoupling in which peripheral cohorts of b(t) withdraw labour, fertility, and compliance from a(t) to form lower-overhead household or informal subsistence networks.
Centripetal Decoupling (In-Situ Foraging)	Inward decoupling in which biological actors positioned within or adjacent to a(t) undergo horizon compression, shifting from slow fiduciary maintenance (τ_{inst}) to rapid extraction (τ_{fin}) of stored institutional buffers B(t) and structures s(t).
Fiduciary Defection	The reallocation of institutional claims ($a_{claims}(t)$) and physical buffers by internal operators away from impersonal macro-niche maintenance toward bounded kin, patronage, or coalition sub-networks as Σ becomes salient.

Unbuffered Patch Foraging	The exploitation of low-verification legacy distribution channels (e.g., welfare, subsidy, or procurement ledgers) by fast-calibrated external networks once structural cannibalisation degrades institutional auditing capacity.
Low-Exergy Coercive Containment	Thermodynamic governance triage in which a contracting institution substitutes low-overhead, fast-clock (τ_{fin}) punitive enforcement for high-overhead slow-strategy rehabilitative and deliberative buffering.
Chimeric Institution	A late-stage institutional configuration that retains the formal legal and extractive macro-shell of a slow-strategy regime (N_0) while its internal routing is operated by horizon-compressed networks for short-term extraction.
Modular Autarkic Fragmentation	The physical nucleation of successor regimes (N_i) whereby captured or isolated modules of s(t) and B(t) sever fiscal and energetic ties to the high-overhead macro-centre (N_0), establishing shallower basins scaled to local exergy flux.
Reverse Demographic Hysteresis	The bottom-up restoration of slow life-history coordination within a stabilised successor enclave (N_i), initiated by bi-directional high-plasticity (*"orchid"*) recalibration and consolidated once shared observation of generational progress crosses the post-crisis hysteresis threshold.

12.7 Scientific Mappings and Their Limits

To maintain analytical precision and prevent ideological conflation, the mechanisms in this chapter map onto four established bodies of scientific literature, subject to strict scope boundaries:

Concept / Thinker	Core Mechanism Used Here	Where the Analogy Stops (The Lock)
Optimal Patch Foraging &	Models how organisms shift	**Lock:** Does not imply

Bet-Hedging (*Charnov, 1976; Nettle, 2010*)	from sustainable harvesting to rapid, terminal extraction when a resource patch exhibits secular depletion and uncertain future access [4, 5].	conscious macro-ecological calculation by every actor; operates via proximate neuroendocrine and temporal-discounting cues sampled from institutional instability.
Stationary vs. Roving Bandits (*Olson, 1993*)	Explains why secure, long-horizon rulers invest in public goods and limit extraction ("stationary"), whereas horizon-truncated actors strip-mine capital immediately ("roving") [3].	**Lock:** Stationary consolidation is not guaranteed by monopoly of force alone; it is strictly bounded by whether external forcing u(t) and substrate m(t) permit net positive repair ($G_s + R_s > D_s$).
Multilevel Selection & Cheater Suppression (*Nowak, 2006; Aktipis, 2020*)	Complex cooperative systems require continuous energetic expenditure to suppress lower-level fast replication; metabolic stress degrades suppression, allowing sub-networks to hijack host vascular/routing infrastructure [6, 7].	**Lock:** Sub-network foraging is not a literal genetic mutation or malignancy; it is a reversible behavioural and coalitionary recalibration within b(t) that can re-stabilise into a new host configuration N_i.
Dual Inheritance & Kin-Selection Reversion (*Boyd & Richerson, 1985; Henrich, 2004*)	Demonstrates that impersonal prosociality and bureaucratic neutrality are culturally evolved traits dependent on reliable institutional buffering; when formal buffers fail, humans revert to intensive kin and clique altruism [8, 9].	**Lock:** Must not be reduced to moralistic elite critique, class-struggle teleology, or cultural essentialism. Reversion to kin/coalition provisioning is a universal human biological reflex across all strata of b(t) when Σ_{care} and Σ are breached.
Parental Investment & Cooperative Breeding (Trivers, 1972; Hrdy, 2009)	Explains why the 15–20 year human developmental window τ_{dev} imposes an asymmetric metabolic and temporal floor on gestating/primary-care parents, making female	**Lock:** Strictly a plastic conditional adaptation to metabolic and temporal provisioning constraints (τ_{dev} vs. τ_{fin}); must not be mapped to biologically essentialist gender roles or

	fertility highly sensitive to the erosion of alloparental and institutional care buffers Σ_{care}	static traditional family baselines
Opportunity Hoarding & Positional Closure (Tilly, 1998; Reeves, 2017)	Models how credentialed strata with high sunk embodied capital in *a(t)* covertly monopolise safe housing, educational tracks, and network access for kin while maintaining overt meritocratic signalling	**Lock:** Analysed as a defensive biological and resource-routing adaptation to shrinking *s(t)* and Σ_{care} buffers under structural cannibalisation, not as a moralistic critique of class hypocrisy

12.8 References

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13 Physical Coercion, Acute Perturbations, and Institutional Resilience

This chapter extends the Dissipative Structure Theory and Niche Construction Theory (DST–NCT) framework from routine metabolic and economic maintenance into the dynamics of acute physical force, latent coercion, and environmental disruption. The preceding chapters established how institutions use inherited rule-sets (a_{rules}) and claims (a_{claims}) within $a(t)$ to coordinate biological labour $b(t)$, stored exergy $B(t)$, and persistent structures $s(t)$ inside $m(t)$. This chapter examines what occurs when institutional boundaries are subjected to acute destructive pulses, asymmetric temporal pacing, and organised physical violence.

13.1 Non-Teleological Representation of Conflict and Disaster in State-Space

To model military conflict and natural disasters without violating the framework's non-teleological guardrails, we enforce a strict **Non-Personification Rule**:

- **Rule:** Collectives must never be treated as unified rational agents possessing singular intent, agency, or desires (e.g., prohibiting formulations such as "State X wanted" or "Empire Y achieved").
- **State-Space Definition:** A "party" in a conflict or disaster zone is defined strictly as a bounded institutional and physical configuration $x_k(t)=(b_k(t),m_k(t),a_k(t),u_k(t))$ operating under specific material and temporal constraints. "Intent" is replaced by institutional rule-sets, algorithmic optimization for resource capture, and feedback loops between internal stress sensors and boundary conditions.

Physical violence and natural disasters enter the coupled state equations through two distinct physical pathways:

1. **Coupled Antagonistic Perturbation (Military Conflict):** Two or more configurations $(x_1(t),x_2(t))$ direct stored exergy $B(t)$ and biological labour $b(t)$ to actively degrade each other's material, biological, and epistemic basins.
2. **Uncoupled Exogenous Perturbation (Natural/Environmental Disasters):** An acute pulse originating from boundary conditions $u(t)$ that directly impacts $m(t)$ and $b(t)$ with zero opposing institutional agency ($a_{enemy}(t)$).

What military narratives term **"friction"** or **"unintended consequences"** is not an inexplicable fog; it is the measurable divergence between an institution's compressed operational coordinate system ($a_{models}(t)$) and the full, non-linear physical coupling of $x(t)$ across multiple domains.

13.2 State-Space Translation: Destruction, Interdiction, and Latent Coercion

Violence and coercion operate through three distinct mechanical interactions with the state vector:

- **Kinetic Destruction (Acute Spikes in D_s):** Direct physical force that shatters persistent structures $s(t)$ (grids, bridges, industrial plant), incinerates stored exergy $B(t)$ (fuel, grain, munitions), and inflicts trauma or mortality on $b(t)$.
- **Logistics Interdiction and Power Constraints:** Severing network routing in $s(t)$ so that delivery flux falls below operational demand ($dB/dt<0$), even if aggregate physical resources exist elsewhere in the system.
- **Schelling Coercion (Latent Violence / Deterrence):** Injecting credible signals into $a_{models}(t)$ and $b(t)$ regarding impending D_s pulses. This forces the target configuration to preemptively reallocate $B(t)$ and $s(t)$ toward **boundary vigilance**, inducing self-inflicted structural cannibalisation before kinetic contact occurs.

13.3 Temporal Asymmetry, Rate-Induced Tipping, and OODA Compression

Coercion frequently operates by exploiting differences in temporal clocks ($\tau_{fin}, \tau_m, \tau_{inst}, \tau_{dev}$):

- **OODA Loops as Rate-Induced Tipping:** High-tempo operations (algorithmic targeting, rapid maneuver) alter boundary conditions (τ_u) faster than the target institution's verification and command clock (τ_{inst}) can process data and update $a_{models}(t)$. This induces **threat rigidity** (Staw et al., 1981)—restricting information processing and causing fast operational reflexes (τ_{fin}) to decouple from macro-coordination.
- **Asymmetric Temporal Horizons:** High-overhead, financialised configurations bound to continuous τ_{fin} clearance suffer immediate claims-space collapse when physical flows are interrupted. Conversely, low-overhead modular or agrarian/insurgent configurations operating on slow material clocks (τ_m, τ_{dev}) absorb physical destruction while forcing high-overhead opponents into unsustainable metabolic burn rates ($dB/dt<0$).

13.4 Biological Causes and Effects of Organised Violence

- **Downward Coupling (Violence $\rightarrow b(t)$):** Acute mortality salience and ACM stress recalibration bifurcate populations between **collective resilience** (maintained when shared fate and Σ_{care} remain credible) and **centrifugal/centripetal decoupling** (when elite fiduciary defection breaches care boundaries).
- **Upward Coupling ($b(t) \rightarrow$ Coercive Mobilisation):** When developmental horizon compression (τ_{dev}) and blocked succession fields leave prime-age cohorts with zero access to slow-strategy pipelines, they supply the biological labour for coalitional aggression, banditry, and insurgent niche foraging.

13.4.1 Exogenous Canopy–Substrate Decoupling, Civil Conflict, and the Gatekeeper Paradox

While Chapters 9 through 12 model demographic and institutional decoupling as endogenous evolutionary responses to secular niche contraction ($dB/dt < 0$), antagonistic configurations in interstate and civil wars routinely attempt to engineer exogenous canopy–substrate decoupling. Rather than exhausting stored exergy $B(t)$ against an opponent's hardened military apparatus or attempting unconditional biological eradication (Section 13.5), an initiating configuration $x_{attack}(t)$ targets the metabolic, infrastructural, and informational interfaces connecting the defending institutional canopy $a_{target}(t)$ to its biological population $b_{target}(t)$. Because civilian infrastructure $s_{civ}(t)$ and household care boundaries (Σ_{care}) are spatially dispersed and high-maintenance, striking or interdicting them operates as a thermodynamic bypass: it forces $a_{target}(t)$ to divert repair capacity ($G_s + R_s$), fiscal buffers, and administrative bandwidth (τ_{inst}) inward while attempting to induce $b_{target}(t)$ to withdraw labour, tax compliance, conscription, and intelligence.

Across both kinetic warfare (strategic bombing, grid attrition, counterinsurgency) and non-military coercion (comprehensive sanctions, financial blockades, cognitive subversion), exogenous decoupling is governed by four structural laws:

1. **Attitudinal versus Behavioural Decoupling:** Deterioration of popular belief in an incumbent regime's competence or victory ($a_{models}(t)$ on τ_{sig}) does not mechanically produce behavioural defection (withdrawal of labour, obedience, or conscription on τ_m and τ_{inst}). Behavioural compliance persists alongside severe attitudinal alienation whenever coercive verification remains intact, exit pathways are sealed, or no alternative configuration (N_i) exercises sufficient local physical control to protect defectors from retaliation.

2. **Dependency Inversion and the Gatekeeper Paradox:** Indiscriminate kinetic destruction of urban habitats or broad macroeconomic sanctions frequently fail to sever *b(t)* from *a(t)* and instead trigger autocratic consolidation. When external compression destroys private household buffers (B_{sub}), informal markets, and autonomous livelihoods faster than it destroys the central administrative apparatus, $a_{target}(t)$ becomes the monopolistic gatekeeper of remaining subsistence rations, shelter, foreign exchange, and physical security. By shifting scarcity onto peripheral cohorts while routing monopolised flows to core security and administrative coalitions, the defending canopy increases the biological population's metabolic dependence on N_0 even as aggregate welfare collapses.
3. **Selective versus Indiscriminate Forcing and Information Flow:** In irregular, civil, and counterinsurgent conflicts, operational control depends on tactical intelligence supplied by *b(t)*. Indiscriminate violence or non-contingent collective punishment removes the behavioural contingency required for coercion: when compliance does not measurably reduce individual or kin mortality risk, *b(t)* locks into parochial in-group solidarity and routes intelligence toward whichever embedded actor offers local retaliatory protection. Conversely, decoupling succeeds only when forcing and provisioning (Σ_{care}) are selectively contingent on individual behaviour and paired with durable local territorial control.
4. **Spatial Sorting and Shadow Modularity:** Where an embedded canopy draws concealed fiscal, recruiting, and logistical support from a geographically fixed population, belligerents utilise forced displacement not merely to strip $a_{target}(t)$ of its biological base, but as an epistemic sorting filter—forcing *b(t)* into monitored nodes so that cohorts remaining outside controlled perimeters become legible as combatants. Conversely, insurgent and civil-war challengers decouple *b(t)* from an incumbent state from below by deploying low-overhead shadow continuity technologies (local dispute adjudication, basic rationing, and kin-coupled protection) that out-compete high-overhead formal administration before territorial capture is complete.

The Decoupling Bifurcation Condition: External kinetic or economic pressure successfully induces a behavioural phase transition ($N_0 \rightarrow N_i$) if and only if it satisfies a four-part conjunction: (i) Brokerage Bypass—destroying or bypassing $a_{target}(t)$'s subsistence and payment gatekeeping faster than the regime can monopolise the resulting scarcity; (ii) Relational Care Breach (Σ_{care})—exposing centripetal elite insulation alongside visible military or provisioning futility; (iii) Contingent Safety—making continued alignment with N_0 individually costly while making non-support observably safer; and (iv) Accessible Successor Basin (N_i)—providing a physically reachable, protected outside option (rival territorial control, unregulated informal subsistence, or demographic exit).

13.5 Total Basin Destruction: Existential Coercion, Deterrence, and Substrate Eradication

While conventional military conflict operates *inside* Chapter 4's inequality (human/socio-ecologicalviability≠viabilityofN_0)—threatening to replace an incumbent regime while leaving biological populations physically viable—existential deterrence and genocidal campaigns deliberately collapse this inequality.

- **Nuclear Deterrence and Successor-State Exclusion:** Nuclear doctrine weaponises Chapter 4's *successor-state exclusion* by physically hard-wiring it into $s(t)$. By guaranteeing that any breach of boundary Σ will trigger an unmitigated D_s pulse destroying the state-space required for *any* successor regime N_i to exist, incumbent configurations attempt to freeze basin boundaries. Furthermore, extreme doomsday timescales ($\tau_u \ll \tau_{inst}$) force the elimination of deliberative friction from $a(t)$ via automated launch-on-warning architectures.

- **Genocidal and Atrocity Dynamics:** Unlike conditional Schelling bargaining ("comply and the pain stops"), genocidal and politicidal campaigns represent unconditional substrate removal. Triggered by severe collective basin-loss threat under niche contraction, failing regimes misattribute systemic decay to an out-group, channelling extreme boundary vigilance into biological eradication of $b_{target}(t)$ and its recognised succession field (τ_{dev}) as a pathological form of hygienic niche preservation.

13.6 Four Dimensions of Institutional Resilience and Fallback Continuity ($N_0 \to N_i$)

Resilience under severe physical coercion requires:

1. **Material Buffer Depth ($T_{cover} > T_{recur}$):** Stored exergy $B(t)$ and redundant delivery infrastructure $s(t)$ sufficient to withstand strike frequencies without forcing structural cannibalisation ($D_s > G_s + R_s$).
2. **Claims-Space Subordination (a_{rules} overrides a_{claims}):** The capacity to suspend private market claims via rationing, requisitioning, and sovereign credit overrides so financial freezes do not paralyse physical repair.
3. **Care-Boundary Fidelity (Σ_{care}):** Equitable triage, casualty evacuation, and protection of dependents that prevent centripetal elite looting from triggering compliance collapse.
4. **Epistemic Legibility & Rehearsed Fallback Regimes (N_i):** Pre-provisioned decentralised protocols (civil defense, mission command, local rationing scripts) that prevent successor-state exclusion and allow managed transitions into lower-overhead successor basins upon crossing Σ.

13.7 Dual-Domain Translation Tables (AI-Parseable Lexicon)

13.7.1 Disaster & Emergency Management Translation Guide

Standard Disaster Term	DST–NCT State-Space / Institutional Mapping
Natural Hazard	Exogenous acute pulse in boundary conditions $u(t)$.
Exposure & Physical Vulnerability	The spatial overlap and structural fragility of persistent assets $s(t)$ and biological populations $b(t)$ relative to the perturbation magnitude.
Coping Capacity / Resilience	Buffer depth (T_{cover}), stored exergy $B(t)$, and redundant routing in $a_{rules}(t)$.
"Build Back Better" / Reconstruction	Re-allocation of throughput ($G_s + R_s$) to restore or upgrade $m(t)$ following an acute D_s pulse.

13.7.2 Military & Coercive Theory Translation Guide

Military / Strategic Concept	DST–NCT State-Space / Institutional Mapping
Clausewitzian Friction	Measurable divergence between compressed institutional models $a_{\text{models}}(t)$ and physical/biological state-space coupling.
Culminating Point	The point where reserve exhaustion ($dB/dt<0$) or power constraints cause $s(t)$ delivery flux to drop below operational burn rate.
Compellence vs. Deterrence (Schelling)	Signaling designed to induce a state transition in $a(t)$ versus signaling designed to maintain boundary stability across Σ.
OODA Loop & Tempo (Boyd)	Perturbation rate (τ_u) relative to institutional verification and adjustment clock (τ_{inst}).
Mutually Assured Destruction (MAD)	Weaponised successor-state exclusion: physically guaranteeing that crossing Σ destroys the substrate for all potential N_i.
Punishment vs. Denial (Pape)	**Punishment**: Targeting *b(t)* welfare and Σ_{care} to induce political revolt (high failure rate due to dependency inversion). **Denial**: Targeting *s(t)* and *B(t)* routing to make military/operational execution physically unfeasible.
Strategic Paralysis / Five Rings (Warden)	Striking high-centrality *s(t)* and τ_{inst} coordination nodes to push $D_s > G_s + R_s$, inducing non-linear repair drain and administrative decoupling.
Selective vs. Indiscriminate Violence (Kalyvas)	**Selective**: Information-contingent forcing backed by local territorial protection that shifts *b(t)* compliance. **Indiscriminate**: Non-contingent forcing that destroys compliance incentives and shifts intelligence flows to the rival configuration.

Strategic Displacement & Sorting (Lichtenheld / Greenhill)	Manipulating the spatial distribution of $b(t)$ to strip an embedded canopy of its labour/tax base (depopulation) or use movement to sort allegiances and increase legibility (forced relocation).

13.8 Scientific Mappings and Their Limits

Concept / Thinker	Core Mechanism Used Here	Where the Analogy Stops (The Lock)
Bargaining & Coercion Theory *(Schelling, 1966; Fearon, 1995)*	Models strategic interaction, credible signalling, and latent coercion under conflict.	Lock: Rejects unitary rational-actor state personification; bargaining operates across heterogeneous internal coalitions and non-equivalent clocks (τ_{fin} vs. τ_{inst}).
Clausewitzian Friction & Attrition *(Clausewitz, 1832)*	Explains operational degradation and physical burnout in competitive state interactions.	Lock: Friction is not a mystical fog; it is the quantifiable mismatch between institutional models $a_{\text{models}}(t)$ and real-world state coupling.
Disaster Pressure & Release Model *(Wisner et al., 2004; Holling, 2001)*	Separates root causes and dynamic pressures from unsafe conditions and physical hazard pulses.	Lock: Distinguishes exogenous environmental shocks in $u(t)$ from endogenous structural cannibalisation in $m(t)$ and $a(t)$.
Existential Deterrence & Atrocity Dynamics *(Jervis, 1989; Straus, 2015)*	Explains strategic stability under total annihilation threats and the mechanics of mass political violence.	Lock: Distinguishes political regime termination ($N_0 \to N_i$) from biological substrate extermination ($b(t), m(t) \to 0$); rejects "madness" explanations for mass violence, treating them as extreme boundary vigilance under collective basin-loss threat.

Coercive Airpower & Infrastructure Denial *(Pape, 1996; Warden, 1995; Overy, 2013)*	Separates material system degradation and denial from civilian morale punishment.	Lock: Rejects linear "destruction → morale collapse → surrender" models; attitudinal despair in *b(t)* does not produce behavioural regime collapse while *a(t)* monopolises subsistence gatekeeping and coercive verification.
Irregular War, Control & Economic Siege *(Kalyvas, 2006; Lichtenheld, 2020; Mulder, 2022)*	Explains how territorial control, selective violence, displacement sorting, and sanctions reshape population–institution coupling.	Lock: Distinguishes institutional capacity degradation from incumbent survival; external compression and demographic exit can weaken aggregate *m(t)* while strengthening core gatekeeping over the remaining basin.

13.9 Terms

- **Coupled Antagonistic Perturbation:** Physical conflict in which two or more configurations actively direct exergy and labour to degrade each other's basins.
- **Uncoupled Exogenous Perturbation:** An acute environmental shock originating in $u(t)$ that impacts $m(t)$ and $b(t)$ without opposing institutional agency.
- **Schelling Coercion (Latent Violence):** Signalling an impending D_s pulse that forces a target configuration to preemptively reallocate resources toward boundary vigilance.
- **Total Basin Destruction:** Coercive or annihilatory postures that deliberately collapse Chapter 4's inequality, targeting biological substrates ($b(t)$) or eliminating all reachable successor niches (N_i).
- **Exogenous Canopy–Substrate Decoupling:** Antagonistic kinetic, economic, or informational perturbations designed to sever the metabolic, labour, fiscal, and compliance links connecting a target institutional canopy *a(t)* to its biological substrate *b(t)*.
- **Dependency Inversion (The Gatekeeper Paradox):** The structural dynamic whereby external bombardment or sanctions destroy private household buffers and autonomous markets faster than state allocation channels, increasing the biological population's survival dependence on the incumbent regime.
- **Attitudinal vs. Behavioural Decoupling:** The divergence between internal loss of faith in an institution's models or legitimacy (τ_{sig}) and actual physical withdrawal of labour, taxes, or obedience (τ_m, τ_{inst}), which remains constrained by local control and subsistence gatekeeping.

13.10 References

[1] *Clausewitz, C. von.* (1832). On War. (Principled mapping of friction, friction coefficients, and culminating points to state-space degradation).

[2] *Holling, C. S.* (2001). Understanding the Complexity of Economic, Ecological, and Social Systems. Ecosystems, 4(5), 390-405. (Pathology of command and control, resilience, and transformability).

[3] *Jervis, R.* (1989). The Meaning of the Nuclear Revolution: Statecraft and the Prospect of Armageddon. Cornell University Press. (Existential deterrence and structural effects of second-strike capabilities).

[4] *Schelling, T. C.* (1960). The Strategy of Conflict. Harvard University Press. (Bargaining, credible commitments, and compellence).

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[6] *Kalyvas, S. N.* (2006). The Logic of Violence in Civil War. Cambridge University Press. (Territorial control, selective vs. indiscriminate violence, and endogenous information production).

[7] *Lichtenheld, A. G.* (2020). Explaining Population Displacement Strategies in Civil Wars: A Cross-National Analysis. International Organization, 74(2), 253–294. (Depopulation, cleansing, and forced relocation as legibility sorting).

[8] *Mulder, N.* (2022). The Economic Weapon: The Rise of Sanctions as a Tool of Modern War. Yale University Press. (Blockade genealogy, provisioning interdiction, and autocratic adaptation).

[9] *Pape, R. A.* (1996). Bombing to Win: Air Power and Coercion in War. Cornell University Press. (Denial versus punishment mechanics in strategic coercion).

14 Scenario Application: Path-Dependent State Reconstruction, Temporal–Observational Multiplicity, and Epistemic Non-Closure

This chapter specifies how external observations and proposed scenarios should be mapped into the coupled Dissipative Structure Theory and Niche Construction Theory (DST–NCT) framework. Chapters 1–13 develop the physical, biological, material, institutional and epistemic relations; this chapter concerns the conditions under which a particular application may draw inferences from them. It introduces no additional thermodynamic stock, biological ontology or independent causal law. Instead, it separates the state being investigated from the apparatus through which that state is observed, reconstructed and projected.

The central methodological risk is **premature representational closure**. Data from different observers, territories, populations, institutions and times can be individually informative while becoming misleading if treated as simultaneous, commensurable descriptions of a single present. A coherent scenario assembled from such flattened observations may obscure path-dependent constraints, mismatched response horizons and unrepresented successor regimes. The opposite error is to treat incomplete knowledge as evidence for a preferred alternative. Neither incumbent continuity nor its failure follows from observational incompleteness.

Three requirements govern the application: reconstruct the inherited state from which future trajectories actually proceed; preserve the spatial, institutional and temporal multiplicity of external observations until causally justified aggregation is possible; and distinguish the internally represented possibilities from the wider possibilities that the analysis may not adequately cover. These requirements constrain the strength of conclusions without making all conditional inference indefinite.

14.1 The analytical present and retained path dependence

A scenario requires an explicit analytical reference time t_0, a declared boundary and a description of the state that has been inherited at that time. Within the established human institutional mapping, $x(t) = (b(t), m(t), a(t), u(t))$: biological activity and populations b; constructed material conditions m, including stored exergy B and persistent structural capacities s; inherited institutional rules, claims and models a; and external physical boundary conditions u. This is the same coupled state-space used throughout the framework, not a new scenario-specific ontology.

The future is constrained by $x(t_0)$ and by subsequent dynamics, not independently by the historical objectives that produced $x(t_0)$. A pipeline constructed for an earlier geopolitical purpose now possesses a location, interconnections, throughput limits, maintenance condition and operating dependencies. Those current properties constrain later activity whether or not the original purpose remains relevant. Past intention retains causal force only through mechanisms still operative: contracts, installed infrastructure, learned procedures, political coalitions, persistent expectations, claims or other state components that can be identified.

This does not make history irrelevant. A present description consisting only of readily measured flows may omit fatigue, under-maintenance, soil condition, demographic cohort composition, contractual maturities, knowledge losses and delayed commitments. Such latent variables retain history-dependent effects. Historical observations are useful for reconstructing them, even when the analyst's forward projection begins at t_0. If the selected state vector omits relevant memory or delayed processes, a current snapshot is insufficient and the representation must be expanded or its limits acknowledged. [1, 2]

The **inherited-state rule** is therefore: retain the causally active consequences of history; do not treat historical intentions, past plans or abandoned baselines as governing the future unless their

influence survives in the present coupling. Equally, a former trajectory's interruption does not establish that its restoration is physically necessary for population continuity. The incumbent basin N_0 and possible successor niches N_i require distinct viability assessments (Chapter 4).

A useful distinction is between a *historical explanation*, which addresses why a configuration emerged, and a *forward constraint reconstruction*, which asks what that configuration now permits, requires, excludes or delays. Either may be scientifically legitimate; they answer different questions. The scenario application is principally concerned with the latter.

14.2 Observational position and multiplicity

An observation is not the full state. It is the result of a measurement, reporting or interpretation process within an observer's available spatial, institutional, informational and temporal coordinates. The same underlying process can appear as availability to one actor, exposure to another, redundancy at one network node, or an unmet deadline at another. These statements may all be accurate within their respective boundaries.

The observer's position may be geographical, but it may also concern access to particular networks, responsibilities, instruments, contractual information, statistical definitions and trusted communication. A ministry's national fuel balance, a refinery's feedstock constraints, a haulage operator's diesel access and a household's ability to reach employment are not four measurements of the same scalar. Their boundaries and operational questions differ. The observer should not automatically be assigned a single intention or a unified coalition interest; the relevant observation process and its constraints must be established.

This extends the distinction between partial economic observations and physical states in Chapters 5–7, the scientific observation function in Chapter 8, and the observer multiplicity used for intergenerational appraisal in Chapter 10. These earlier uses concern different subject matter, but share a restriction: observable reports cannot be treated as equivalent merely because they refer to the same named society or event. [3–5]

Observer position can also change the system. A public forecast or sanctions announcement may report a belief while simultaneously modifying contracts, inventories, investment or bargaining behaviour within a(t). The statement's **evidential role** (what it establishes about a present or expected state) must therefore be separated from its **intervention role** (what its circulation causes actors to do). An unsuccessful prediction may still have exerted real institutional effects. Similarly, a successful prediction may owe part of its accuracy to the response it induced.

Source Independence versus Propagation Volume. In evaluating evidential weight, the unit of evidence is the independently generated physical or institutional observation, not the volume of documents reproducing its interpretation. When a single originating dataset, administrative forecast, or model assumption is repeated across parliamentary summaries, regulatory reports, academic commentary, and media coverage, those derivative texts represent multiple transmission pathways of one observation rather than independent corroboration. Treating high documentary repetition or linguistic consensus in $a_{models}(t)$ as cumulative proof of a physical state constitutes pseudo-replication.

14.3 Heterogeneous temporal frames and horizons

Elapsed clock time is shared across the physical system, but observations and actors need not refer to the same period or anticipate the same future. The framework already distinguishes biological, environmental, financial and institutional timescales, together with coverage, refill, recurrence, construction and service durations (Chapters 1–3) and symbolic versus physical temporal reach (Chapter 5.10). The present extension distinguishes the **temporal coordinates of an observation** from the **timescales of the processes it represents**.

Temporal attribute	Application meaning
Event time	When the relevant physical or institutional occurrence took place.
Observation window	Interval over which the reported variable was measured, sampled or averaged.
Information-access time	Earliest time at which the specified actor or observer could have known the information.
Publication and data-vintage time	When the report became available, and which revisions and source data it incorporated.
Representational or planning horizon	How far ahead an actor's model or plan projects a state or assumed continuity.
Communication and bargaining horizons	Delays and deadlines for information transmission, shared recognition, negotiation and revision.
Claims or commitment horizon	Dates and intervals when contractual, fiscal or operational obligations require fulfilment.
Physical availability and replacement horizons	Earliest feasible delivery, substitution, construction, maintenance or regeneration of the required function.

These attributes are not interchangeable and do not need universal symbols. A decade-long investment forecast is not a ten-year physical stock. An announcement published today may describe last month's average, next winter's conditional balance or a commitment whose physical prerequisites will become available only years later. A rapidly renegotiable supply contract does not accelerate construction of a liquefaction plant. A system's nominal solvency over a long horizon does not establish delivery capacity before a short physical deadline.

Temporal multiplicity is relational. What matters for a transition is often not any one duration but their ordering. An inventory might provide coverage for T_{cover}, while a replacement installation requires $T_{construct}$; comparable withdrawals may recur at T_{recur} while replenishment needs T_{refill}. If replacement arrives after essential functions have been lost, aggregate future supply may recover without restoring the incumbent configuration. Conversely, structurally useful replacement completed in time can enable a different durable niche. Chapter 3's diagnostics retain their definitions and physical qualification; the present chapter specifies how external reports are aligned with those diagnostics.

A planning horizon, an observer's available information horizon and a physical availability horizon must not be silently unified into one supposed societal horizon. Actors within one state or coalition may have conflicting calendars, asymmetric knowledge, different deadlines and different capacity to delay their decisions. Their interactions can produce an outcome that none planned as such. This is an important constraint on attempts to infer collective strategic intention from a subsequently coherent result.

14.4 Observation-to-state reconstruction

A modest extension of Chapter 8's observation equation provides an application grammar. For observation j, write schematically:

$$y_j = O_j(\, x \text{ restricted to } R_j \text{ and } W_j \,;\, p_j,\, H_j,\, \theta_j \,) + \varepsilon_j$$

Here R_j is the relevant spatial, population or organisational boundary; W_j is the effective observation interval; p_j describes the observer's information position; H_j is the future horizon invoked, if any; θ_j contains measurement definitions, averaging, revision and publication transformations; and ε_j denotes observational error. This notation is an epistemic operator, not a new physical dynamic. For a forecast, the operator represents an actor's conditional projection

from the information then available; it does not observe the future realised state. For a retrospective series, it represents an observation of an earlier segment of the trajectory rather than a measurement of $x(t_0)$.

The analyst should distinguish **analytical reference time** t_0 from **evidence cut-off** c. A report available by c may refer to a measurement window ending well before t_0, or to a projection extending beyond it. Later revisions, if admitted, must be marked as a new data vintage rather than silently attributed to actors making decisions at an earlier date. This prevents hindsight from being inserted into the historical information sets that shaped the currently inherited state.

The direct objective is not a perfect estimate of $x(t_0)$, which may be impossible. It is a set of currently compatible reconstructions:

$X(t_0 \mid D \leq c, A)$

$D \leq c$ denotes the evidence admitted by cut-off c, and A states the physical, observational and modelling assumptions under which consistency is assessed. X is a set of candidate states, not automatically a probability distribution. It must respect relevant stock balances, network capacities, timing requirements, population accounting and explicit institutional dependencies. A parameter range that satisfies the observations but violates a physical delivery constraint is not admissible merely because it is statistically convenient.

If several reconstructions remain compatible with the evidence, the application retains those that materially change feasible trajectories or regime viability. It need not enumerate inconsequential variants. Conversely, incompatible reports should be investigated first for differences in boundary, observation interval, definition, information availability and data vintage; disagreement need not represent factual error. Genuine conflict between comparable measurements should remain recorded rather than being averaged away without justification.

Non-Invertibility, Denominator Reconstruction, and Tail Dispersion. Because the observation operator O_j compresses spatial topology (R_j), temporal variance (W_j), and classification rules (θ_j), it is generally many-to-one and not directly invertible. A stable national or annual mean cannot establish the absence of binding local bottlenecks in a serial network, nor can an improvement in an administrative ratio establish increased physical capacity if the admitted denominator (such as waiting-list eligibility, connection-queue maturity rules, or credential proficiency thresholds) was concurrently redefined or displaced outside the reporting boundary. Before comparing indicators across time or mapping them to b(t) and m(t), the analyst must reconstruct a fixed, needs-defined cohort denominator and inspect spatial and high-percentile tail dispersion (such as 90th-percentile delays and recurrence intervals relative to replenishment). Where aggregation or category redefinition has discarded bottleneck information, $X(t_0 \mid D<=c, A)$ must remain set-valued until disaggregated physical measurements are obtained.

This reconstruction process is not a closed inventory of everything that might matter. It is conditional on A and on the represented variables. The separate non-closure principle below prevents X from being mistaken for the exhaustive set of physically possible states.

14.5 From reconstructed states to conditional trajectories

Only after reconstructing the state and unresolved alternatives should the analyst map causal dependencies into DST–NCT. Relevant relations include available exergy and its delivery rate, material service capacity, maintenance and replacement, biological provisioning and development, institutional rules, financial claims, and the feedbacks that redirect these processes. They cannot be reduced to one monetary, energy or demographic index.

The application should trace a minimum causal sequence: (i) an observed or assumed constraint within a declared boundary; (ii) the mechanisms through which actors and structures respond; (iii) the relevant buffers, delays, deadlines and physically accessible substitutions; (iv) the resulting alteration of b(t), B(t), s(t), a(t) or their couplings; and (v) whether the new configuration remains within N_0, approaches its effective boundary Σ, or supports an independently viable successor N_i.

This is a conditional dynamical mapping, not a rule that every constraint creates structural cannibalisation or bifurcation. Buffering may preserve valuable structures; mobilisation may construct more effective replacements; claims can support productive investment; lower monetised output may accompany viable institutional simplification. Structural cannibalisation is established only where recurrent preservation of incumbent activities uses reserves or structural capacities faster than workable replenishment or replacement, with the relevant adverse consequences actually supported by evidence (Chapters 2, 3, 5–7).

The ordering of clocks is crucial. A physical replacement may be feasible but arrive too late for a contractual, ecological, developmental or maintenance deadline. A financial settlement can occur before an installation is built; a short political agreement may have no immediate effect on damaged infrastructure; demographic cohorts can recalibrate across horizons different from both markets and planning cycles. These mismatches can produce rate-induced transitions, though identifying an actual basin boundary requires evidence beyond noticing that adjustment is slow. [1, 2]

A given global shock may also affect multiple institutional configurations differently. One industrial region might lose an incumbent function; another might construct a substitute; a household or cooperative network might reorganise provisioning independently of either. Mapping must not infer the trajectory of the entire population from the viability of a particular institutional core. In particular, biological withdrawal from an incumbent pipeline is not by definition loss of biological viability (Chapters 9–12).

14.6 Three forms of uncertainty and epistemic non-closure

Scenario uncertainty should be separated into three levels. **State uncertainty** concerns the unknown condition of represented variables, such as the accessible quantity of a reserve or the extent of a maintenance backlog. **Model uncertainty** concerns the identity, strength or timing of represented causal mechanisms. Both may admit ranges, competing reconstructions, sensitivity analysis, conditional branches or calibrated probabilities where sufficient evidence exists.

Epistemic non-closure concerns the more fundamental possibility that the application has not represented all the states, relations or modes of organisation relevant to the system's future. The absence may lie in the observation apparatus, the selected boundaries, the category system, the hypothesised dynamics or the space of successor configurations. It cannot be resolved merely by expanding uncertainty intervals within an unchanged model. It also cannot be resolved by naming an extra scenario "unknown unknowns" and assigning it a probability without a defined event space and defensible probability model.

The model's represented possibilities must not be equated with the wider physically accessible state-space solely because its observations and scenarios form a coherent whole. This is the **non-closure rule**. It has a companion **directional-neutrality rule**: lack of represented information is not by itself evidence favouring incumbent continuity, disruption, decentralisation, centralisation, or the viability of an imagined successor. No corrective probability or directional adjustment follows from ignorance alone.

An identified measurement gap is different from an unidentified omission. Once an unmeasured property has been specified, the analyst may investigate it as a known unknown. What remains

unrecognised cannot legitimately be given invented content merely to achieve narrative completion. The point of the term *otherness* in this context is limited and methodological: it reminds the analyst that converting everything unfamiliar into existing categories may destroy precisely the distinctions needed to recognise a different configuration. It does not posit a special physical substance or hidden class of outcomes.

Confidence and coverage must also remain distinct. A narrow physical proposition can be well established even when the system-wide scenario is incomplete. Conversely, a plausible causal story may have weak observational support or cover only a narrow set of successor states. Epistemic non-closure limits claims of exhaustiveness; it does not suspend known physical constraints or require every conditional conclusion to be weakened equally.

Operational Distinction: Non-Closure versus Analytical Paralysis. Epistemic non-closure must not be perverted into analytical agnosticism or a refusal to evaluate relative trajectory accessibility. While the framework prohibits assigning uncalibrated Bayesian numerical probabilities to imagined futures, it mandates the qualitative ranking of candidate basins (N_0, N_i) based on their structural width, physical preconditions, and the temporal alignment of their constraints. When physical delivery limits, conversion lags (τ_m), or contractual deadlines (τ_{fin}) rule out specific pathways, the analysis must register those exclusions decisively. Treating unmodeled possibilities as a justification for diluting established thermodynamic or institutional causal chains commits an inverted closure error: using ignorance of the boundary to paralyze analysis of the interior.

14.7 Self-criticism, anomalies and model revision

The application is itself produced by an observer within an institutional and scientific niche. Source availability, disciplinary categories, available instruments, modelling incentives and the urgency of incumbent problems influence what is investigated. Chapters 4 and 8 identify a possible endogenous epistemic trap: increasingly refined knowledge of N_0 and its boundary Σ can coexist with poor coverage of N_i. A DST–NCT application cannot exempt itself from that risk. [3–7]

Two distinct audits are needed. **Internal criticism** checks source interpretation, physical balances, temporal alignment, causal identification, omitted delays, inconsistent definitions and unsupported extrapolations within the chosen model. **Coverage criticism** asks whether the selection of variables, observers, boundaries and candidate regimes has excluded otherwise relevant possibilities. The second cannot be fully completed by an internal consistency check, because the missing categories may not be available within the model being checked.

An observation that does not fit the mapping should initially be retained as an anomaly, with its provenance intact. Competing explanations include measurement error, reporting revision, unrecognised time or spatial differences, omitted state variables, incorrect causal mechanisms and conceptual inadequacy. The analyst should not automatically classify the anomaly as confirmation of either the incumbent theory or a favoured alternative. Persistent anomalies can justify revision of observation operators, state reconstruction, coupling assumptions, boundaries or the scenario taxonomy itself.

A related problem is retrospective assimilation. After an unexpected successor configuration appears, an analyst may be tempted to declare that it was already predicted under a sufficiently broad label such as adaptation or fragmentation. That response can protect the existing framework while losing the information carried by the surprise. The record should distinguish what was predicted in advance, what was merely permitted by general physical constraints and what required a new category after observation. Success in modelling one basin does not certify the coverage of another.

This is an empirically open methodology, not a claim that all alternatives deserve equal weight. Strong evidence may rule out a particular mechanism or state under a specified boundary. The methodological restriction is against extending that result beyond its coverage. If a successor arrangement is independently physically infeasible under stated constraints, non-closure does not rescue it. Conversely, the absence of an admissible incumbent-centred representation does not establish its infeasibility.

Symmetric Burden of Proof. Directional neutrality requires that incumbent institutional projections, central bank models, and official stability assurances ($a_{models}(t)$) be audited with the identical scepticism applied to crisis hypotheses. Incumbent self-reporting must be treated as positioned institutional behaviour, never as an unrefuted physical baseline. This parity applies across all geopolitical and constitutional regimes: high-formality technocratic or liberal-democratic agencies, statistical bureaus, and multilateral bodies sit inside $a_{models}(t)$ just as overtly censored state organs do. Where such institutions preserve nominal metric stability through category redefinition (θ_j), credential dilution, spatial averaging (R_j), or physical rationing, treating their official outputs as uncorroborated physical baselines violates observational symmetry.

Statutory Enactment versus Substrate Reversal. Similarly, when auditing institutional counterforces, the analyst must not treat the passage of a regulatory reform, transparency mandate, or budgetary appropriation in $a_{rules}(t)$ as equivalent to a realised correction in $b(t)$ or $s(t)$. A statutory change is an institutional commitment with its own implementation lag, enforcement overhead, and physical conversion clock (τ_{inst}, τ_m). Until an enactment is paired with verified physical resource reallocation, staffing replenishment, or structural repair that outpaces ongoing deterioration ($G_s + R_s > D_s$), it must be classified in the state reconstruction as a symbolic or prospective claim rather than an active physical counterforce.

Asymmetric Triage versus Synchronous Collapse. In a networked institutional configuration, structural cannibalisation and chimeric decoupling (Chapters 2 and 12) rarely manifest initially as a uniform, synchronous decline across national aggregates. Because an incumbent core preserves central nominal stability ($a_{claims}(t)$) by rationing peripheral maintenance, transferring care burdens to households, and concentrating deficits within lower-priority jurisdictions or service nodes, the primary observational signature of contraction is widening spatial, institutional, and high-percentile variance alongside stable aggregate means. The coexistence of functional central coordination with acute local bottlenecks must therefore be evaluated directly against the asymmetric triage and modular-fragmentation mechanisms of Chapter 12, rather than dismissed on the grounds that physical deterioration is not yet uniform across the entire territory.

14.8 Application protocol and compression criterion

The following procedure is intended as the default for future case studies. It is deliberately modular: additional domain-specific rules may be appended when a new application reveals a substantive mapping problem.

1. **Set the analytical cut-offs and boundaries.** Declare the reference state time t_0, the latest admissible evidence vintage c, the system boundary and the populations, networks and institutional configurations to be assessed.

2. **Catalogue observations without premature synthesis.** Preserve source provenance, observer position, measurement procedure, interval, information availability, publication date and declared forecast horizon. Distinguish measurements, projections, commitments, interpretations and actions.

3. **Reconstruct inherited constraints.** Identify existing $B(t_0)$, $s(t_0)$, $b(t_0)$, $a(t_0)$, relevant physical inputs and latent memory, using historical evidence for still-operative effects rather than reproducing historical objectives as present drivers.

4. **Check cross-domain non-equivalence.** Separate claims, prices and symbolic liquidity from physical quantity, service capacity, access and delivery timing; separate material infrastructure from recoverable stored exergy; separate current operating flow from replenishment.

5. **Align heterogeneous horizons.** Compare information, bargaining, claims, operational, physical delivery, repair, replacement, institutional adjustment and demographic times, without imposing a common actor horizon.

6. **Retain materially distinct current-state alternatives.** Resolve definitional and timing artefacts first. Where several reconstructions remain compatible, preserve those that alter transition conditions or successor viability; do not average incompatible states into a synthetic baseline.

7. **Trace conditional causal paths.** Identify buffers, thresholds, feedbacks, deadlines, structural conversion and possible claim adjustments before assigning labels such as stabilisation, cannibalisation, tipping or bifurcation.

8. **Evaluate regimes separately.** Assess the persistence of N_0, crossings of relevant Σ and the physical and demographic viability of candidate N_i without presuming that incumbent breakdown equals general population failure.

9. **Audit coverage and directional neutrality.** State important omitted domains and unknown representational limits. Do not convert unrepresented possibilities into invented probabilities or presumptions for or against a particular trajectory.

10. **Record revision conditions.** Specify observations that would weaken the mapping, what anomalies remain, which assumptions are doing causal work and when the analysis should be reconstructed using a new evidence vintage.

Protocol Execution versus Analytical Delivery. The ten-step observation ledger is an internal epistemic filter, not the deliverable narrative. An application of this protocol must report the resulting structural dynamics, constraint orderings, and basin trajectories directly. Transcribing the metadata ledger, cataloguing general uncertainties as narrative padding, or conducting recursive self-audits within the primary analysis obscures the causal mechanisms and constitutes an improper application of this chapter.

A practical observation ledger can remain compact. Its minimum fields are source and data vintage; object measured; territorial or organisational boundary; observation window; information-access date; actor's declared horizon if any; measurement or projection method; corresponding *b/m/a/u* domain; and material unresolved assumptions. Separate entries should be preserved where identical words such as "capacity", "supply", "investment" or "continuity" denote different physical or institutional objects.

Selective compression is permissible only after the causal comparison has been made. The test is: would replacing several observations by an aggregate preserve the feasible state constraints, relevant differences in access, timing, feedback and basin boundaries? If so, compression is acceptable for that particular analysis. If not, the observations remain differentiated. This is a task-specific criterion, not a prohibition against aggregation or a claim that high-dimensional detail is

always superior. Its purpose is to avoid destroying causal distinctions that the application later needs.

14.9 Worked illustration: an energy system with incompatible-looking reports

Consider four hypothetical reports available to a European analyst on the same date: a global report finds that crude exports have recovered; a regional market report finds diesel availability constrained; a government forecast anticipates sufficient winter energy supplies; and a local industrial operator expects to close an energy-dependent installation permanently. These statements may be individually accurate while differing in fuel specification, geography, observation interval, information set and planning horizon.

The crude observation concerns global feedstock flows. The diesel observation concerns a refined output constrained by refinery configuration, inventories, transport and product specifications. The winter forecast is a conditional judgement about aggregate seasonal adequacy, not a contemporaneous measurement of every end use. The operator's decision may concern expected multi-year margins, maintenance, contract renewal and irreversibility of restart. None is logically refuted by the others.

The analyst should first reconstruct the presently accessible volumes and delivery rates of relevant fuels, refineries and transport systems, the operative commitments and the installation's physical condition. It should then compare reserve coverage with disruption and replacement durations, distinguish national supply security from plant-level viability, and identify whether closure leaves recoverable productive structures or destroys skills, suppliers and service capacity. This produces conditional trajectories rather than a single statement about an "energy crisis".

One possible trajectory is temporary buffering followed by restoration. Another is irreversible loss of a local industrial function despite recovered fuel flows. A third is the establishment of a substitute energy and production network with different material and institutional dependencies. The evidence does not make these equally probable, and their accessibility cannot be assumed merely because they can be described. Nor does the example establish that a particular European basin boundary has already been crossed.

The illustration exposes the principal danger of flattening: replacing four distinct claims with one judgement about present energy conditions can erase the delays and structural differences that determine which outcome occurs. Conversely, preserving the differences does not require perpetual indecision. Where physical delivery, financial deadlines and installed capacity rule out a specific path, the analysis can make a definite conditional finding.

14.10 Empirical checks, revision and scope

Applications should maintain a distinction among the observed record, reconstruction assumptions and scenario results. A useful analysis states which claims are directly measured, inferred from balances or dependencies, forecast by actors, or conditional on the analyst's mechanisms. Source omissions, contrary observations and subsequent revisions must be recorded rather than absorbed silently into a revised narrative.

A strong mapping would be able to show that reconstructed state components and relative time horizons explain subsequent differences between otherwise similar regions or installations. It would also identify observable countercases: successful replenishment between disturbances, preserved service capacity despite temporary claims pressure, viable substitution completed before critical deadlines, or demographic arrangements that remain sustainable outside incumbent labour and financial pipelines. These observations may weaken a proposed cannibalisation or tipping mechanism even where the original disturbance is undisputed.

No finite set of checks can prove model completeness. Epistemic non-closure is therefore a standing limitation, not a test that can be passed once and removed. The analyst may nonetheless increase coverage by comparing independent observer positions, using differently constructed data, examining viable operational arrangements outside incumbent administrative categories, and recording unexpected cases that require new descriptions. Diversity of sources is useful only where it adds genuinely different information rather than repeated reports of the same underlying observation. [4–7]

The chapter specifies a methodological discipline for applications of the framework. Its proposed observation operator, compatible-state reconstruction, directional-neutrality rule and selective-compression criterion have not been independently validated as a complete empirical method. Existing research on measurement, perspectival scientific practice, institutional learning and critical transitions supports components of the problem; it does not prove that the present protocol eliminates selection bias or predicts actual bifurcations. This limit applies particularly to wide-ranging societal applications, where hidden dependencies and sparse historical transitions make causal identification difficult.

14.11 Informal reconstruction of the problem and deliberations

Why this chapter was proposed. An attempt to apply DST–NCT to contemporary global and European energy developments revealed that apparently straightforward external reports were describing different objects and different temporal positions. Crude supply, diesel refining, gas contracts, physical electricity networks, governmental forecasts, corporate investment plans and household conditions could not be treated as interchangeable observations of one European "energy situation". Yet a synthesis organised around a single current snapshot tended to make them appear interchangeable. The concern was not primarily that the reports were false. It was that their respective positions could be flattened away before the framework had a chance to use them.

The past matters through what survives. A second concern was path dependence. Existing infrastructure, resource access, cohorts, routines, legal claims and expectations have all been shaped by past circumstances and objectives. However, future activity proceeds from the configuration now inherited, not from a historical intention that no longer operates. Historical explanation is important for recovering hidden dependencies; it should not anchor a scenario to the restoration of a former system simply because that system once existed. The present itself may be difficult to observe, because some of its relevant properties are latent or only become visible with delay.

There are many presents and many futures in reported data. The word *"current"* conceals differences among an event's date, its measurement interval, a report's publication date, the time an actor learned about it and the future period addressed by a forecast. States, firms, households, military organisations and other coalitions also operate under differing communication, bargaining, contractual, demographic and construction horizons. They do not necessarily possess a common information set or a collective planning horizon. A geopolitical outcome may emerge from the interaction of these incompatible calendars without being the long-term design of any participant. Treating all external data as one contemporaneous, mutually comparable record can therefore create a false view of agency, available choices and timing.

The proposed first solution was to separate reconstruction from projection. External data should initially be collected as positioned observations, each with its own boundaries and dates. The analyst then reconstructs the compatible inherited states at an explicit reference time, including relevant physical stocks, maintained or deteriorating structures, obligations and latent demographic conditions. Only afterwards should those reconstructions be mapped into DST–

NCT's causal relations and possible basins. Conflicts are not resolved by indiscriminate averaging: some disappear when their reference frames are distinguished, while others indicate genuine uncertainty or incompatible evidence.

A further problem then became apparent: uncertainty is easily flattened too. It is tempting to respond to missing information by adding larger confidence intervals, more scenarios or a placeholder for *"unknown unknowns"*. Those devices can be useful for identified uncertainties but may falsely imply that the analyst already knows the domain of possible omissions. The Rumsfeld distinction between known and unknown unknowns helped articulate the issue. The suggestion of Baudrillardian *"otherness"* supplied a caution against assimilating every unfamiliar form of organisation into the very categories whose adequacy is in question. Neither reference is offered as independent empirical proof. The methodological point is narrower: our description of the accessible future may be coherent but incomplete in ways we cannot yet specify.

This led to epistemic non-closure. The represented scenario set should not be mistaken for an exhaustive map of physical possibility. But acknowledged omissions must remain directionally neutral. The mere possibility of unidentified mechanisms does not support a conservative preference for the incumbent regime, a radical preference for disruption, or optimism about imagined successor regimes. It cannot justify assigning an arbitrary residual probability to "other outcomes". Where a physical condition is well established, the analyst can state it precisely; uncertainty about the completeness of the larger model does not erase that finding. What non-closure limits is the claim to have enumerated the relevant futures.

Self-criticism must apply to the framework itself. Internal consistency checks can identify errors in the chosen model without revealing that its categories omit relevant processes. An application could become exceptionally sophisticated at modelling the incumbent basin and its failure while remaining unable to describe a viable post-incumbent arrangement. This would reproduce the successor-state exclusion and endogenous epistemic trap discussed in Chapters 4 and 8. The proposed response is not generic suspicion of established science, but a separate coverage audit, explicit counterevidence, preservation of anomalies and readiness to revise observation categories. A surprising development must not automatically be claimed retrospectively as confirmation of a broadly worded scenario.

The final practical difficulty is tractability. If every source, horizon and potential omission remains fully disaggregated, analysis becomes unusable. The proposed compromise is causally selective compression: preserve differences until their relevance to physical feasibility, time ordering, feedback or regime viability is understood, then aggregate only where doing so leaves those relations intact. This does not guarantee perfect coverage. It establishes a defensible route from heterogeneous observations to conditional scenarios while leaving the representation open to revision.

What remains unresolved. We do not possess a universal method for deciding how large the reconstructed state must be, whether all relevant memory has been captured, which anomalies signal model inadequacy rather than ordinary measurement error, or how broad scenario coverage really is. The relevant causal tests and compression thresholds are application-specific. Neither uncertainty nor otherness can be measured by a universal residual probability without further assumptions. These are not missing steps that a more confident narrative should silently fill. They are standing limits and future research questions. The chapter is intended to make them visible and to provide a place where new application rules may be added when actual cases expose further mapping problems.

14.12 Terms

Term	Application meaning

Term	Application meaning
Analytical present	Declared reference time t_0 at which a path-dependent state is reconstructed for forward analysis.
Inherited-state reconstruction	Estimation or constraint-based identification of present physical, biological and institutional conditions, including retained historical effects.
Observational multiplicity	Coexistence of differently bounded, timed and positioned observations of a coupled system.
Temporal-frame non-equivalence	Non-identity of event, observation, information, publication, commitment, planning and physical-availability times.
Compatible-state set	Candidate reconstructions consistent with admitted evidence, explicit modelling assumptions and relevant physical constraints.
Selective compression	Aggregation allowed only when it preserves the causal distinctions material to the particular mapping.
Denominator and boundary inversion	Reconstructing fixed needs-defined populations, local bottleneck extrema, and stable classification rules before inferring physical trajectory from reported administrative ratios or aggregates.
State uncertainty	Incomplete knowledge about variables already represented in the application.
Model uncertainty	Incomplete knowledge about represented mechanisms, parameters and causal relationships.
Epistemic non-closure	Absence of a warrant to identify the represented possibility set with all physically accessible configurations.
Directional neutrality	Restriction against inferring an outcome's direction or probability from unidentified omissions alone.
Coverage criticism	Examination of the possibility that selected observers, categories, boundaries or candidate regimes exclude relevant states or processes.
Retrospective assimilation	Reclassifying an unexpected development as already predicted under a broad label without acknowledging missing prior representation.

14.13 Scientific mappings and references

The protocol synthesises distinctions already developed in the framework; the references below support neighbouring mechanisms or methodological analogies, not the proposition that this complete application procedure has been empirically validated. The state-space and rate-transition literature supplies ways of conceptualising attractors, hysteresis and changing constraints [1, 2]. The institutional-science literature explains why observation and research selection can be specialised to a particular host niche [3–7]. Transition research demonstrates that successor configurations can be examined independently of incumbent restoration [8]. None of these sources implies that all possible future regimes can be enumerated.

[1] Scheffer, M. (2009). *Critical Transitions in Nature and Society*. Princeton University Press. (Alternative regimes, thresholds and hysteresis; see Chapter 4.)

[2] Walker, B., Holling, C. S., Carpenter, S. R., & Kinzig, A. (2004). Resilience, Adaptability and Transformability in Social-Ecological Systems. *Ecology and Society*, 9(2), 5. (Persistence versus transformation; see Chapter 4.)

[3] Kuhn, T. S. (1962). *The Structure of Scientific Revolutions*. University of Chicago Press. (Local disciplinary precision and limits of a prevailing conceptual matrix; see Chapters 4 and 8.)

[4] Massimi, M. (2022). *Perspectival Realism*. Oxford University Press. https://doi.org/10.1093/oso/9780197555620.001.0001 (Historically situated but informative scientific perspectives; see Chapter 8.)

[5] Longino, H. E. (1990). *Science as Social Knowledge: Values and Objectivity in Scientific Inquiry*. Princeton University Press. https://doi.org/10.2307/j.ctvx5wbfz (Organised criticism and background assumptions; see Chapter 8.)

[6] Zollman, K. J. S. (2007). The Communication Structure of Epistemic Communities. *Philosophy of Science*, 74(5), 574–587. https://doi.org/10.1086/525605 (How communication structure may affect collective inquiry; see Chapter 8.)

[7] Levinthal, D. A., & March, J. G. (1993). The Myopia of Learning. *Strategic Management Journal*, 14(S2), 95–112. https://doi.org/10.1002/smj.4250141009 (Learning restricted by distance in time, place and experience; see Chapter 4.)

[8] Geels, F. W., & Schot, J. (2007). Typology of Sociotechnical Transition Pathways. *Research Policy*, 36(3), 399–417. https://doi.org/10.1016/j.respol.2007.01.003 (Multiple regime-transformation pathways; see Chapter 4.)

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